

Credit Allocation Mix in Peer-to-Peer Lending: A Network Study on Bondora

Social Network Analysis - Group 7

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I Abstract

In this study we investigate the behavioural phenomena that influence individual borrowers' choice of loan uses. In the first study, we evaluate how loan counts within each loan use choice are influenced by borrowers' individual characteristics by asking "*Do borrowers vary in their loan use type and frequency based on the individual and loan characteristics?*". Using the Quadratic Assignment Procedure (QAP) linear regression framework, our first hypothesis asserts that borrowers sharing the same credit rating should choose similar loan uses, but with a preference for a lower loan count. We found that while this is a significant effect, the impact on loan count is minimal. The second asserts that borrowers sharing the same occupation area should share loan uses with a lower count as well. Here, we do not find a meaningful effect. The second study concerns which behavioural phenomena affect how borrowers decide which combination of loan purposes to request. We ask, "*What behavioural mechanisms affect the structure of borrower-to-loan-use connections*". With the use of Exponential Random Graph Models (ERGM), we first determine whether there is co-affiliation between borrowers overlapping on loan choices. We find that there is no systematic aggregate overlapping effect. Further, we evaluate whether there is an individual behavioural effect where borrowers minimise their loan types. The results indicate that there is a systematic individual behavioural bias. Assessing whether younger borrowers have different loan use selections, we fail to find evidence for this assertion. Similarly, we fail to find support for the idea that there are gender differences in how borrowers choose loan types.

1 Introduction

With the advent of online peer-to-peer (P2P) lending platforms, the traditional methods of financial intermediation have been usurped by individual choice; with no stringent third parties involved in transactions, individuals have greater power in sourcing credit. This presents new realities for individuals disenfranchised by the traditional financial system (Alistair Milne & Paul Parboteeah, 2017). Traditional financial literature shows that individually, investors exhibit behavioural biases (Agrawal, 2012; Ayal, Hochman, & Zakay, 2011) in how they choose their investments, however, numerous studies demonstrate that these behavioural biases are present among non-professional P2P borrowers. For example, Herzenstein, Dholakia, & Andrews (2011) and Lee & Lee (2012) suggest that users cluster around popular loans, exhibiting ‘herding’ behaviour. Literature focuses predominantly on funding success (Yao, Chen, Wei, Chen, & Yang, 2019), however, Ayal, Bar-Haim, & Ofir (2018) suggests that there is insufficient attention on how borrowers form their loan portfolios. As the authors indicate, this area of research is important in understanding why less professional investors deviate from known models of rational investor behaviour. Our study aims to model and understand how P2P borrowers’ behaviours influence what kinds of loans they acquire. In subsequent discussions, we consider investors as rational participants in credit markets, as described by theory, whereas borrowers refer to the users of P2P platforms.

Investigating behavioural biases, Ayal et al. (2018) find that investors’ familiarity with specific assets can make them seem less risky, making recognisable assets more attractive. However, such assets lead to undiversified portfolios. Evidence with P2P platforms supports the familiarity bias in lenders, as Galak, Small, & Stephen (2010) demonstrate that lenders choose borrowers who are similar across demographic attributes such as gender, occupation, and ethnicity. Moreover, Amar, Ariely, Ayal, Cryder, & Rick (2011) explains that borrowers show an aversion to holding a *large number of debts*, preferring to first reduce debt count, even if they had other debts with higher interest rates. This irrational behaviour demonstrates that individual borrowers face behavioural biases on the *type* and *number* of debts they select. Based on these notions, we formulate our first study, asking:

Research Question 1: *Do borrowers vary in their loan use type and frequency based on their individual and loan characteristics?*

This question is answered using linear Quadratic Assignment Procedure (QAP) regressions, as we can represent similarities across loan uses and attributes as matrices, weighted by loan use frequency, as inputs in the model.

Using data from LendingClub, Serrano-Cinca, Gutiérrez-Nieto, & López-Palacios (2015) study funding success, finding that the reported loan use is a significant factor explaining differences in default rates. Specifically, the authors argue that there is heterogeneity in the credit riskiness of individuals seeking specific loan purposes, indicating that credit rating can explain some variation in borrowers’ choice of loan use. Considering the preference for lower loan counts as well, we formulate the following hypothesis:

Hypothesis 1: *Borrowers with the same credit ratings choose fewer loans within their shared loan purpose*

Supporting the familiarity bias, Ravina (2019) studies personal characteristics in P2P markets, finding that loans tend to be awarded to those sharing personal characteristics such as occupation.

As Ayal et al. (2018) implies, this can be generalised to suggest that lenders' loan grants reflect a wider network of similar borrower preferences, which can reflect the homogeneity in loan use observed by Serrano-Cinca et al. (2015). Therefore, we expect occupation to be a meaningful predictor, but per Amar et al. (2011), we expect a preference for smaller loan counts. Thus,

Hypothesis 2: Borrowers sharing the same occupation area tend to share fewer loans for their shared loan purpose

Further, behavioural finance suggests that individuals' financial decisions are not independent, but rather, shaped by bounded rationality and psychological preferences (Kahneman & Tversky, 1979). In the P2P context, Ayal et al. (2018) shows that these preferences can influence how borrowers choose loan types and how this can overlap with other borrowers' decisions. Using the Exponential Random Graph Model (ERGM) framework, we can identify how these behaviours manifest as structural regularities in a borrower-to-loan-use network. Therefore, we ask:

Research Question 2: *What behavioural mechanisms affect the structure of borrower-to-loan-use connections?*

The aforescribed section outlined that the familiarity bias can lead to similarities across borrower portfolios in terms of their characteristics. However, in P2P settings, this can manifest itself as familiarity *clustering*, wherein borrowers who have previously been granted certain loan types are more likely to repeat their choices, or select other categories that appear familiar to different borrowers. Herzenstein et al. (2011), studying borrower decisions on Prosper.com, identifies a 'strategic herding' behaviour, where individuals, **in aggregate**, gravitate towards popular or socially-validated loans. Considering that loan popularity is directly observable through its number of bids, we assert that the social validation of a loan purpose for one borrower positively reinforces the selection of that same loan purpose for other, **co-affiliated** borrowers.

In network terms, this aggregate herding behaviour generates a structural propensity for four-cycles to form, resulting in closed paths between overlapping nodes of each of the bipartite partitions, as suggested by Koskinen & Edling (2012). This structure represents the fact that two borrowers connected to a common loan use have a higher likelihood of also connecting to a *second* common loan use. We capture this specific form of bipartite closure through the ERGM term `cycle(4)`, formulating:

Hypothesis 3: Co-affiliated borrowers are more likely to overlap on a second loan purpose

Further, Ayal & Zakay (2009), expands on the theory of Debt Account Aversion with the Theory of Perceived Diversification, explaining that **individual** borrowers' distress increases with not just by the number of debts, but also the perceived *distinctiveness* of each debt type, suggesting that in the P2P context, debt mentally differentiated by purpose would also be seen as distressing. Therefore, we assert that if borrowers do not diversify, then they try to minimise loan distinctiveness by requesting loans of a single type. This can be captured by the `bddegree(1)` term, which models whether borrowers are only linked to one loan purpose. Therefore, we formulate:

Hypothesis 4: Borrowers exhibit preferences for minimal debt distinctiveness, selecting one loan use

Notwithstanding, credit choices relate to non-structural factors. Cooper, Gorbachev, & Luengo-Prado (2023) show that younger borrowers face constraints in the types of loans they can access in typical credit markets. The authors argue that this constraint shapes how these borrowers form

loan portfolios, often concentrating around different use cases such as education, personal consumption, or small business loans. Considering the systemic differences in how age affects credit consumption, we assert that younger borrowers have different loan use portfolios. The ERGM term `b1cov("age")` can capture this by assessing how changes in age affect how borrowers form connections with different loan types. Therefore:

Hypothesis 5: Younger borrowers have different loan use mixes than older borrowers

Alongside age, the borrower’s gender is known to be a significant factor in influencing credit use decisions. Aliano, Alnabulsi, Cestari, & Ragni (2023) study the role of gender and other factors on borrowers’ probability of default on the Bondora.com, finding a persistent gender effect on different loan uses; women tend to default less on health, home, and business loans. Croson & Gneezy (2009) shows that these differences can be due to risk perception and self-selection effects, where women exhibit greater risk aversion, leading to different borrowing patterns. We expect that these differences in risk tolerance and perceived creditworthiness by lenders affect loan use composition in terms of gender. The term `b1factor("gender")` captures this effect by assessing whether borrowers’ connections with loan uses differ across genders. We would find support for this idea given a significant, non-zero effect estimate. Therefore:

Hypothesis 6: Gender differences lead to different loan use mixes

Table 1: Hypotheses related to Study 2

Hypothesis	Term	Explanation
H_3^a : Co-affiliated borrowers are more likely to overlap on a second loan purpose	<code>cycle(4)</code>	This captures the tendency of borrowers who are already affiliated through their links to the same loan type to have links to another type. With this term we capture aggregate borrower behaviour.
H_4^a : Borrowers exhibit preferences for minimal debt distinctiveness, selecting one loan use	<code>b1degree(1)</code>	This captures the tendency of borrowers to prefer minimal distinctiveness in their loan uses, preferring to hold only one distinct type to minimise their perceived risk. With this term, we capture individual borrower behaviour rather than an aggregation phenomenon.
H_5^a : Younger borrowers have different loan use mixes than older borrowers	<code>b1cov("age")</code>	This captures the tendency for younger borrowers to choose different kinds of loans based on how differently they consume credit.

Hypothesis	Term	Explanation
H_6^a : Gender differences lead to different loan use mixes	<i>b1factor("gender")</i>	This captures the idea that there are systemic differences in what kinds of loan uses are preferred the genders due to their perceived riskiness and creditworthiness.

Following this section, we outline the methodology, explaining our dataset and network construction. To facilitate this, descriptive statistics are analysed for both the dataset and resulting network. Subsequently, we elaborate on our choice of models, discussing how network models can be used to conduct the study. Then, modelling results are shown and explained to evaluate our hypotheses. Simultaneously, the robustness of the study is evaluated through the models’ diagnostics. Finally, we arrive at our conclusions, summarising the study, its results, and implications. Supporting items such as the source code and specific data processing steps are shown in section [A](#).

2 Dataset

The study utilises publicly-available data from Bondora, a European P2P lending platform primarily operating in the Baltics and Spain. The dataset contains detailed information on defaulted and non-defaulted loans granted to users between February 2009 and July 2021. Specifically, contained is a range of numeric, binary, categorical, and time-series attributes across 85,087 unique users and 179,235 individual loans. Since the company’s API is no longer accessible, we utilised a publicly available repository compiled by Manu Siddhartha ([2021](#)) on kaggle.com. The user made a series of API calls to collect the data. Following this, the dataset was processed according to Appendix Section [A.1](#).

2.1 Pre-Processing & Network Formation

Initially, the dataset was filtered loans between 2014 and 2016 since the data is incomplete during other periods. Since we are studying interactions across two distinct set of nodes, a bipartite network was formed with unique users as the first partition and the purpose of loans as the second. This allows us to evaluate how borrowers interact with loan purposes; a one-mode projection onto users is not meaningful for P2P networks as it produces no discernible structure. For model efficiency and performance, we randomly sample 500 individuals from the larger sample. Ultimately, we have 500 and 9 nodes in the first and second partitions, respectively.

2.2 Descriptive Statistics & Preliminary Analysis

Table [2](#) summarises the numeric variables between the reduced and larger samples. First, borrowers take out relatively small loans with fairly high interest rates, possibly reflecting that the average borrower is deemed fairly risky. The average loan is quite long, at approximately 46 months out of a maximum 60. Figure [11](#) shows that the largest credit category is HR¹, which is the low-

¹Bondora uses their own proprietary credit rating system where the best rating is AA and worst is HR ([Bondora, 2024](#)). The company bases these ratings on their calculated expected probability of loss on the loan. For example, someone rated

est possible rating. So, the typical borrower is relatively uncreditworthy. Moreover, the largest categories of loan use are Other and Home Improvement, suggesting these are largely personal. Additionally, Figure 13 shows that the biggest borrower group is from Estonia, however, Spain and Finland are also highly represented. Conversely, Slovakia has minimal representation. We observe minimal differences between the random sampling and the larger sample of individuals, per Table 2 and Figure 11, however, the study may be biased by the fact that the dataset contains only granted loans, rather than all bids for loans. This means our inferences about how borrowers behave is limited to successful bids. Further, the time period is relatively distant, meaning that recent structural changes to the P2P sector cannot be accounted for.

Table 2: Descriptive Statistics of Numeric Variables

DataFrame	Variable	Mean	Median	Std. Dev.	Min	Max
Initial Sample	Amount	2652.01	2125	2151.28	115	10630
Initial Sample	Interest	35.94	31	24.77	7.62	263.63
Initial Sample	Age	38.53	37	11.4	19	70
Initial Sample	LoanDuration	45.5	60	17.53	3	60
Reduced Sample	Amount	2637.64	2112.5	2112.2	170	10630
Reduced Sample	Interest	35.85	30	27.24	10.17	253.08
Reduced Sample	Age	38.73	37	11.34	21	69
Reduced Sample	LoanDuration	46.23	60	17.35	3	60

Table 3 shows a network density of 0.12, indicating that only 12% of all possible borrower–loan-type connections exist. This density suggests that borrowers typically engage with few loan types, reflecting specialization in borrowing behaviour. The centralization value of 0.47 further indicates an uneven distribution across loan types, where a few popular categories attract many borrowers while most remain peripheral. The mean distance of approximately three implies that borrowers are closely connected, typically separated by only three-to-four loan-use steps.

Centrality distributions reinforce this structure Figure 2. Betweenness is highest among loan-type nodes, though several borrowers also bridge distinct loan clusters. Closeness centrality remains low but scattered, suggesting localized clustering constrained by the bipartite structure. Degree distributions show wide variation among loan types, consistent with a hub-like structure dominated by a few popular uses. However, the second partition has relatively small degree dispersion. Finally, the homogeneous eccentricity distribution indicates that no borrowers are highly isolated, reflecting a compact and moderately cohesive network overall.

Table 3: Basic Network Descriptive Statistics

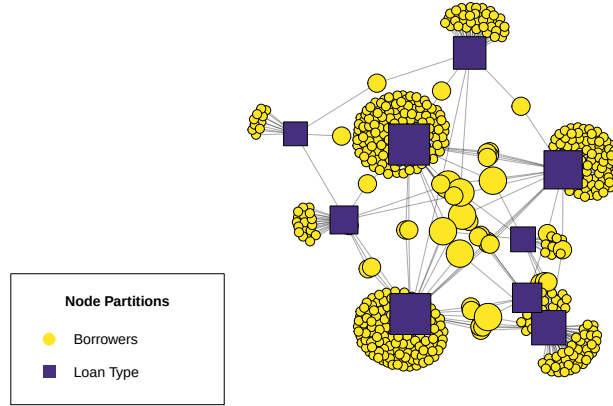
Statistic	Measure
Vertex Count	509.00
Edge Count	547.00
Density	0.12
Centralization	0.47

AA ranges from a expected loss of 0% to 2%, whereas for HR this is 25% - >25%.

Table 3: Basic Network Descriptive Statistics

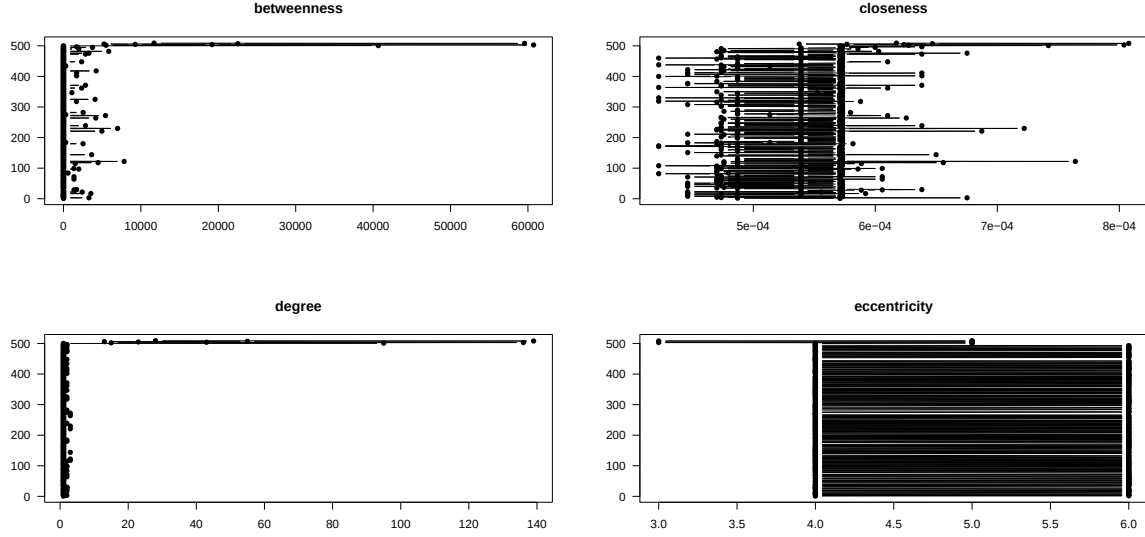
Statistic	Measure
Mean Distance	3.66

Figure 1: Network Visualisation



Note: Each colour represents a different partition of the network. The partitions' nodes are scaled to their relative degrees. Users with many different loan types are larger than those with single loan uses. Similarly, the size of second partition nodes represents how many users belong to each.

Figure 2: Network Centrality Distribution Plots



3 Research Rationale

Our overall study analyses borrowers' behaviours, however, the studies are differentiated in their scope; the first study observes dyad-level relations among borrowers to understand how shared attributes affect borrowers' behaviours. This is inherently useful because behavioural finance aims to understand how individuals behave in *aggregate* (Ayal et al., 2018). Typically, the Quadratic Assignment Procedure is selected for dyad-level observations because basic linear regressions such as Ordinary Least Squares are fundamentally flawed and biased when using non-independent observations (Simpson, 2001).

Recognising this, studies such as Zuo (n.d.) assume that groups of borrowers have dependencies, and accordingly, use clustering methods such as K-Means or Fuzzy Clustering to understand behaviour. However, these methods are unsuitable when we require a distribution of estimates to make inferences from our results. Moreover, studying individuals as dyads rather than large clusters provides finer granularity in results. Therefore, QAP regression is the most appropriate choice for this study: it allows us to control for the non-independence among dyads, derive a statistical distribution of estimates through matrix permutation, and draw valid inferences about the relationship between borrower similarities and loan-use behaviour. This makes it both theoretically and methodologically well-suited to our research objectives and dataset.

The second study extends the analysis from a dyadic similarities to the structural formation of borrower-to-loan-use connections, which can identify the behavioural mechanisms that influence borrowers' specific loan use mixes. While QAP can account for exogenous influences, it cannot analyse higher-order network patterns such as clustering while accounting for simultaneous interactions between borrowers and loan uses. Using the Exponential Random Graph Model allows us to analyse how local behavioural tendencies aggregates to a global network structure.

Recent evidence highlights the need to focus on structural approaches in P2P lending. Liu, Baals,

Osterrieder, & Hadji-Misheva (2024) studies borrowers' centrality with the Bondora P2P dataset, modelling edges through borrower-to-borrower similarity metrics. Ultimately, they find that a borrower's position is highly significant for their probability of default, using centrality as an exogenous variable in a logistic regression. In contrast, ERGM allows us to move away from predictive association and instead account for endogenous factors, allowing us to understand the likelihood that borrowers' selection of loan uses are random. Consequently, we can better understand how behavioural phenomena such as familiarity and debt account aversion can translate into *systemic* patterns of credit use.

As we aim to understand how two distinct groups of nodes interact, a bipartite approach is highly suitable. Specifically, Koskinen & Edling (2012) shows that in bipartite networks, terms such as `cycle(4)` can be effective in capturing the idea of grouping across partitions, as in hypothesis 3, whereas it would fail in a fully-connected one-mode projected networks. Furthermore, the flexibility of ERGM also allows for modelling exogenous impacts such gender-based differences in loan use mixes, as in hypothesis 6, and covariate age effects as in hypothesis 5.

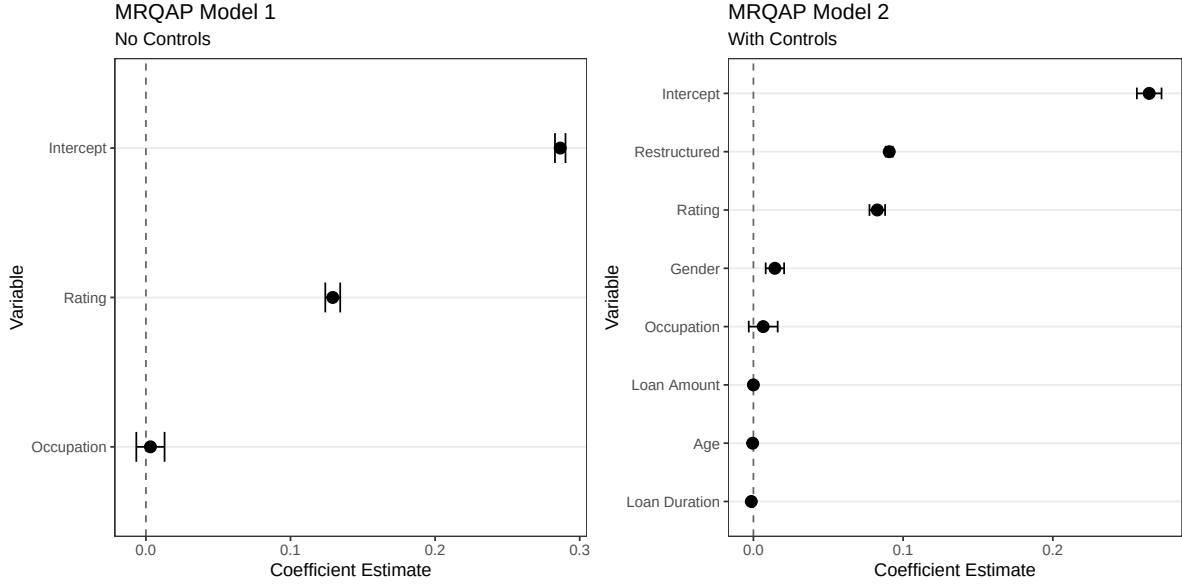
4 Results

4.1 Study 1 - QAP Linear Regression

The first study is assessed with two QAP linear regression models, as shown in Figure 3 and Table 4. The first model includes only the main predictors whereas the second controls for borrowers' age, gender, loan amount, loan duration, and whether the loan was restructured. Differences across these models can indicate the strength of omitted variable bias.

Assessing hypothesis 1, in the unrestricted model, borrowers sharing the same credit rating once leads to a 0.083 increase in the number of loans they share within the same loan use category, holding all other effects fixed. While this effect is statistically significant at 1%, notably, it is not economically significant because the effect magnitude is very small. In essence, having the same credit rating as another borrower does not mean that the borrower will have meaningfully different amounts of loans taken out for the same reason. Without controls, this effect was slightly larger, meaning that other predictors capture some of this effect. Our results corroborate aspects of Serrano-Cinca et al. (2015); similarly, we find that there is a non-zero effect of creditworthiness on loan use, however, they find a stronger effect. This can be due to borrowers' aversion to holding large numbers of debt, as explained by Ayal et al. (2011), where borrowers attempt to minimize debt counts independently of the loan use type.

Figure 3: QAP Coefficient Plots



Further, assessing hypothesis 2, borrowers sharing the same occupation area does not have a significant impact on the outcome. This indicates that the number of loans taken out for the same reason as another borrower does not vary given that pairs of borrowers have the same kind of job. The omission of control variables does not change this effect, meaning that its insignificance is not due to other correlated effects. The result is contrary to Ravina (2019), who finds that certain occupation areas have a higher likelihood of loan success. As the author states, these effects are likely present in markets with sophisticated decision makers. It is possible that Bondora lenders do not conform to this standard. Alternatively, the familiarity bias lenders exhibit on shared personal characteristics may not generalise to similarities across borrower characteristics, as we have argued.

Observing the controls, we see that loan amounts and whether the loan were restructured are statistically significant at 5%, but not economically meaningful. Moreover, the model diagnostics show an $R^2 = 0.035$, suggesting that the linear QAP only explains 3.5% of the variation observed in the data. This is rather small, indicating that the linear model is not a good fit for this specific study. We argue that borrowers are not biased against large debt counts per loan use type as *dyads*, but rather, as part of a more complex network structure. This is indicated by the low R^2 , showing that this dyadic relationship cannot be captured as a linear effect.

4.2 Study 2 - ERGM

The second study is assessed with four iterative models shown in Figure 4 and Table 5; the fourth is considered to be the full, unrestricted model. In each iteration, additional terms are added and diagnosed in Appendix Section A.4 by the Goodness of Fit (GOF) and Markov chain Monte Carlo (MCMC) simulations. We first interpret model estimates, their relation to our hypotheses, and conclude by evaluating the validity of the fitted models.

In Model 4, the inclusion of `bifactor("gender")` has pushed the edges term towards zero, making

Table 4: QAP Modelling Results

	M1 No Controls	M2 Controls
Intercept	0.287*** (0.002)	0.264*** (0.004)
Rating	0.129*** (0.003)	0.083*** (0.003)
Occupation	0.003 (0.005)	0.007 (0.005)
Loan Amount		−0.000** (0.000)
Age		−0.001 (0.000)
Gender		0.014 (0.003)
Loan Duration		−0.001** (0.000)
Restructured		0.091*** (0.001)
R-squared	0.010	0.035

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

it insignificant. This suggests that the combination of individual borrowers’ loan use formation is likely random in the presence of other terms, indicating that the edge formation effect is instead captured jointly by the rest of the terms. We consider this a good result since it suggests that our final model may be fully saturated.

Evaluating hypothesis 4, we determine whether borrowers exhibit a preference for minimal debt distinctiveness by choosing only one loan type. In Model 4, `b1degree(1)` is highly significant and positive with a log-odds of 3.271, suggesting that the odds of a borrower connected to exactly one loan use is approximately 26.34 times higher than the odds of observing a configuration where the degree is not one. In other words, in this Bondora sample, it is highly likely that borrowers select only one loan type. This effect size or error has not deviated strongly across the four model permutations, indicating its stability. This result shows support for the Theory of Perceived Diversification by Ayal & Zakay (2009) as see that on an **individual level**, borrowers prefer to hold minimal *distinct* debt types.

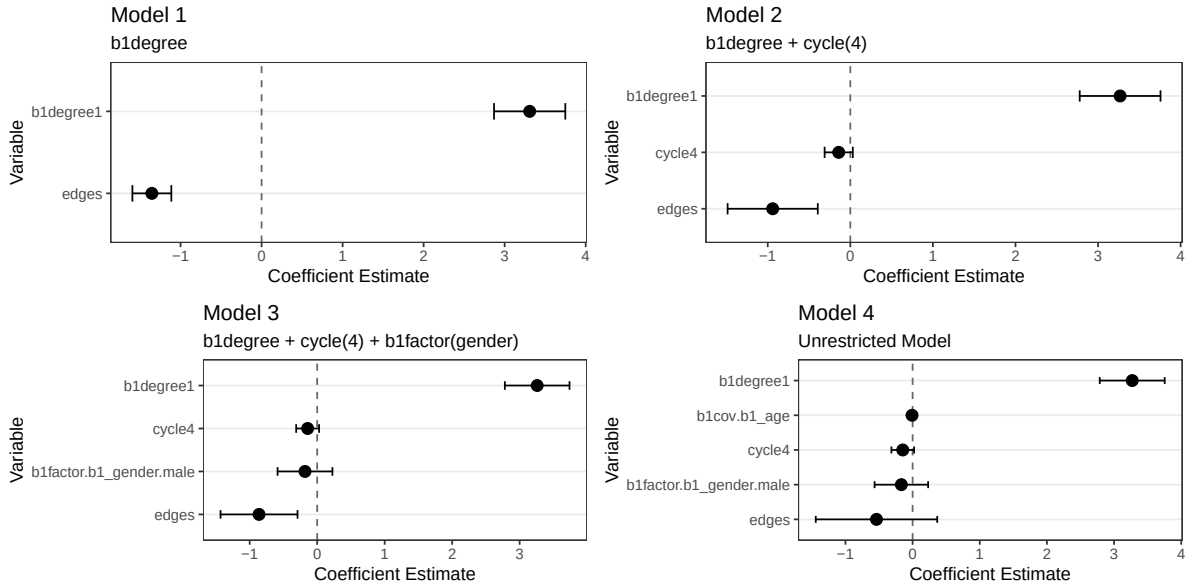
Further, the third hypothesis asserts that borrowers affiliated by a common loan purpose overlap on a second loan purpose, reflecting familiarity bias on an **aggregated** level. In Model 4, four-cycles are insignificant, meaning that we have not found sufficient evidence to reject the null hypothesis that there are no systematic co-affiliation effects. In other words, borrowers who are affiliated by the same loan use do not overlap on a second loan purpose. This finding is contrary to Herzenstein et al. (2011), who argues that borrowers aggregate around popular, or familiar loans. This could be explained by the fact that the authors examine auction bids, rather than winning bids, as we have. The behaviours or characteristics of bidders, compared to winners, may be fundamentally

different.

Evaluating attribute-level effects, we see that neither age nor gender has a significant impact on loan use selection by borrowers. Specifically, we do not find sufficient evidence to conclude that borrowers being younger has any influence over how they select their loan uses. This indicates a lack of support for hypothesis 5, which asserts that younger borrowers should have different loan use combinations than older borrowers. In this vein, we cannot corroborate Cooper et al. (2023), however, this is likely due to the fact that their study focuses on credit consumption of U.S. young adults, which is fundamentally than credit consumption in Europe Jentzsch & Riestra (2006). Moreover, they study traditional credit markets such as personal credit lines from banks rather than peer-to-peer lending services, which also vary fundamentally.

In the same vein, we do not observe sufficient evidence to determine that borrowers' gender affects how they select their specific loan uses. Similarly, this is contrary to hypothesis 6, which states that there is a gender difference in how borrowers select loan uses. The lack of effect is contrary to Aliano et al. (2023), who finds gender to be a significant predictor of loan use heterogeneity. They argue that gender differences across credit risk perception and self-selection effects are responsible, however, it is likely that these behavioural factors are already captured by the $b1degree(1)$ term; as Ayal & Zakay (2009) argues, similar risk-related phenomena are responsible for Prospect Theory, which in turn drives the Theory of Perceived Diversification.

Figure 4: ERGM Coefficient Plots



Evaluating model fit, we must first evaluate the MCMC simulations seen in Appendix Section A.4. Observing Model 4, we see that all terms have mixed sufficiently well. While edges, $b1degree(1)$, $b1factor("gender")$, and $b1cov("age")$ are well centred at zero, $cycle(4)$ is slightly right-skewed, indicating minor mixing issues. Additionally, the trace plots appear to resemble a “fuzzy” shape, supporting this. We find this diagnostic acceptable.

Further, the GOF measures indicate how well simulation iterations match our observed network.

Across the GOF measures, p-values close to 1 resemble a close match with reality. All geodesic distances except for Inf and 2 are well-modelled. The former is expected, as our bipartite mode restricts any possibility of links within each partition, meaning that Inf distances can never be modelled. For `b1degree(1)`, all degrees except for zero and two are highly insignificant. It is expected that zero degrees is poorly modelled because our observed network has no isolates. Degree three is arguably poorly modelled since there are only 7 observations of it, limiting the simulations. The term `b2degree()` is not modelled well, however, this term is very difficult to capture statistically; there is no meaningful statistical variation with respect how many borrowers each loan use has, so the GOF test cannot reliably evaluate it. Moreover, edge-wise shared partners are all well modelled. Overall, we consider the GOF tests to be sufficient.

Finally, we evaluate the optimal statistical model configuration by observing the minimum AIC/BIC values across the models. AIC prefers Model 3 whereas BIC chooses Model 1; this is expected, as BIC penalizes less informative model terms more strongly.

Table 5: ERGM Modelling Results

	Base ERGM	b1degree(1)	cycle(4)	Age	Gender
edges	-1.978*** (0.046)	-1.354*** (0.123)	-0.940*** (0.279)	-0.862** (0.291)	-0.538 (0.462)
b1degree1		3.309*** (0.224)	3.267*** (0.250)	3.260*** (0.244)	3.271*** (0.247)
cycle4			-0.141 (0.087)	-0.141 (0.086)	-0.147 (0.086)
b1factor.b1_gender.male				-0.181 (0.207)	-0.168 (0.203)
b1cov.b1_age					-0.008 (0.009)
AIC	3325.828	2674.435	2676.439	2672.013	2674.852
BIC	3332.238	2687.255	2695.668	2697.652	2706.901
Log Likelihood	-1661.914	-1335.218	-1335.219	-1332.006	-1332.426

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

5 Conclusion

This study aimed to assess the prevalence of behavioural biases in a peer-to-peer lending context by assessing how connections across borrowers and loan types varied; by answering the question “Do borrowers vary in their loan use type and frequency based on their individual and loan characteristics”, we can determine the extent that borrowers have an aversion to large loan counts within loan use categories, per Ayal et al. (2011), and by answering “What behavioural mechanisms affect the structure of borrower-to-loan-use connections?” in a network setting, we can see how loan use choices vary with structural factors.

To evaluate how loan counts are affected within loan use selections, we utilised the QAP linear regression framework. Within this study, borrowers are studied as dyads rather than a whole network. Firstly, it was found that borrowers having the same credit rating was significant in explaining variations in the number of loans they took out for a shared purpose, however, the impact was small in size. Further, we found that a shared occupation area between borrowers

does not have a meaningful impact on loan counts. Assessing the models, we find a poor linear fit, which could indicate that dyads are not an optimal means for studying these effects. These findings do not necessarily corroborate Ayal et al. (2011) because we do not observe strong negative effects on loan counts, however, the regression diagnostics make this conclusion quite unreliable.

To evaluate specific borrower behaviours independent of loan counts, a network statistical model such as ERGM is more suitable. In this study, we differentiate between **aggregate** and **individual** borrower level effects. The results indicate very strongly that borrowers prefer to hold one loan type. This yields significant support to the Theory of Perceived Diversification by Ayal & Zakay (2009), capturing the idea that on an **individual** level, borrowers try to have the least amount of distinctiveness possible. Specifically, by holding only one loan type, they can place them in the same mental ‘account’, which prevents distress (Ayal & Zakay, 2009). However, we do not find that, in **aggregate**, borrowers have co-affiliation by being related to two distinct loan types. In other words, borrowers do not overlap with their choices of pairs of loan types. These two findings highlight that borrowers are more prone to individual behavioural biases rather than a ‘herding’ effect as described by Herzenstein et al. (2011). However, as we do not utilise auction data, it is possible that aggregate effect does not extend to how *winners* of auctions behave.

Further, examining borrowers’ attribute effects, we found that neither gender nor age have significant effects on how borrowers choose loan types. Specifically, gender differences do not account for a difference in loan use formation. We argue that this could be due to the idea that the same gender differences that affect risk perception are also responsible for the underlying behavioural phenomena behind the Theory of Perceived Diversification. Consequently, the use of `b1degree(1)` had already captured this process. Finally, we do not find that younger borrowers have a different loan use mix than older borrowers. This theory was derived from Cooper et al. (2023), who studied young adults in the U.S. However, considering the fundamental differences across credit markets between the U.S. and Europe, it is likely that these behavioural effects may not apply to European borrowers across our sample of three countries. Evaluating the ERGM models, we find them to be well constructed and our diagnostics show that they largely conform to our expectations. Therefore, we consider these findings to be robust to modelling or specification error.

Arguably, the main beneficiaries of our findings are credit regulatory agencies in Finland, Estonia, and Spain, to which the majority of our borrowers fall under. Some studies indicate shortcomings in how these specific credit markets are regulated, owing their young age and online nature (Lenz, 2016). By understanding the behavioural phenomena driving borrowers’ behaviour, these institutions can construct policies to minimise the risks of users utilising credit inappropriately, for example, by systematically maximizing credit counts. Further, while we have studied how credit counts and borrower selections are influenced, we have been unable to combine these phenomena together by investigating loan counts within the ERGM framework. This extension would be valuable because it would account for the shortcomings of the QAP regressions’ linearity whilst also allowing for more complex behavioural phenomena to be modelled. In general, however, we have observed that researchers focus on singular P2P platforms. We recommend that studies focus on concurrent analyses of a combination of P2P platforms because individual platforms may have idiosyncrasies that make analyses difficult to generalise to the wider P2P lending market.

A Appendix

A.1 Data Preprocessing Steps

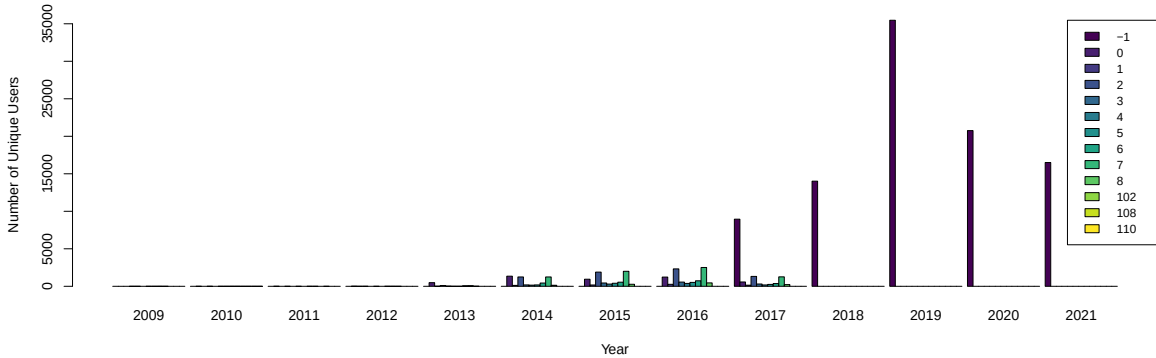
This section outlines the specific steps taken to process the data into the network object that was used in later analysis, outlining the rationale, where needed.

- Following the data import, only the following attributes were kept to make our processing steps more focused

```
keep_cols <- c("LoanId", "UserName", "Age", "Gender",  
              "Country", "Amount", "Interest", "LoanDuration",  
              "UseOfLoan", "Rating", "Restructured", "MonthlyPayment",  
              "OccupationArea", "BiddingStartedOn")
```

- We removed rows with any NA values for the selected attributes to ensure that we have a complete dataset
- Observing Figure 5, we see that between 2017-2021, there exists many -1 values for UseofLoan. This is because Bondora no longer provided this detail in later datasets. However, between 2014 and 2016 this data is available. Because of this, we restrict the time period to have the least noisy data. BiddingStartedOn was chosen to record the time, because the auction start date can be a good indication of activity around the loan.

Figure 5: Number of Unique Users per Loan Type per Year



- Then, we sampled 500 individuals out of the filtered dataset to ensure that our models run sufficiently well and converge. By choosing 500 individuals, we preserve importance structures within the network while adding sufficient statistical power.
- When adding labels to each loan use and occupation area, we keep any possible missing values because these do not indicate lower quality data. Instead, they can carry information about lenders distribute loans according to the quality of information presented by borrowers (Yao et al., 2019). Additionally, keeping missing labels is important to ensure that the edges are not superficially limited and that nodes can be isolates if they are indeed that way in reality.
- To create the network object for ERGM models:

- We first create an incidence matrix of size $n \times m$, borrower usernames by the loan use, where the values are the number of loans the user obtained for each loan use category. These values are then turned into binary because our modelling does not support the use of weighted edges.
- A bipartite network is created from this incidence matrix, where the first partition are unique users, and the second partition is the loan use associated among all loans of the user. Users can share more than more loan use type.
- We add age and gender attributes to each node of the first partition
- To create the relevant networks/matrices for QAP Regressions:
 - We first re-create the loan use incidence matrix, however, we keep the counts as the linear regression supports this.
 - * We then convert the loan use incidence matrix into an adjacency matrix of shape $n \times n$ through the transformation $\mathbf{X} \cdot \mathbf{X}^T$, which results in a matrix of shared loan uses weighted by the frequency of each user’s loan count within each loan use category.
 - * We then set the diagonal values of $\mathbf{X}$ to be zero to ensure that there are no loops within the network.
 - In accordance with this procedure, we create adjacency matrices for Credit Rating and Occupation Area which will be used as the main predictors in the QAP models.
 - * Credit Rating is a weighted adjacency matrix based on the loan count belonging to each user for each credit rating category
 - * Occupation Area is a binary adjacency matrix based on which occupation each user has reported to have held in the past when applying for each of their loans.
 - Additionally, we create a set of control variables to improve the validity and performance of the linear regression using Loan Amounts, Age, Gender, Loan Duration, and whether the loans have been restructured.
 - * Loan Amounts is a weighted adjacency matrix based on five bins across the range of possible loan amounts
 - * Age is an adjacency matrix based on differences in users’ ages
 - * Gender is a binary adjacency matrix based on users’ shared gender.
 - * Loan Duration is an adjacency matrix based on the differences in users’ loans average durations
 - * Restructured is a binary adjacency matrix based on whether the users share the fact that they have defaulted on any loans in the past. The users can either have shared both defaults and non-defaults, or both.

A.2 Distributions of Variables across Samples

Figure 6: Distribution of Loan Amount across Samples

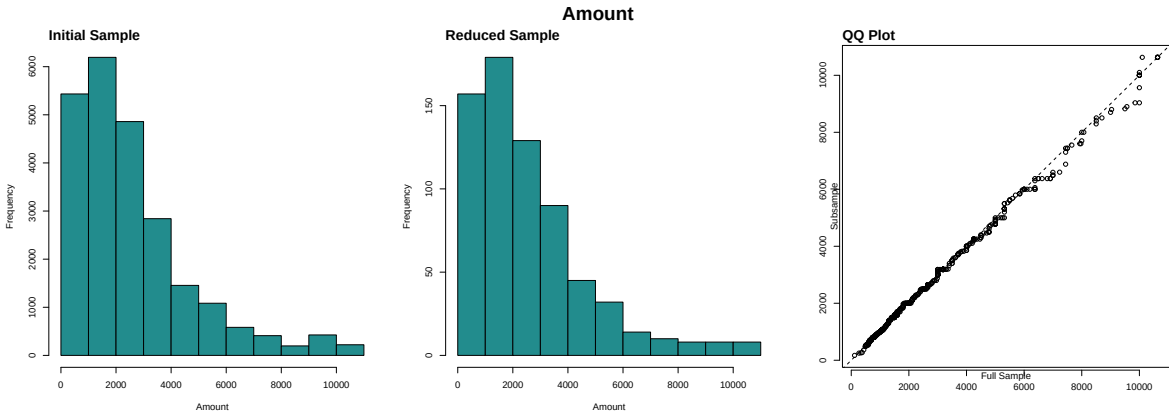


Figure 7: Distribution of Interest Rates across Samples

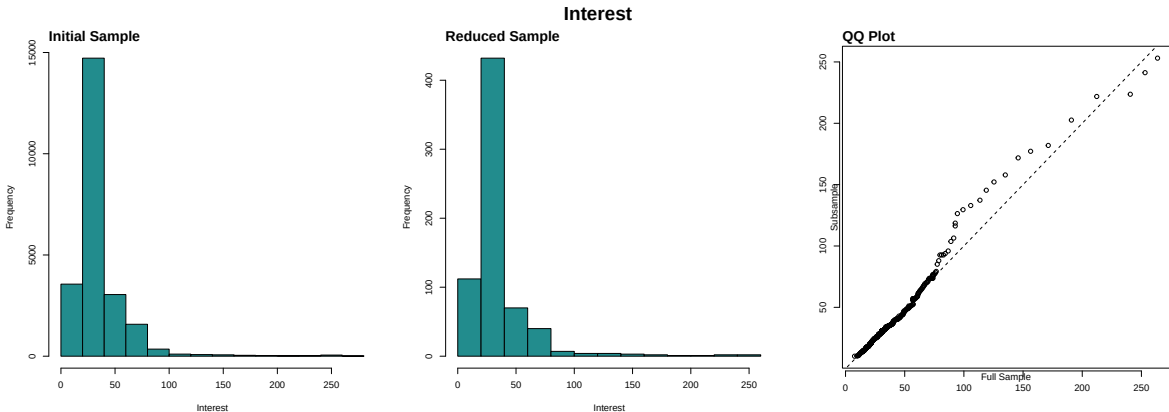


Figure 8: Distribution of Age across Samples

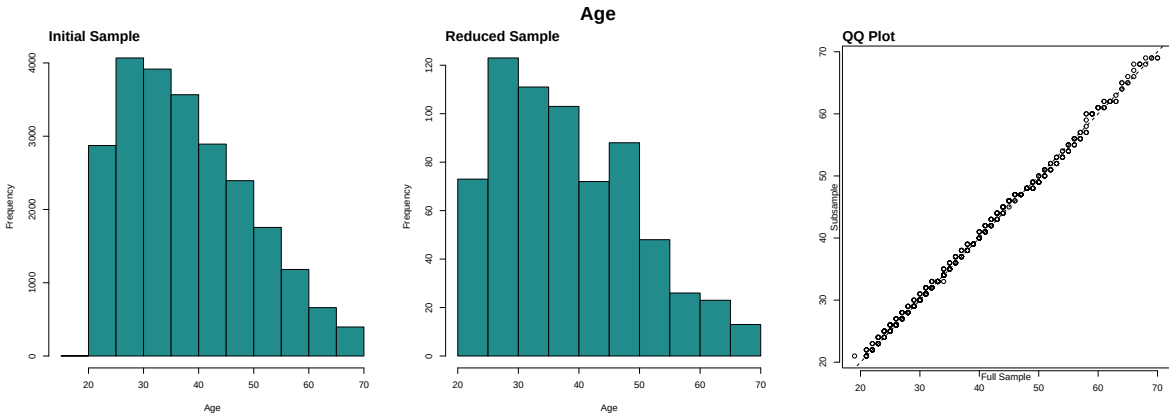


Figure 9: Distribution of Loan Duration across Samples

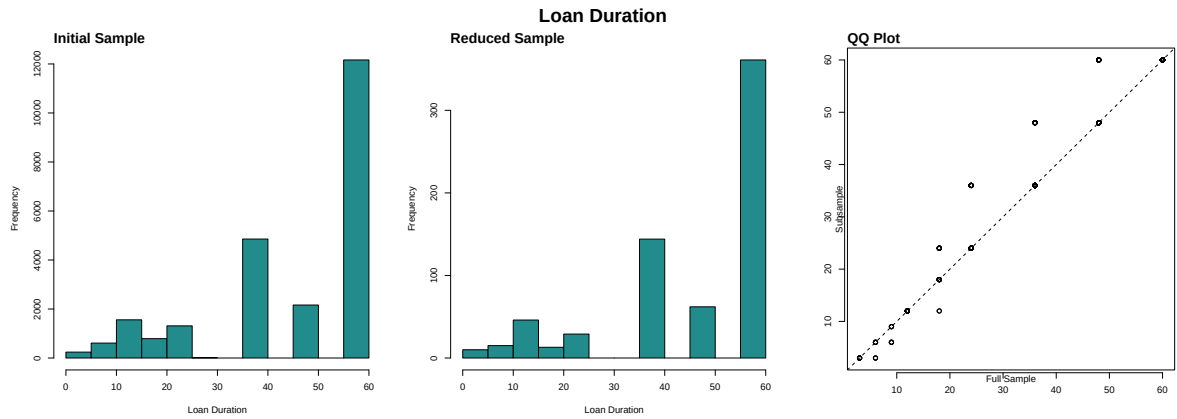


Figure 10: Distribution of Loan Purpose across Samples

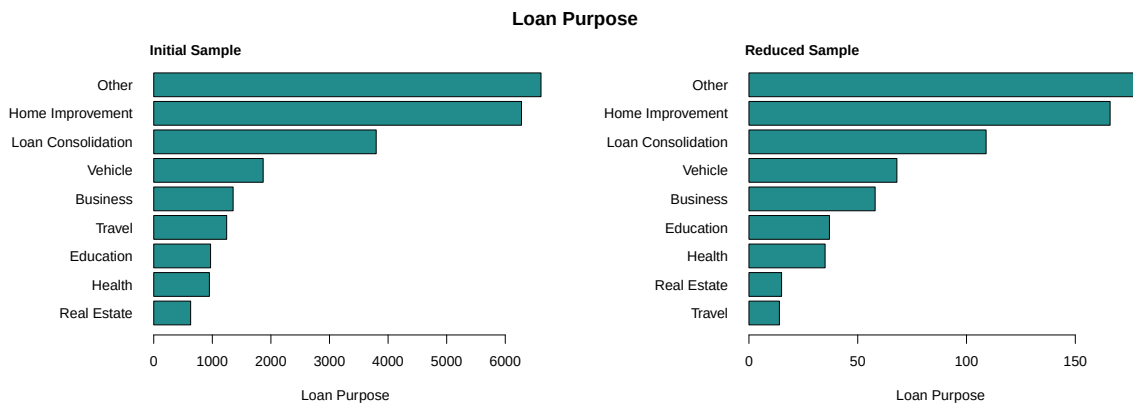


Figure 11: Distribution of Credit Ratings across Samples

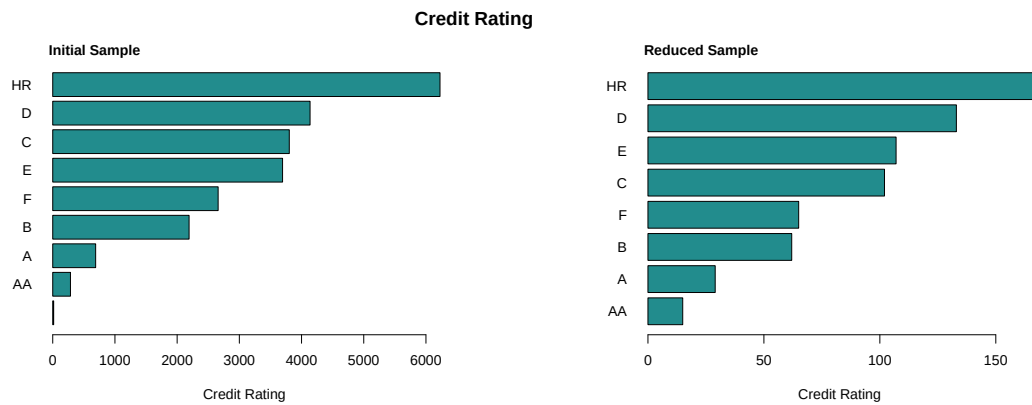


Figure 12: Distribution of Occupation Types across Samples

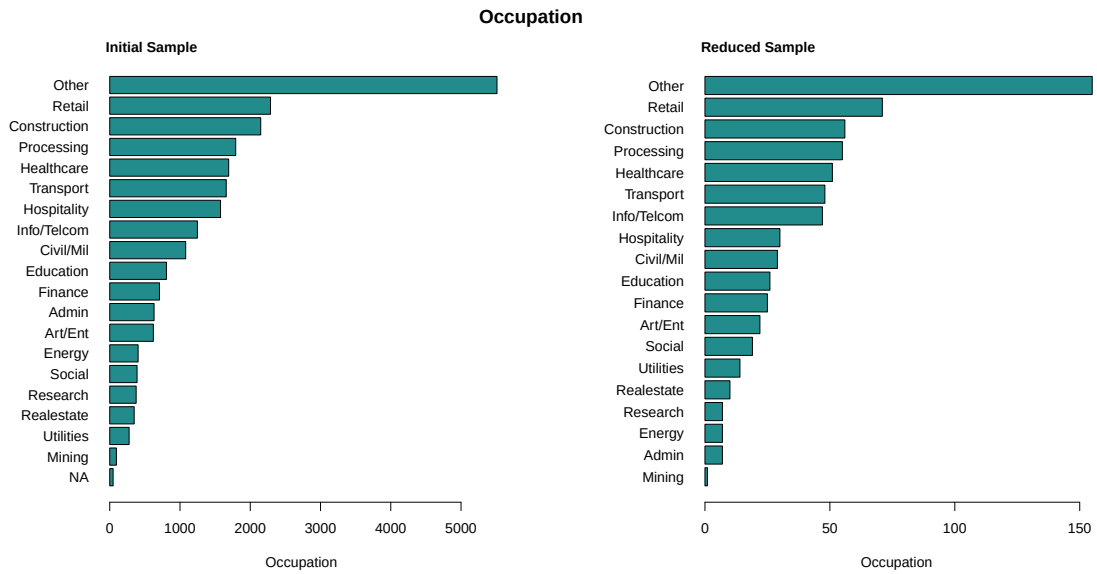
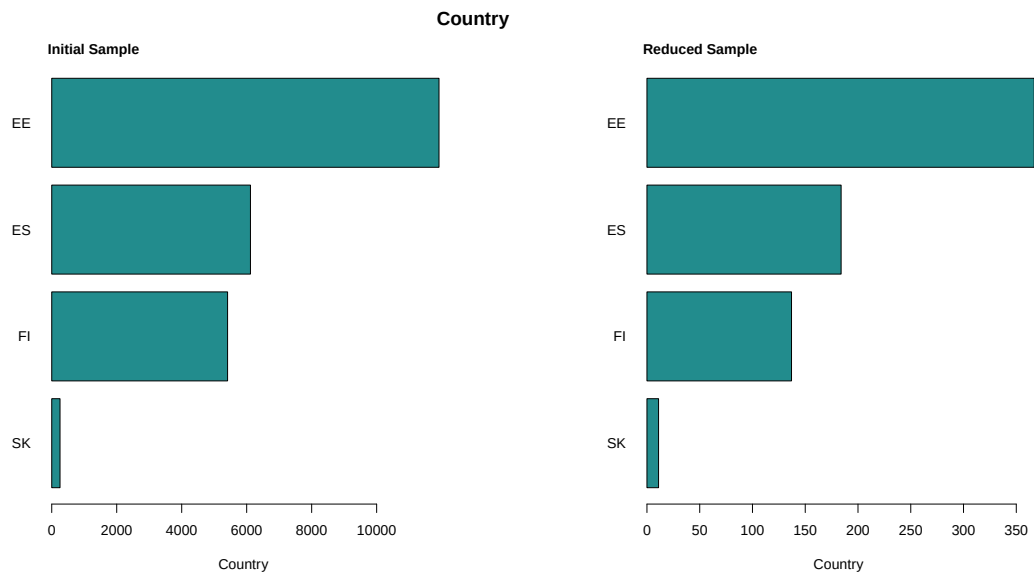


Figure 13: Distribution of Borrower Origins



A.3 ERGM Results - Supplementary Tables

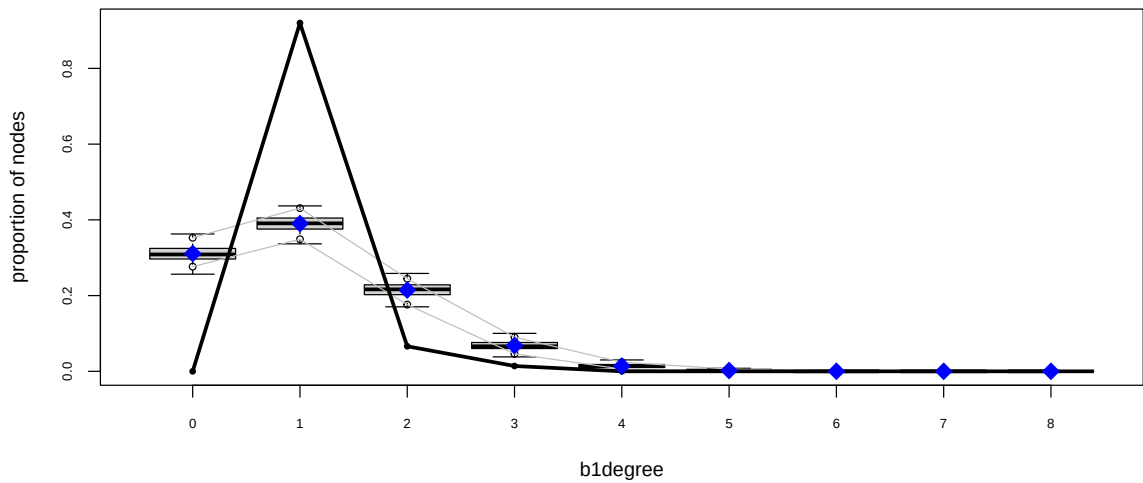
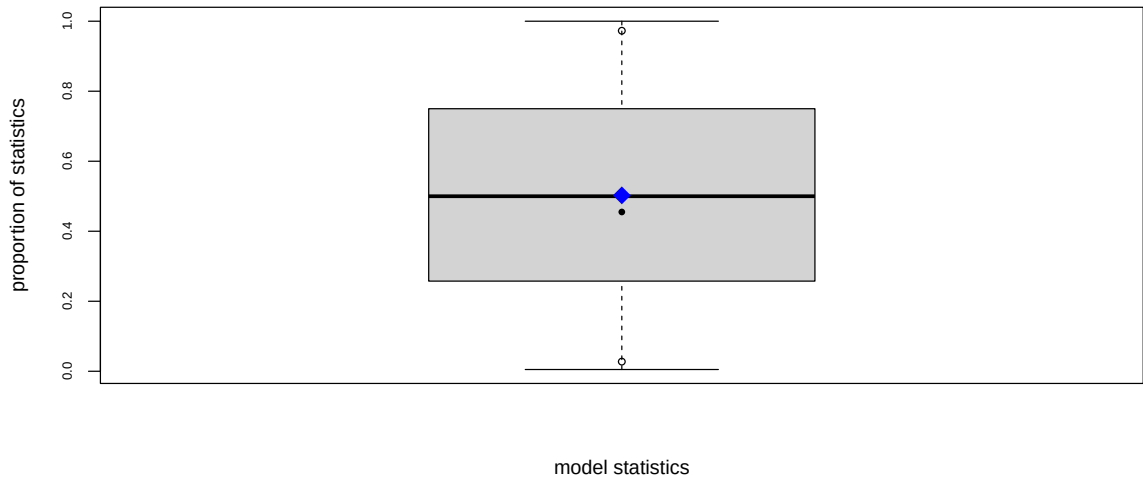
Table 6: ERGM Results in Probability Scale

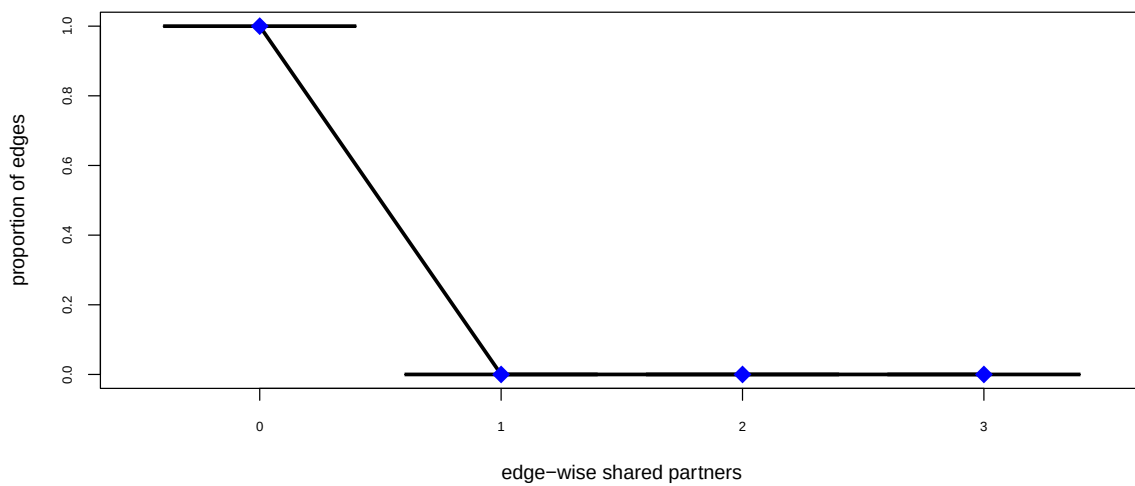
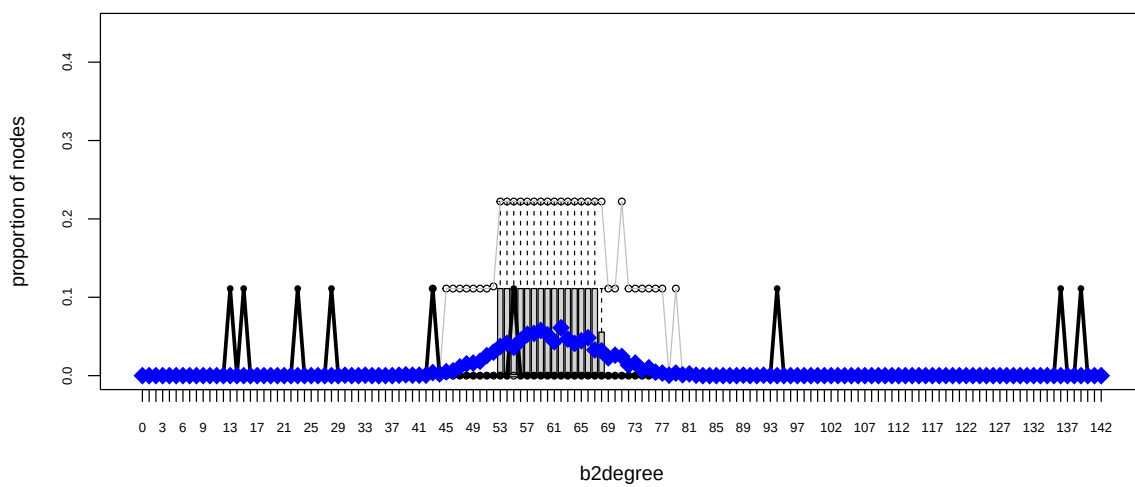
Terms	Base	b1degree.1.	cycle.4.	Age	Gender
edges	0.122	0.205	0.281	0.297	0.369
b1degree1	-	0.965	0.963	0.963	0.963
cycle4	-	-	0.465	0.465	0.463
b1factor.b1_gender.male	-	-	-	0.455	0.458
b1cov.b1_age	-	-	-	-	0.498

A.4 ERGM Diagnostics Plots

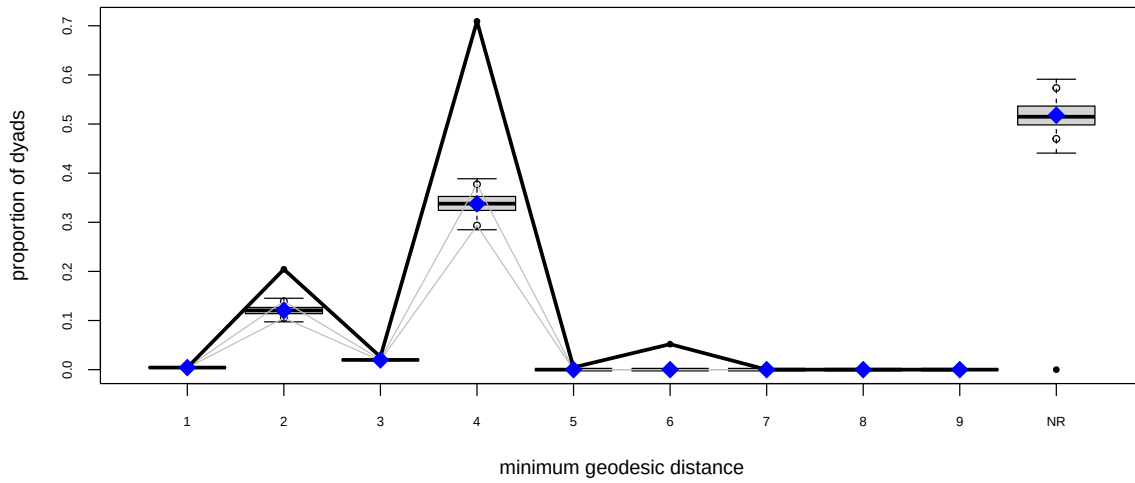
A.4.1 Base Model

```
Registered S3 methods overwritten by 'snafun':  
  method      from  
  plot.network network  
  print.network network
```





Goodness-of-fit diagnostics



Goodness-of-fit for model statistics

obs	min	mean	max MC	p-value
546.00	487.00	543.86	602.00	0.91

Goodness-of-fit for minimum geodesic distance

	obs	min	mean	max MC	p-value
1	546	487	543.860	602	0.91
2	26301	12531	15520.635	18711	0.00
3	3351	2348	2549.430	2749	0.00
4	91309	36680	43470.795	50047	0.00
5	594	0	0.460	48	0.00
6	6677	0	5.255	567	0.00
Inf	0	56768	66687.565	76128	0.00

Goodness-of-fit for bipartition 1 degree

	obs	min	mean	max MC	p-value
0	0	128	155.250	183	0.00
1	459	168	194.705	230	0.00
2	33	73	107.065	129	0.00
3	7	19	34.040	50	0.00
4	0	2	6.865	15	0.00
5	0	0	1.005	4	0.75
6	0	0	0.070	1	1.00

Goodness-of-fit for bipartition 2 degree

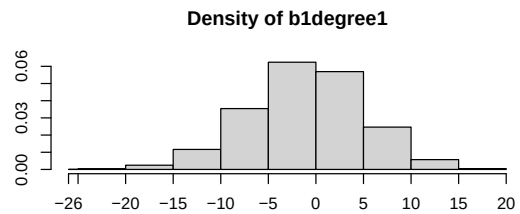
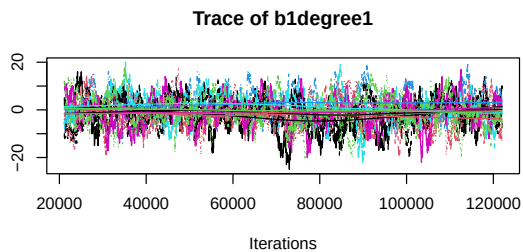
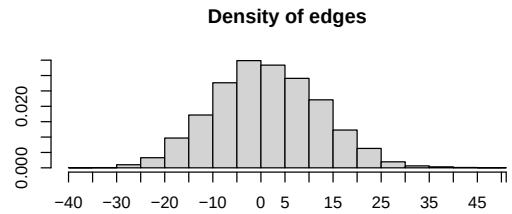
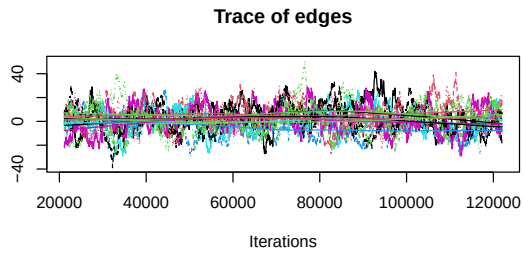
	obs	min	mean	max MC	p-value
13	1	0	0.000	0	0.00
15	1	0	0.000	0	0.00
23	1	0	0.000	0	0.00
28	1	0	0.000	0	0.00
30	0	0	0.005	1	1.00
33	0	0	0.005	1	1.00
38	0	0	0.005	1	1.00
39	0	0	0.010	1	1.00
40	0	0	0.005	1	1.00
41	0	0	0.005	1	1.00
42	0	0	0.005	1	1.00
43	1	0	0.035	1	0.07

44	0	0	0.020	1	1.00
45	0	0	0.045	1	1.00
46	0	0	0.055	1	1.00
47	0	0	0.100	1	1.00
48	0	0	0.135	2	1.00
49	0	0	0.145	2	1.00
50	0	0	0.165	2	1.00
51	0	0	0.230	2	1.00
52	0	0	0.270	2	1.00
53	0	0	0.335	3	1.00
54	0	0	0.370	3	1.00
55	1	0	0.325	2	0.56
56	0	0	0.405	3	1.00
57	0	0	0.475	3	1.00
58	0	0	0.485	2	1.00
59	0	0	0.520	3	1.00
60	0	0	0.470	3	1.00
61	0	0	0.390	2	1.00
62	0	0	0.550	3	1.00
63	0	0	0.420	3	1.00
64	0	0	0.370	3	1.00
65	0	0	0.400	3	1.00
66	0	0	0.435	4	1.00
67	0	0	0.295	3	1.00
68	0	0	0.285	3	1.00
69	0	0	0.205	2	1.00
70	0	0	0.235	2	1.00
71	0	0	0.220	2	1.00
72	0	0	0.120	2	1.00
73	0	0	0.145	3	1.00
74	0	0	0.065	1	1.00
75	0	0	0.090	1	1.00
76	0	0	0.040	1	1.00
77	0	0	0.030	1	1.00
78	0	0	0.005	1	1.00
79	0	0	0.030	1	1.00
80	0	0	0.010	1	1.00
81	0	0	0.020	1	1.00
82	0	0	0.005	1	1.00
89	0	0	0.005	1	1.00
92	0	0	0.005	1	1.00
94	1	0	0.000	0	0.00
136	1	0	0.000	0	0.00
139	1	0	0.000	0	0.00

Goodness-of-fit for edgewise shared partner

obs	min	mean	max MC	p-value
546.00	487.00	543.86	602.00	0.91

A.4.2 Model 1



Sample statistics summary:

Iterations = 21136:121992
 Thinning interval = 28
 Number of chains = 9
 Sample size per chain = 3603

1. Empirical mean and standard deviation for each variable,
 plus standard error of the mean:

	Mean	SD Naive	SE Time-series	SE
edges	1.727	11.066	0.06145	0.6592
b1degree1	-0.554	6.039	0.03353	0.2513

2. Quantiles for each variable:

	2.5%	25%	50%	75%	97.5%
edges	-19	-6	1	9	24
b1degree1	-13	-4	0	3	11

Are sample statistics significantly different from observed?

	edges	b1degree1	(Omni)
diff.	1.726955932	-0.55404447	NA
test stat.	2.577412722	-2.16661869	5.80269221
P-val.	0.009954302	0.03026394	0.05651482

Sample statistics cross-correlations:

	edges	b1degree1
edges	1.0000000	-0.6736048
b1degree1	-0.6736048	1.0000000

Sample statistics auto-correlation:

Chain 1

	edges	b1degree1
Lag 0	1.0000000	1.0000000
Lag 28	0.9826388	0.9701176
Lag 56	0.9664557	0.9439196
Lag 84	0.9498574	0.9180981
Lag 112	0.9335225	0.8938654
Lag 140	0.9173206	0.8721985

Chain 2

```

edges bidegree1
Lag 0 1.0000000 1.0000000
Lag 28 0.9829174 0.9611738
Lag 56 0.9663965 0.9239480
Lag 84 0.9502249 0.8879717
Lag 112 0.9333931 0.8535116
Lag 140 0.9178077 0.8205518
Chain 3
edges bidegree1
Lag 0 1.0000000 1.0000000
Lag 28 0.9883845 0.9692833
Lag 56 0.9772397 0.9397716
Lag 84 0.9662077 0.9126215
Lag 112 0.9556828 0.8871831
Lag 140 0.9453678 0.8626915
Chain 4
edges bidegree1
Lag 0 1.0000000 1.0000000
Lag 28 0.9825875 0.9674543
Lag 56 0.9666999 0.9364074
Lag 84 0.9515609 0.9049248
Lag 112 0.9362933 0.8735428
Lag 140 0.9216203 0.8433670
Chain 5
edges bidegree1
Lag 0 1.0000000 1.0000000
Lag 28 0.9817720 0.9637732
Lag 56 0.9642654 0.9307404
Lag 84 0.9471522 0.8990125
Lag 112 0.9306044 0.8686574
Lag 140 0.9141629 0.8394052
Chain 6
edges bidegree1
Lag 0 1.0000000 1.0000000
Lag 28 0.9840958 0.9657033
Lag 56 0.9685792 0.9335799
Lag 84 0.9533266 0.9036673
Lag 112 0.9378353 0.8764645
Lag 140 0.9228011 0.8498216
Chain 7
edges bidegree1
Lag 0 1.0000000 1.0000000
Lag 28 0.9852593 0.9626353
Lag 56 0.9707885 0.9284125
Lag 84 0.9574817 0.8962185
Lag 112 0.9445881 0.8654185
Lag 140 0.9318521 0.8363137
Chain 8
edges bidegree1
Lag 0 1.0000000 1.0000000
Lag 28 0.9806314 0.9686263
Lag 56 0.9615573 0.9383344
Lag 84 0.9432512 0.9091995
Lag 112 0.9247775 0.8805019
Lag 140 0.9061377 0.8532768
Chain 9
edges bidegree1
Lag 0 1.0000000 1.0000000
Lag 28 0.9795136 0.9606722
Lag 56 0.9603946 0.9238283
Lag 84 0.9417626 0.8881744
Lag 112 0.9231795 0.8544565
Lag 140 0.9036287 0.8209240

```

Sample statistics burn-in diagnostic (Geweke):
Chain 1

Fraction in 1st window = 0.1

```

Fraction in 2nd window = 0.5

      edges  bidegree1
-0.8251148  1.1021806

Individual P-values (lower = worse):
      edges bidegree1
0.4093064 0.2703832
Joint P-value (lower = worse):  0.5497544
Chain 2

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

      edges  bidegree1
-0.6130584  1.2345011

Individual P-values (lower = worse):
      edges bidegree1
0.5398377 0.2170163
Joint P-value (lower = worse):  0.3088801
Chain 3

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

      edges  bidegree1
-0.5254825  1.8245035

Individual P-values (lower = worse):
      edges bidegree1
0.59924780 0.06807598
Joint P-value (lower = worse):  0.3048263
Chain 4

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

      edges  bidegree1
2.1763745 -0.6342556

Individual P-values (lower = worse):
      edges bidegree1
0.02952727 0.52591401
Joint P-value (lower = worse):  0.1381523
Chain 5

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

      edges  bidegree1
0.05826116 -0.14359472

Individual P-values (lower = worse):
      edges bidegree1
0.9535406 0.8858205
Joint P-value (lower = worse):  0.9955405
Chain 6

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

      edges  bidegree1
-2.2393205  0.4044089

Individual P-values (lower = worse):
      edges bidegree1
0.02513507 0.68591207

```

Joint P-value (lower = worse): 0.07543831
Chain 7

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

edges bidegree1
0.6404571 -0.4060245

Individual P-values (lower = worse):
edges bidegree1
0.5218755 0.6847246
Joint P-value (lower = worse): 0.9039975
Chain 8

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

edges bidegree1
-0.1195638 -0.2165394

Individual P-values (lower = worse):
edges bidegree1
0.9048287 0.8285673
Joint P-value (lower = worse): 0.7900773
Chain 9

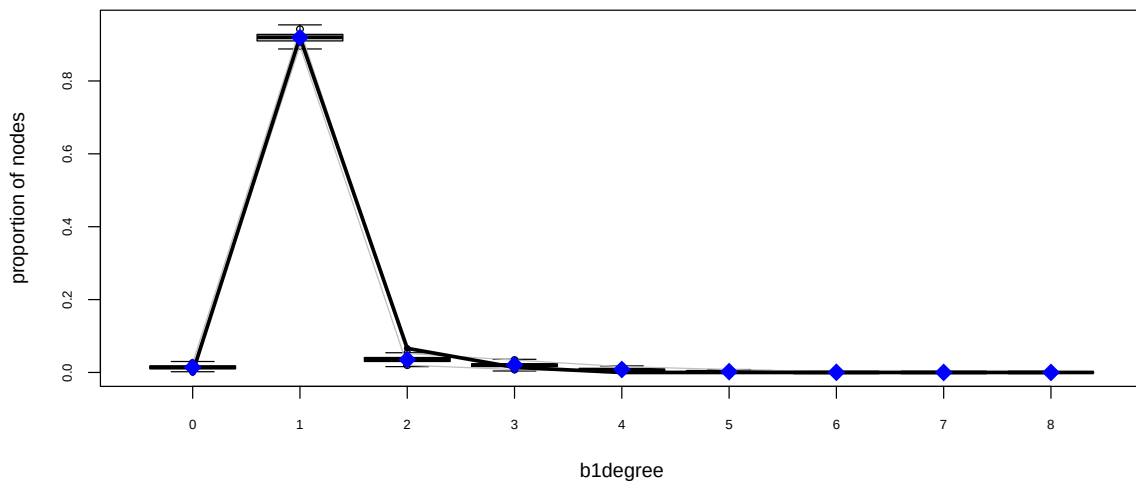
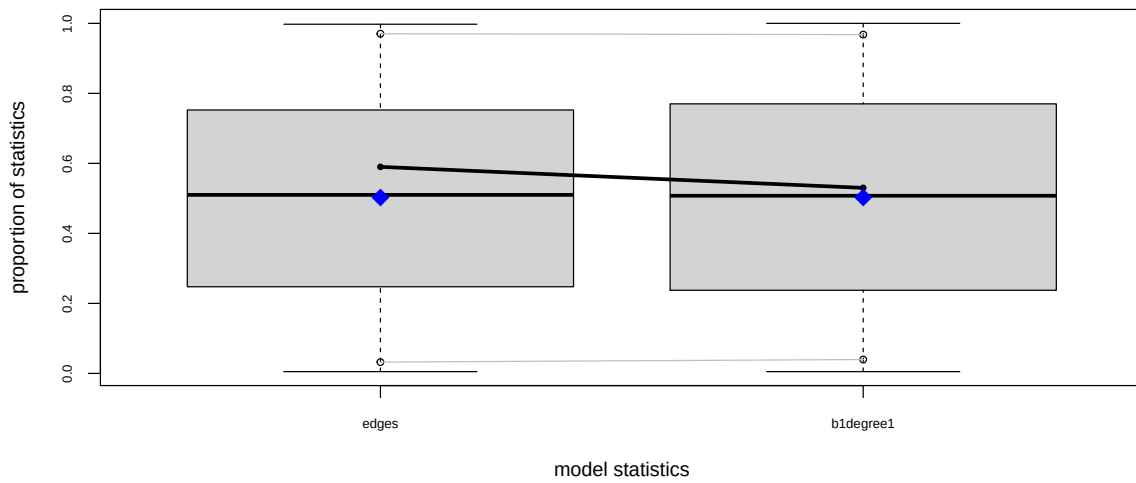
Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

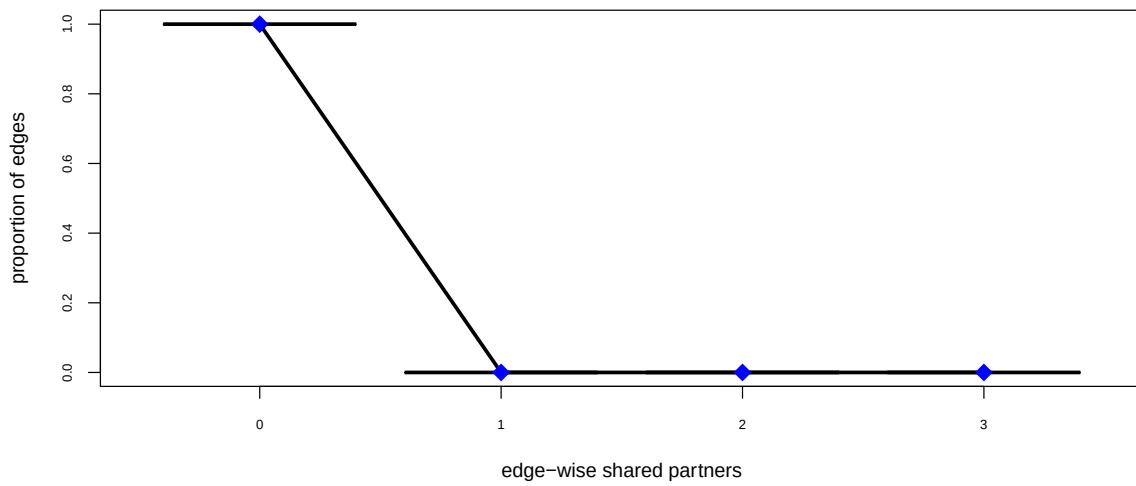
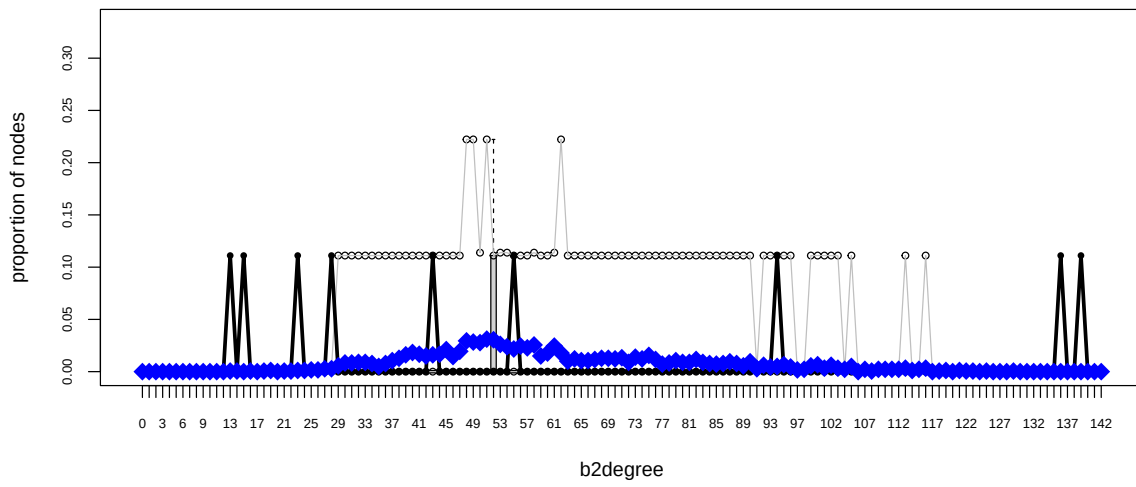
edges bidegree1
1.1789189 0.2317552

Individual P-values (lower = worse):
edges bidegree1
0.2384305 0.8167281
Joint P-value (lower = worse): 0.4940679

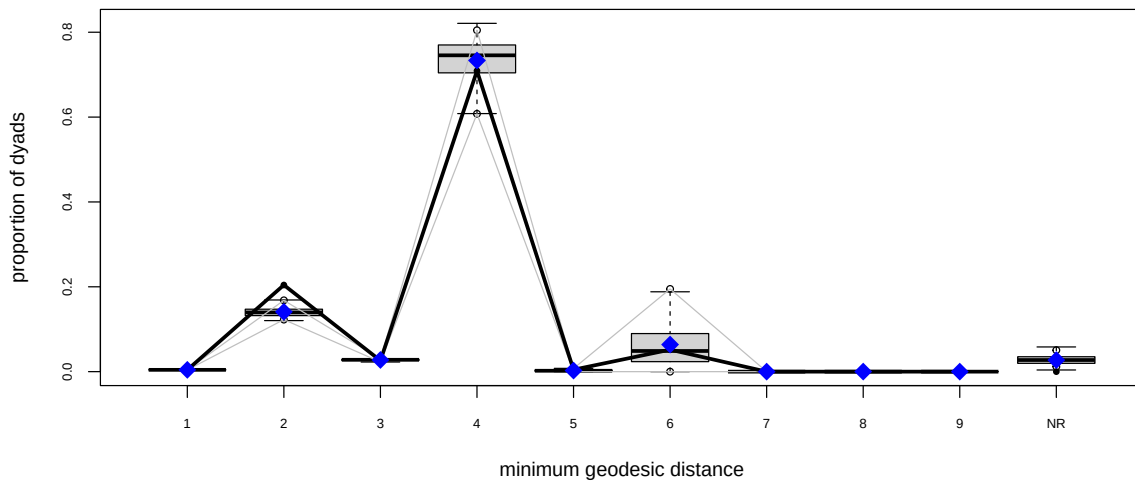
Note: To save space, only one in every 7 iterations of the MCMC sample
used for estimation was stored for diagnostics. Sample size per chain
was originally around 25221 with thinning interval 4.

Note: MCMC diagnostics shown here are from the last round of
simulation, prior to computation of final parameter estimates.
Because the final estimates are refinements of those used for this
simulation run, these diagnostics may understate model performance.
To directly assess the performance of the final model on in-model
statistics, please use the GOF command: `gof(ergmFitObject,
GOF=~model)`.





Goodness-of-fit diagnostics



Goodness-of-fit for model statistics

	obs	min	mean	max	MC	p-value
edges	546	525	548.185	574		0.86
bidegree1	459	437	458.670	477		1.00

Goodness-of-fit for minimum geodesic distance

	obs	min	mean	max	MC	p-value
1	546	525	548.185	574		0.86
2	26301	15493	18153.770	23058		0.00
3	3351	2676	3547.225	3909		0.43
4	91309	74402	94453.820	105721		0.54
5	594	0	332.150	1251		0.32
6	6677	0	8222.595	31969		0.90
7	0	0	0.800	160		1.00
8	0	0	15.000	3000		1.00
Inf	0	507	3504.455	7992		0.00

Goodness-of-fit for bipartition 1 degree

	obs	min	mean	max	MC	p-value
0	0	1	6.960	16		0.00
1	459	437	458.670	477		1.00
2	33	8	17.630	31		0.00
3	7	2	10.340	24		0.39
4	0	0	4.040	10		0.05
5	0	0	1.120	4		0.68
6	0	0	0.210	2		1.00
7	0	0	0.025	2		1.00
8	0	0	0.005	1		1.00

Goodness-of-fit for bipartition 2 degree

	obs	min	mean	max	MC	p-value
13	1	0	0.005	1		0.01
14	0	0	0.005	1		1.00
15	1	0	0.000	0		0.00
16	0	0	0.005	1		1.00
18	0	0	0.005	1		1.00
19	0	0	0.010	1		1.00
21	0	0	0.005	1		1.00

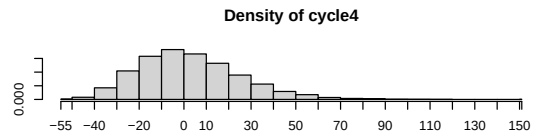
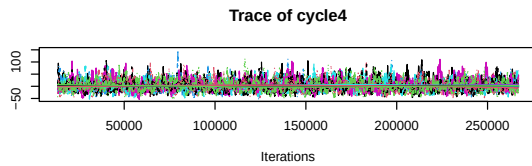
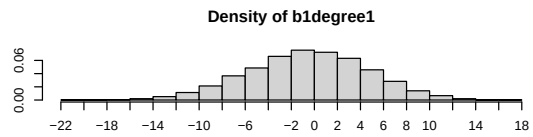
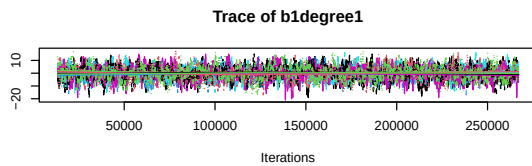
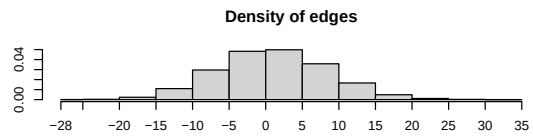
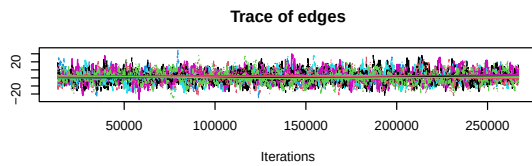
22	0	0	0.005	1	1.00
23	1	0	0.010	1	0.02
24	0	0	0.010	1	1.00
25	0	0	0.015	1	1.00
26	0	0	0.015	1	1.00
27	0	0	0.025	2	1.00
28	1	0	0.025	1	0.05
29	0	0	0.050	2	1.00
30	0	0	0.075	1	1.00
31	0	0	0.075	1	1.00
32	0	0	0.080	2	1.00
33	0	0	0.080	1	1.00
34	0	0	0.070	1	1.00
35	0	0	0.045	1	1.00
36	0	0	0.070	1	1.00
37	0	0	0.095	2	1.00
38	0	0	0.115	1	1.00
39	0	0	0.145	2	1.00
40	0	0	0.165	2	1.00
41	0	0	0.150	2	1.00
42	0	0	0.135	1	1.00
43	1	0	0.145	2	0.27
44	0	0	0.150	1	1.00
45	0	0	0.190	2	1.00
46	0	0	0.130	2	1.00
47	0	0	0.175	2	1.00
48	0	0	0.265	3	1.00
49	0	0	0.255	2	1.00
50	0	0	0.250	2	1.00
51	0	0	0.280	3	1.00
52	0	0	0.275	2	1.00
53	0	0	0.235	2	1.00
54	0	0	0.215	2	1.00
55	1	0	0.195	2	0.38
56	0	0	0.220	2	1.00
57	0	0	0.205	2	1.00
58	0	0	0.230	3	1.00
59	0	0	0.135	2	1.00
60	0	0	0.160	2	1.00
61	0	0	0.220	3	1.00
62	0	0	0.160	2	1.00
63	0	0	0.090	2	1.00
64	0	0	0.110	2	1.00
65	0	0	0.095	2	1.00
66	0	0	0.095	2	1.00
67	0	0	0.105	2	1.00
68	0	0	0.115	1	1.00
69	0	0	0.115	2	1.00
70	0	0	0.110	3	1.00
71	0	0	0.120	2	1.00
72	0	0	0.080	1	1.00
73	0	0	0.125	1	1.00
74	0	0	0.110	2	1.00
75	0	0	0.140	2	1.00
76	0	0	0.105	1	1.00
77	0	0	0.065	1	1.00
78	0	0	0.075	1	1.00
79	0	0	0.095	2	1.00
80	0	0	0.080	2	1.00
81	0	0	0.080	1	1.00
82	0	0	0.105	2	1.00
83	0	0	0.080	1	1.00
84	0	0	0.070	1	1.00
85	0	0	0.065	1	1.00
86	0	0	0.070	1	1.00
87	0	0	0.085	2	1.00
88	0	0	0.070	1	1.00
89	0	0	0.050	2	1.00

90	0	0	0.085	1	1.00
91	0	0	0.025	1	1.00
92	0	0	0.055	2	1.00
93	0	0	0.040	2	1.00
94	1	0	0.045	1	0.09
95	0	0	0.060	1	1.00
96	0	0	0.040	1	1.00
97	0	0	0.015	1	1.00
98	0	0	0.020	1	1.00
99	0	0	0.050	1	1.00
100	0	0	0.060	2	1.00
101	0	0	0.030	1	1.00
102	0	0	0.055	1	1.00
103	0	0	0.030	1	1.00
104	0	0	0.020	2	1.00
105	0	0	0.045	1	1.00
107	0	0	0.020	1	1.00
108	0	0	0.005	1	1.00
109	0	0	0.020	2	1.00
110	0	0	0.020	1	1.00
111	0	0	0.020	1	1.00
112	0	0	0.020	1	1.00
113	0	0	0.030	1	1.00
114	0	0	0.010	1	1.00
115	0	0	0.020	1	1.00
116	0	0	0.030	1	1.00
118	0	0	0.005	1	1.00
119	0	0	0.010	1	1.00
121	0	0	0.010	1	1.00
122	0	0	0.005	1	1.00
123	0	0	0.005	1	1.00
125	0	0	0.005	1	1.00
129	0	0	0.005	1	1.00
136	1	0	0.000	0	0.00
139	1	0	0.000	0	0.00

Goodness-of-fit for edgewise shared partner

obs	min	mean	max MC	p-value
546.000	525.000	548.185	574.000	0.860

A.4.3 Model 2



Sample statistics summary:

Iterations = 13376:267072
Thinning interval = 64
Number of chains = 9
Sample size per chain = 3965

1. Empirical mean and standard deviation for each variable,
plus standard error of the mean:

	Mean	SD	Naive SE	Time-series SE
edges	1.44649	7.475	0.03957	0.1970
bldegree1	-0.05955	5.224	0.02766	0.1202
cycle4	3.02365	22.559	0.11942	0.5202

2. Quantiles for each variable:

	2.5%	25%	50%	75%	97.5%
edges	-13	-4	1	7	16
bldegree1	-10	-4	0	4	10
cycle4	-34	-13	1	17	53

Are sample statistics significantly different from observed?

	edges	bldegree1	cycle4	(Omni)
diff.	1.446490e+00	-0.05954883	3.023651e+00	NA
test stat.	7.380226e+00	-0.49639643	5.879148e+00	8.382182e+01
P-val.	1.580205e-13	0.61961475	4.123823e-09	1.072914e-17

Sample statistics cross-correlations:

	edges	bldegree1	cycle4
edges	1.0000000	-0.644952	0.8593833
bldegree1	-0.6449520	1.000000	-0.5739620
cycle4	0.8593833	-0.573962	1.0000000

Sample statistics auto-correlation:

Chain 1

	edges	bldegree1	cycle4
Lag 0	1.0000000	1.0000000	1.0000000
Lag 64	0.9207720	0.8939305	0.8908135
Lag 128	0.8477613	0.7974903	0.7902465
Lag 192	0.7824516	0.7176731	0.7027372
Lag 256	0.7199936	0.6447281	0.6311525
Lag 320	0.6593805	0.5796761	0.5623123

Chain 2

	edges	bldegree1	cycle4
Lag 0	1.0000000	1.0000000	1.0000000
Lag 64	0.9127709	0.8916141	0.8699662
Lag 128	0.8323720	0.7990969	0.7576879
Lag 192	0.7585718	0.7152578	0.6632909
Lag 256	0.6928837	0.6408781	0.5816583
Lag 320	0.6307202	0.5757939	0.5101108

Chain 3

	edges	bldegree1	cycle4
Lag 0	1.0000000	1.0000000	1.0000000
Lag 64	0.9247032	0.9053829	0.8947725
Lag 128	0.8626373	0.8250421	0.8072021
Lag 192	0.8036997	0.7535264	0.7285900
Lag 256	0.7514687	0.6915726	0.6666843
Lag 320	0.7033289	0.6395590	0.6111047

Chain 4

	edges	bldegree1	cycle4
Lag 0	1.0000000	1.0000000	1.0000000
Lag 64	0.9194227	0.8882619	0.8836353
Lag 128	0.8460630	0.7872335	0.7865274
Lag 192	0.7760913	0.6949317	0.7035127
Lag 256	0.7112373	0.6160141	0.6286306
Lag 320	0.6561840	0.5565084	0.5674450

```

Chain 5
      edges bidegree1   cycle4
Lag 0   1.0000000 1.0000000 1.0000000
Lag 64   0.9216747 0.8899834 0.8896689
Lag 128  0.8516064 0.7949439 0.7975814
Lag 192  0.7891529 0.7120355 0.7220413
Lag 256  0.7350742 0.6384775 0.6582887
Lag 320  0.6902214 0.5758803 0.6053976
Chain 6
      edges bidegree1   cycle4
Lag 0   1.0000000 1.0000000 1.0000000
Lag 64   0.9286170 0.9040554 0.8938356
Lag 128  0.8635580 0.8200027 0.8061285
Lag 192  0.8054181 0.7433293 0.7345318
Lag 256  0.7518688 0.6798884 0.6733401
Lag 320  0.7022343 0.6197807 0.6207549
Chain 7
      edges bidegree1   cycle4
Lag 0   1.0000000 1.0000000 1.0000000
Lag 64   0.9146513 0.8928942 0.8829731
Lag 128  0.8378383 0.7951906 0.7895102
Lag 192  0.7709455 0.7122812 0.7088251
Lag 256  0.7106735 0.6369230 0.6390087
Lag 320  0.6506091 0.5680243 0.5719387
Chain 8
      edges bidegree1   cycle4
Lag 0   1.0000000 1.0000000 1.0000000
Lag 64   0.9169727 0.8948412 0.8835476
Lag 128  0.8384772 0.7985448 0.7808854
Lag 192  0.7696477 0.7101403 0.6929739
Lag 256  0.7056378 0.6336532 0.6158042
Lag 320  0.6465219 0.5651076 0.5422954
Chain 9
      edges bidegree1   cycle4
Lag 0   1.0000000 1.0000000 1.0000000
Lag 64   0.9153677 0.8944287 0.8843268
Lag 128  0.8355754 0.7943154 0.7820059
Lag 192  0.7669759 0.7137630 0.6998537
Lag 256  0.7064500 0.6496772 0.6291565
Lag 320  0.6520078 0.5914224 0.5688262

Sample statistics burn-in diagnostic (Geweke):
Chain 1

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

      edges bidegree1   cycle4
-0.9336618 -0.3072449  0.1164942

Individual P-values (lower = worse):
      edges bidegree1   cycle4
0.3504784 0.7586570 0.9072609
Joint P-value (lower = worse):  0.1702401
Chain 2

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

      edges bidegree1   cycle4
0.009917489 -0.849595199  0.751238982

Individual P-values (lower = worse):
      edges bidegree1   cycle4
0.9920871 0.3955502 0.4525088
Joint P-value (lower = worse):  0.4049945
Chain 3

```

```

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

      edges bidegree1    cycle4
-0.2985873  0.6818755 -1.1698052

Individual P-values (lower = worse):
      edges bidegree1    cycle4
0.7652550  0.4953177  0.2420794
Joint P-value (lower = worse):  0.5700744
Chain 4

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

      edges bidegree1    cycle4
0.1115741 -0.9752093  0.1768475

Individual P-values (lower = worse):
      edges bidegree1    cycle4
0.9111611  0.3294564  0.8596282
Joint P-value (lower = worse):  0.5713822
Chain 5

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

      edges bidegree1    cycle4
-0.1818034  0.3848732 -0.4954972

Individual P-values (lower = worse):
      edges bidegree1    cycle4
0.8557370  0.7003313  0.6202492
Joint P-value (lower = worse):  0.8975539
Chain 6

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

      edges bidegree1    cycle4
-0.4751854  0.8079716 -0.2877042

Individual P-values (lower = worse):
      edges bidegree1    cycle4
0.6346548  0.4191069  0.7735732
Joint P-value (lower = worse):  0.6088046
Chain 7

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

      edges bidegree1    cycle4
-0.8753117 -0.1067233 -0.7792244

Individual P-values (lower = worse):
      edges bidegree1    cycle4
0.3814043  0.9150085  0.4358475
Joint P-value (lower = worse):  0.5840519
Chain 8

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

      edges bidegree1    cycle4
-0.4947983  1.4777652 -1.9407711

Individual P-values (lower = worse):
      edges bidegree1    cycle4

```

0.62074253 0.13947063 0.05228605
 Joint P-value (lower = worse): 0.1057777
 Chain 9

Fraction in 1st window = 0.1
 Fraction in 2nd window = 0.5

edges	b1degree1	cycle4
0.7249734	0.1051507	0.8464796

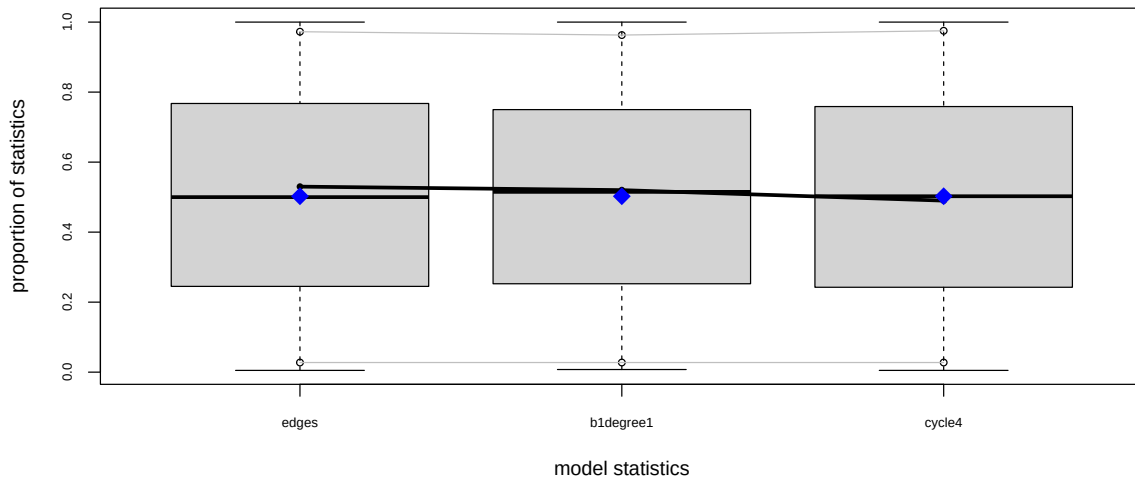
Individual P-values (lower = worse):

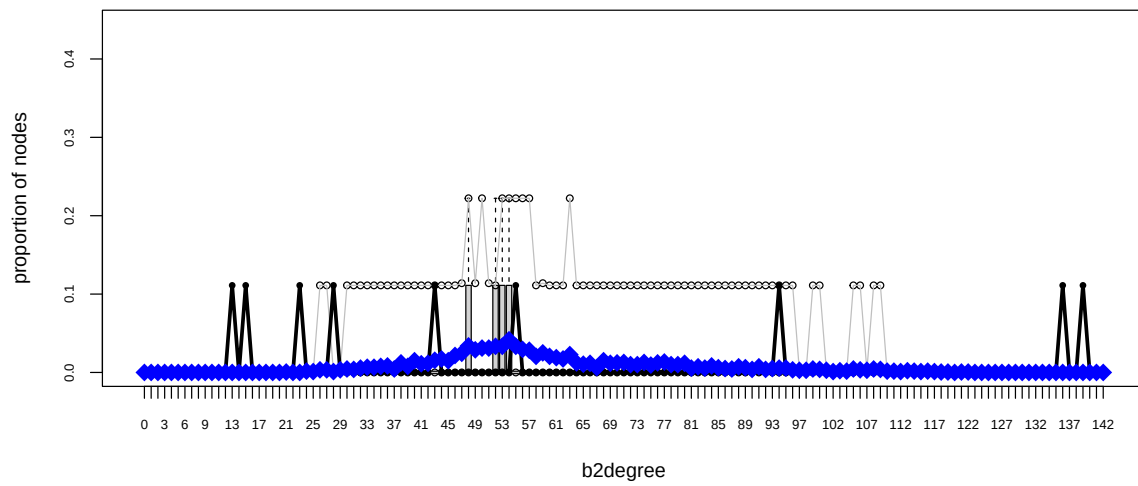
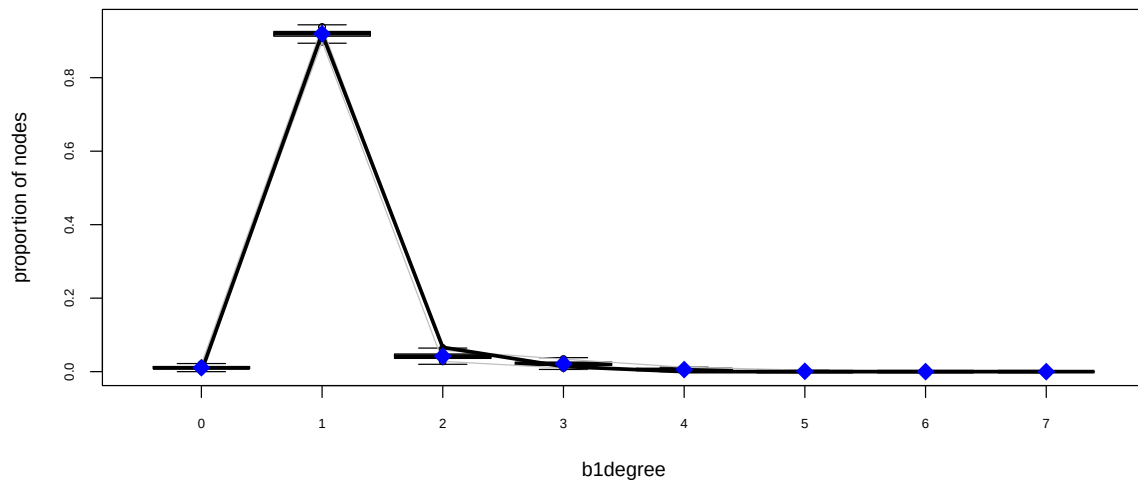
edges	b1degree1	cycle4
0.4684684	0.9162563	0.3972853

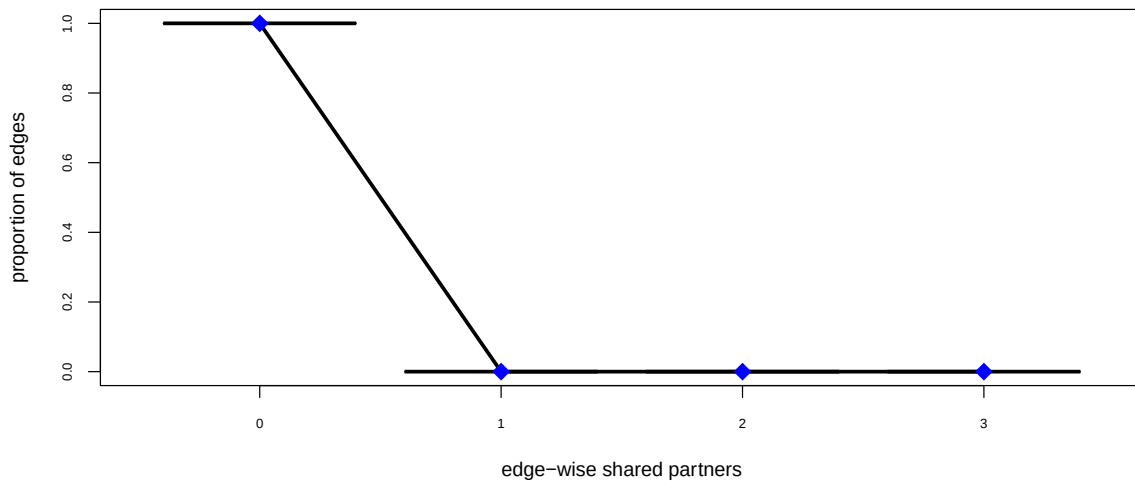
 Joint P-value (lower = worse): 0.725172

Note: To save space, only one in every 4 iterations of the MCMC sample used for estimation was stored for diagnostics. Sample size per chain was originally around 15860 with thinning interval 16.

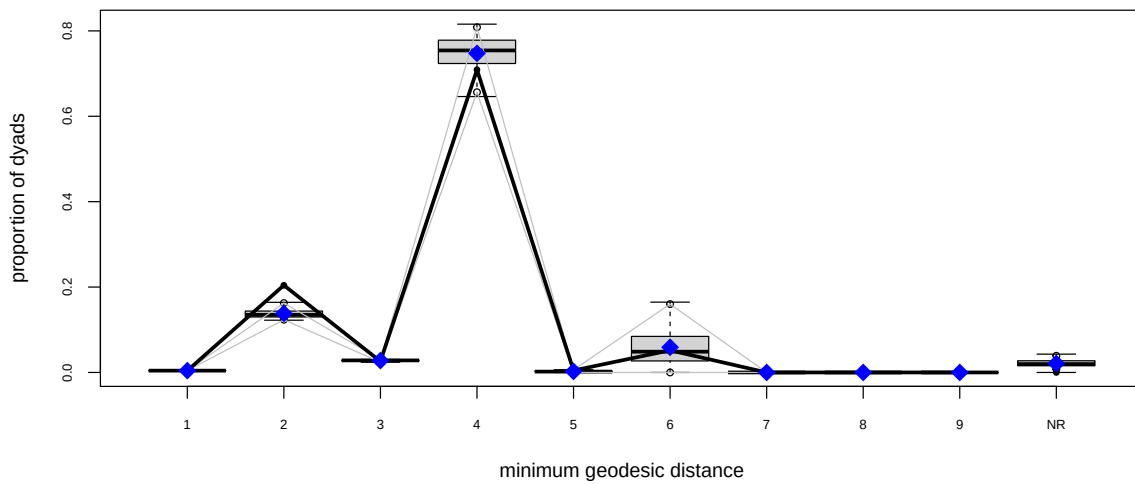
Note: MCMC diagnostics shown here are from the last round of simulation, prior to computation of final parameter estimates. Because the final estimates are refinements of those used for this simulation run, these diagnostics may understate model performance. To directly assess the performance of the final model on in-model statistics, please use the GOF command: `gof(ergmFitObject, GOF=~model)`.







Goodness-of-fit diagnostics



Goodness-of-fit for model statistics

	obs	min	mean	max	MC	p-value
edges	546	528	546.330	568		1.00
bldegree1	459	446	458.785	473		1.00
cycle4	67	26	68.885	157		0.98

Goodness-of-fit for minimum geodesic distance

	obs	min	mean	max	MC	p-value
1	546	528	546.330	568		1.00
2	26301	15759	17794.830	21669		0.00
3	3351	2845	3587.910	3920		0.33
4	91309	77582	96233.180	105053		0.37
5	594	0	308.520	1061		0.25
6	6677	0	7603.865	28137		0.96
Inf	0	0	2703.365	6018		0.01

Goodness-of-fit for bipartition 1 degree

	obs	min	mean	max	MC	p-value
0	0	0	5.360	12		0.01
1	459	446	458.785	473		1.00
2	33	10	20.880	32		0.00
3	7	3	10.670	21		0.30
4	0	0	2.790	7		0.10
5	0	0	0.475	3		1.00
6	0	0	0.040	2		1.00

Goodness-of-fit for bipartition 2 degree

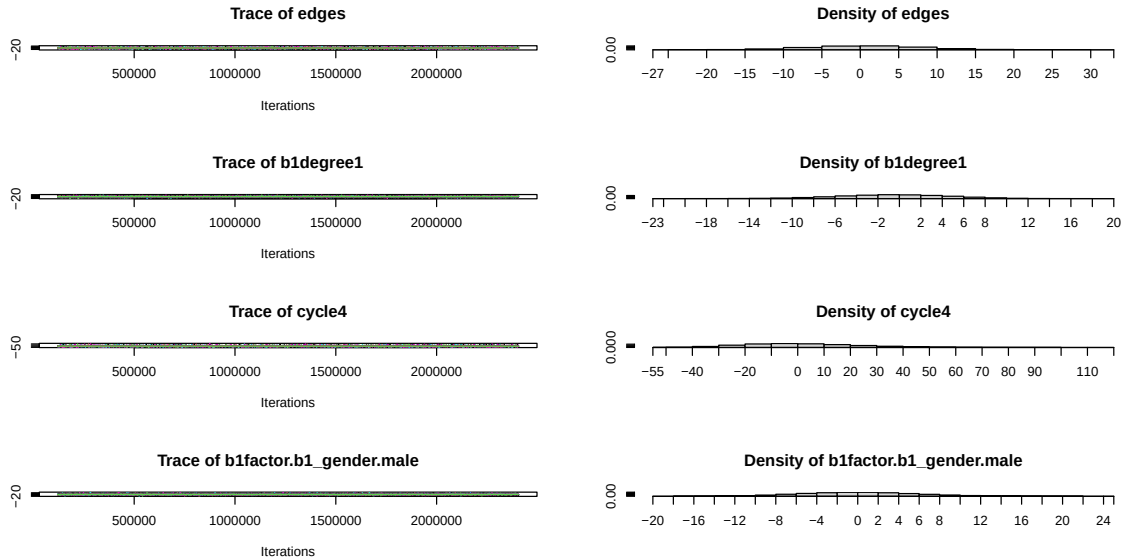
	obs	min	mean	max	MC	p-value
13	1	0	0.000	0		0.00
15	1	0	0.000	0		0.00
21	0	0	0.005	1		1.00
23	1	0	0.000	0		0.00
24	0	0	0.015	2		1.00
25	0	0	0.010	1		1.00
26	0	0	0.030	1		1.00
27	0	0	0.030	1		1.00
28	1	0	0.010	1		0.02
29	0	0	0.030	2		1.00
30	0	0	0.040	2		1.00
31	0	0	0.035	1		1.00
32	0	0	0.050	2		1.00
33	0	0	0.060	1		1.00
34	0	0	0.060	1		1.00
35	0	0	0.070	2		1.00
36	0	0	0.075	1		1.00
37	0	0	0.040	1		1.00
38	0	0	0.110	1		1.00
39	0	0	0.070	2		1.00
40	0	0	0.135	1		1.00
41	0	0	0.100	3		1.00
42	0	0	0.105	3		1.00
43	1	0	0.140	2		0.27
44	0	0	0.155	2		1.00
45	0	0	0.135	2		1.00
46	0	0	0.195	2		1.00
47	0	0	0.220	2		1.00
48	0	0	0.305	2		1.00
49	0	0	0.260	3		1.00
50	0	0	0.280	3		1.00
51	0	0	0.280	3		1.00
52	0	0	0.300	2		1.00
53	0	0	0.300	2		1.00
54	0	0	0.375	2		1.00
55	1	0	0.300	3		0.48
56	0	0	0.265	4		1.00
57	0	0	0.255	2		1.00
58	0	0	0.185	3		1.00
59	0	0	0.225	2		1.00
60	0	0	0.185	3		1.00
61	0	0	0.170	2		1.00
62	0	0	0.155	3		1.00
63	0	0	0.205	3		1.00
64	0	0	0.110	2		1.00
65	0	0	0.095	1		1.00
66	0	0	0.105	2		1.00
67	0	0	0.055	2		1.00
68	0	0	0.135	2		1.00
69	0	0	0.105	2		1.00
70	0	0	0.110	2		1.00
71	0	0	0.115	2		1.00
72	0	0	0.090	2		1.00

73	0	0	0.090	2	1.00
74	0	0	0.115	2	1.00
75	0	0	0.085	2	1.00
76	0	0	0.110	2	1.00
77	0	0	0.120	1	1.00
78	0	0	0.090	2	1.00
79	0	0	0.090	2	1.00
80	0	0	0.105	2	1.00
81	0	0	0.050	1	1.00
82	0	0	0.065	1	1.00
83	0	0	0.050	1	1.00
84	0	0	0.075	1	1.00
85	0	0	0.055	1	1.00
86	0	0	0.045	1	1.00
87	0	0	0.040	2	1.00
88	0	0	0.065	2	1.00
89	0	0	0.055	1	1.00
90	0	0	0.035	1	1.00
91	0	0	0.065	1	1.00
92	0	0	0.035	2	1.00
93	0	0	0.045	1	1.00
94	1	0	0.050	2	0.09
95	0	0	0.050	1	1.00
96	0	0	0.030	1	1.00
97	0	0	0.025	1	1.00
98	0	0	0.025	1	1.00
99	0	0	0.040	1	1.00
100	0	0	0.035	1	1.00
101	0	0	0.025	1	1.00
102	0	0	0.010	1	1.00
103	0	0	0.020	1	1.00
104	0	0	0.015	1	1.00
105	0	0	0.040	1	1.00
106	0	0	0.030	1	1.00
107	0	0	0.025	1	1.00
108	0	0	0.040	1	1.00
109	0	0	0.035	1	1.00
110	0	0	0.015	1	1.00
111	0	0	0.015	1	1.00
112	0	0	0.010	1	1.00
113	0	0	0.020	1	1.00
114	0	0	0.015	1	1.00
115	0	0	0.010	1	1.00
116	0	0	0.015	1	1.00
117	0	0	0.010	1	1.00
118	0	0	0.005	1	1.00
119	0	0	0.005	1	1.00
121	0	0	0.005	1	1.00
122	0	0	0.005	1	1.00
136	1	0	0.000	0	0.00
139	1	0	0.000	0	0.00

Goodness-of-fit for edgewise shared partner

obs	min	mean	max MC	p-value
546.00	528.00	546.33	568.00	1.00

A.4.4 Model 3



Sample statistics summary:

Iterations = 120448:2406528
 Thinning interval = 640
 Number of chains = 9
 Sample size per chain = 3573

1. Empirical mean and standard deviation for each variable,
 plus standard error of the mean:

	Mean	SD	Naive SE	Time-series SE
edges	1.25307	7.431	0.04144	0.06826
b1degree1	-0.05458	5.252	0.02929	0.04472
cycle4	2.61648	22.356	0.12467	0.18772
b1factor.b1_gender.male	0.71207	5.713	0.03186	0.05854

2. Quantiles for each variable:

	2.5%	25%	50%	75%	97.5%
edges	-13	-4	1	6	16
b1degree1	-11	-4	0	4	10
cycle4	-34	-13	0	16	52
b1factor.b1_gender.male	-10	-3	1	4	12

Are sample statistics significantly different from observed?

	edges	b1degree1	cycle4	b1factor.b1_gender.male
diff.	1.253071e+00	-0.05457599	2.616475e+00	7.120689e-01
test stat.	1.837232e+01	-1.22226000	1.394755e+01	1.216653e+01
P-val.	2.188414e-75	0.22160932	3.256092e-44	4.685805e-34

(Omni)

	NA
diff.	NA
test stat.	4.981756e+02
P-val.	2.180719e-104

Sample statistics cross-correlations:

	edges	b1degree1	cycle4
edges	1.0000000	-0.6273622	0.8549354
b1degree1	-0.6273622	1.0000000	-0.5698004
cycle4	0.8549354	-0.5698004	1.0000000
b1factor.b1_gender.male	0.5314239	-0.3247677	0.4217988

b1factor.b1_gender.male

edges	0.5314239
bidegree1	-0.3247677
cycle4	0.4217988
bifactor.b1_gender.male	1.0000000

Sample statistics auto-correlation:

Chain 1

	edges	bidegree1	cycle4	bifactor.b1_gender.male
Lag 0	1.00000000	1.00000000	1.00000000	1.00000000
Lag 640	0.46219220	0.36123841	0.36808454	0.56555488
Lag 1280	0.22083629	0.13173911	0.15681895	0.32971930
Lag 1920	0.10106233	0.05114607	0.06465253	0.19505666
Lag 2560	0.04911140	0.02504884	0.03203135	0.10243262
Lag 3200	0.01765148	-0.01676550	0.03047180	0.04651129

Chain 2

	edges	bidegree1	cycle4	bifactor.b1_gender.male
Lag 0	1.00000000	1.00000000	1.00000000	1.00000000
Lag 640	0.437858208	0.38674645	0.35549994	0.54439105
Lag 1280	0.170365470	0.17315370	0.11391348	0.30861060
Lag 1920	0.077068682	0.07864354	0.04139655	0.19067311
Lag 2560	0.014848922	0.03272962	0.01527084	0.11125867
Lag 3200	-0.002735924	0.02799260	0.01029926	0.06692818

Chain 3

	edges	bidegree1	cycle4	bifactor.b1_gender.male
Lag 0	1.0000000000	1.00000000	1.00000000	1.00000000
Lag 640	0.4638303468	0.38499285	0.36101793	0.56588893
Lag 1280	0.2486813189	0.18855567	0.18681347	0.33965285
Lag 1920	0.1317563283	0.10151423	0.09475260	0.19857586
Lag 2560	0.0475566842	0.05226031	0.04985226	0.09518536
Lag 3200	0.0003148158	0.02992456	0.02826998	0.01726816

Chain 4

	edges	bidegree1	cycle4	bifactor.b1_gender.male
Lag 0	1.0000000e+00	1.00000000	1.00000000	1.00000000
Lag 640	4.474375e-01	0.40827558	0.36602612	0.56673910
Lag 1280	1.999899e-01	0.18376229	0.16108594	0.33121846
Lag 1920	7.810462e-02	0.06819393	0.06662650	0.17515468
Lag 2560	2.214036e-02	0.03229893	0.02759365	0.07527084
Lag 3200	1.327434e-05	0.03020447	0.01418929	0.04068117

Chain 5

	edges	bidegree1	cycle4	bifactor.b1_gender.male
Lag 0	1.000000000	1.00000000	1.00000000	1.00000000
Lag 640	0.437103052	0.38423058	0.34784445	0.56856874
Lag 1280	0.216170483	0.17224360	0.15877064	0.33285961
Lag 1920	0.105458281	0.07302384	0.09177188	0.19717275
Lag 2560	0.044444789	0.02504269	0.04868227	0.08931366
Lag 3200	-0.009955747	-0.01250088	-0.01686271	0.04036875

Chain 6

	edges	bidegree1	cycle4	bifactor.b1_gender.male
Lag 0	1.00000000	1.00000000	1.00000000	1.00000000
Lag 640	0.45435011	0.361268517	0.35812343	0.56960682
Lag 1280	0.21566779	0.153981676	0.15965314	0.33619428
Lag 1920	0.12654181	0.072802022	0.09339018	0.18847000
Lag 2560	0.05006227	0.007440694	0.03046906	0.09862146
Lag 3200	0.02497662	-0.008779587	0.01561748	0.02754053

Chain 7

	edges	bidegree1	cycle4	bifactor.b1_gender.male
Lag 0	1.000000000	1.00000000	1.00000000	1.00000000
Lag 640	0.463362102	0.362904627	0.363305483	0.56274889
Lag 1280	0.207687456	0.144460485	0.156659826	0.30735588
Lag 1920	0.089675932	0.044654202	0.078071052	0.17663718
Lag 2560	0.036751121	0.023897805	0.035773901	0.10232906
Lag 3200	0.003172149	0.006837233	-0.001948617	0.07518207

Chain 8

	edges	bidegree1	cycle4	bifactor.b1_gender.male
Lag 0	1.00000000	1.00000000	1.00000000	1.00000000
Lag 640	0.44705209	0.40603636	0.341088943	0.55333751
Lag 1280	0.22469509	0.18881850	0.160573033	0.31794405
Lag 1920	0.10221498	0.08973941	0.058634998	0.19089898

```

Lag 2560 0.03224681 0.04772215 0.027979386      0.10318507
Lag 3200 0.01803967 0.02104698 0.001062492      0.05033198
Chain 9
      edges    bidegree1      cycle4 bifactor.b1_gender.male
Lag 0    1.00000000 1.00000000 1.00000000      1.00000000
Lag 640  0.44119476 0.365874321 0.343237943      0.54715732
Lag 1280 0.21378712 0.165901421 0.149457193      0.29771860
Lag 1920 0.09606213 0.056225445 0.054445652      0.15065216
Lag 2560 0.03676788 0.018721924 -0.002869194      0.06468348
Lag 3200 0.03169008 0.002035173 0.007041516      0.02444377

```

Sample statistics burn-in diagnostic (Geweke):
Chain 1

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

```

      edges      bidegree1      cycle4
-0.8376964      -0.4855318      -0.5127717
bifactor.b1_gender.male
-0.3354309

```

Individual P-values (lower = worse):

```

      edges      bidegree1      cycle4
0.4022012      0.6272992      0.6081110
bifactor.b1_gender.male
0.7373001

```

Joint P-value (lower = worse): 0.8182054
Chain 2

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

```

      edges      bidegree1      cycle4
-2.2283677      -0.1273937      -1.7492323
bifactor.b1_gender.male
-1.1146404

```

Individual P-values (lower = worse):

```

      edges      bidegree1      cycle4
0.02585601      0.89862881      0.08025087
bifactor.b1_gender.male
0.26500453

```

Joint P-value (lower = worse): 0.09741188
Chain 3

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

```

      edges      bidegree1      cycle4
0.12534550      -0.54659601      -0.09595021
bifactor.b1_gender.male
-0.46754728

```

Individual P-values (lower = worse):

```

      edges      bidegree1      cycle4
0.9002500      0.5846563      0.9235601
bifactor.b1_gender.male
0.6401084

```

Joint P-value (lower = worse): 0.8205227
Chain 4

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

```

      edges      bidegree1      cycle4
-0.8731893      0.4147736      -0.6828871
bifactor.b1_gender.male

```

-1.0626114

Individual P-values (lower = worse):

edges	bidegree1	cycle4
0.3825599	0.6783077	0.4946782

bifactor.b1_gender.male
0.2879582

Joint P-value (lower = worse): 0.8571621

Chain 5

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

edges	bidegree1	cycle4
-0.7303596	0.4544881	-0.7449774

bifactor.b1_gender.male
-0.7221944

Individual P-values (lower = worse):

edges	bidegree1	cycle4
0.4651704	0.6494776	0.4562854

bifactor.b1_gender.male
0.4701749

Joint P-value (lower = worse): 0.9284842

Chain 6

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

edges	bidegree1	cycle4
-0.4456012	-0.3455419	-0.7847546

bifactor.b1_gender.male
-0.2160909

Individual P-values (lower = worse):

edges	bidegree1	cycle4
0.6558853	0.7296870	0.4325975

bifactor.b1_gender.male
0.8289169

Joint P-value (lower = worse): 0.7642974

Chain 7

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

edges	bidegree1	cycle4
-0.3819243	0.7027031	-0.7415056

bifactor.b1_gender.male
1.4539718

Individual P-values (lower = worse):

edges	bidegree1	cycle4
0.7025175	0.4822408	0.4583870

bifactor.b1_gender.male
0.1459541

Joint P-value (lower = worse): 0.5891684

Chain 8

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

edges	bidegree1	cycle4
-0.104999617	-0.968118379	-0.004073132

bifactor.b1_gender.male
-0.599161923

Individual P-values (lower = worse):

edges	bidegree1	cycle4
-------	-----------	--------

```

0.9163761      0.3329853      0.9967501
b1factor.b1_gender.male
0.5490649
Joint P-value (lower = worse): 0.6925802
Chain 9

```

```

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

```

```

edges      b1degree1      cycle4
1.2922259  -0.9945804      1.0602979
b1factor.b1_gender.male
1.4268769

```

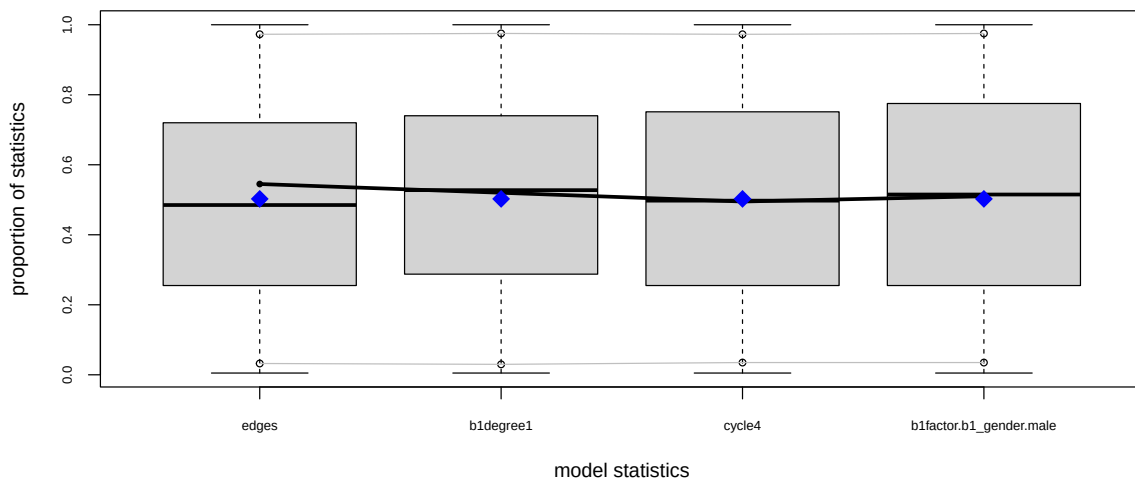
```

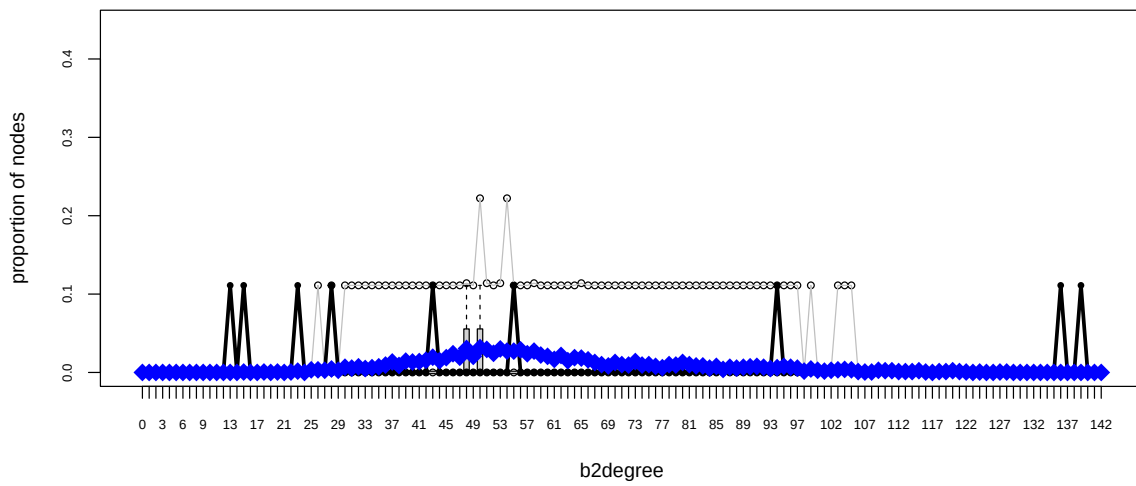
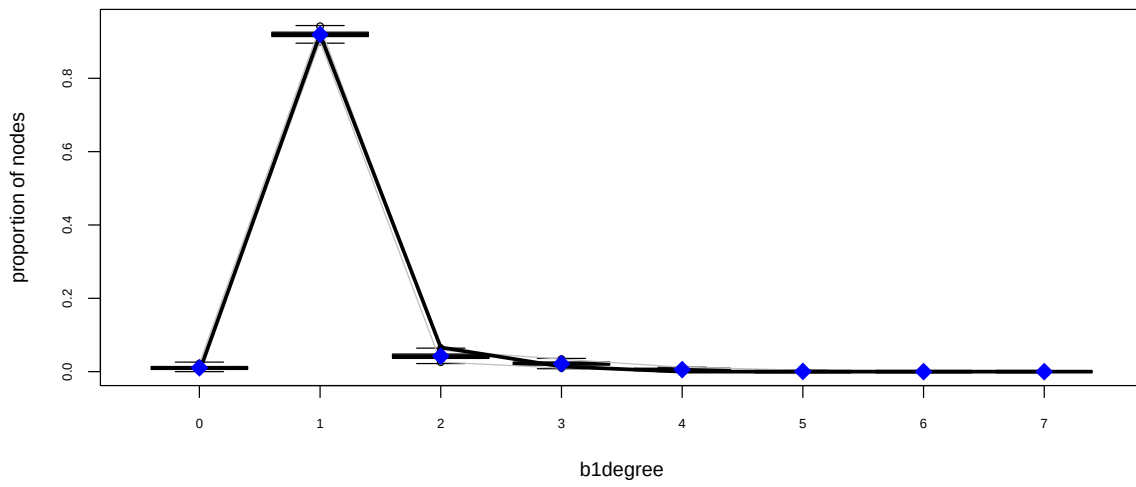
Individual P-values (lower = worse):
edges      b1degree1      cycle4
0.1962789  0.3199404      0.2890091
b1factor.b1_gender.male
0.1536154
Joint P-value (lower = worse): 0.5098452

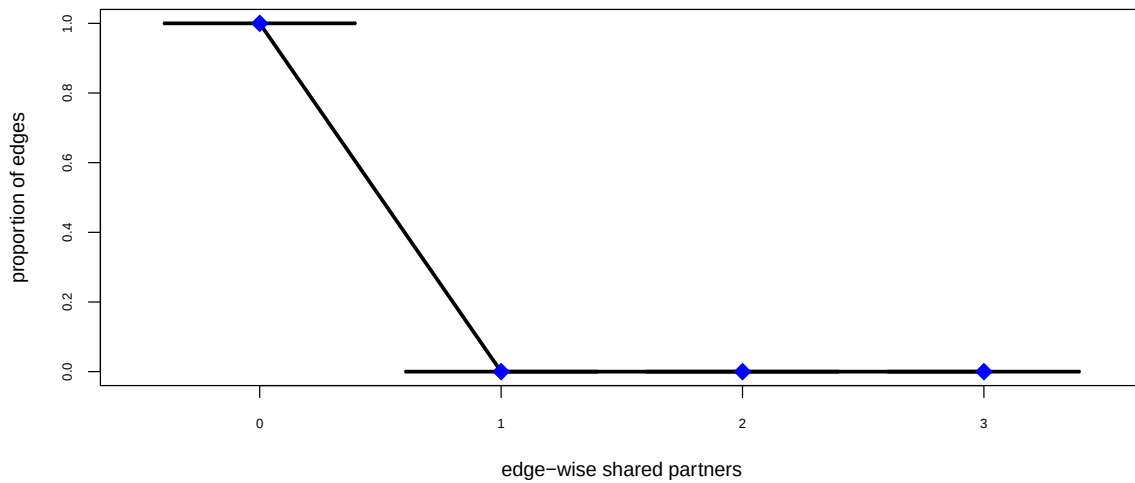
```

Note: To save space, only one in every 5 iterations of the MCMC sample used for estimation was stored for diagnostics. Sample size per chain was originally around 17865 with thinning interval 128.

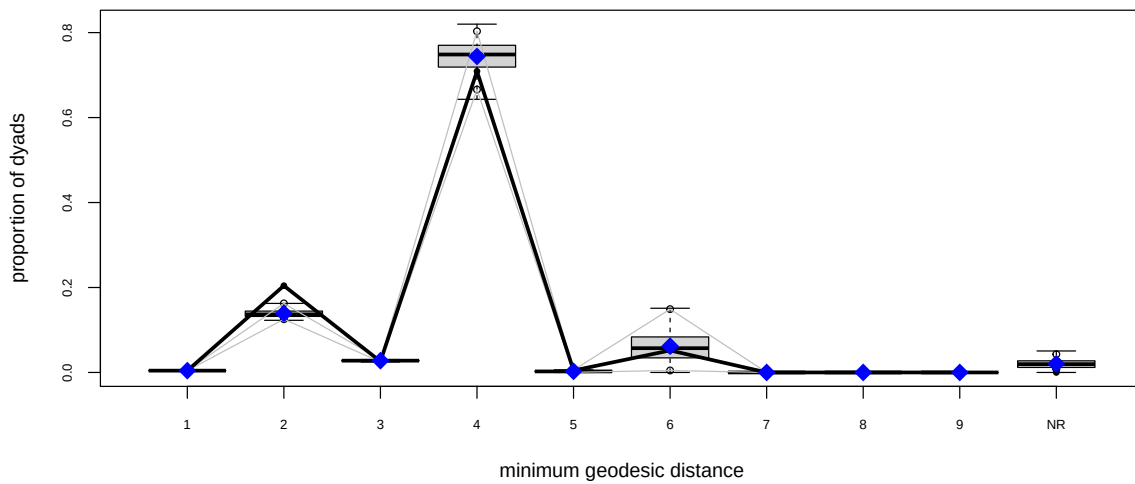
Note: MCMC diagnostics shown here are from the last round of simulation, prior to computation of final parameter estimates. Because the final estimates are refinements of those used for this simulation run, these diagnostics may understate model performance. To directly assess the performance of the final model on in-model statistics, please use the GOF command: `gof(ergmFitObject, GOF=~model)`.







Goodness-of-fit diagnostics



Goodness-of-fit for model statistics

	obs	min	mean	max	MC	p-value
edges	546	527	546.070	570		1.00
bldegree1	459	441	458.810	474		1.00
cycle4	67	20	67.250	144		0.99
bifactor.b1_gender.male	257	243	256.995	274		1.00

Goodness-of-fit for minimum geodesic distance

	obs	min	mean	max	MC	p-value
1	546	527	546.070	570		1.00
2	26301	15809	17922.080	22664		0.00
3	3351	2921	3563.925	3912		0.26
4	91309	79282	95788.770	105587		0.34
5	594	0	333.755	953		0.18
6	6677	0	7976.075	24134		0.92

Inf 0 0 2647.325 6513 0.01

Goodness-of-fit for bipartition 1 degree

	obs	min	mean	max	MC	p-value
0	0	0	5.250	13		0.01
1	459	441	458.810	474		1.00
2	33	9	21.095	32		0.00
3	7	3	10.715	18		0.30
4	0	0	2.755	7		0.13
5	0	0	0.345	2		1.00
6	0	0	0.030	1		1.00

Goodness-of-fit for bipartition 2 degree

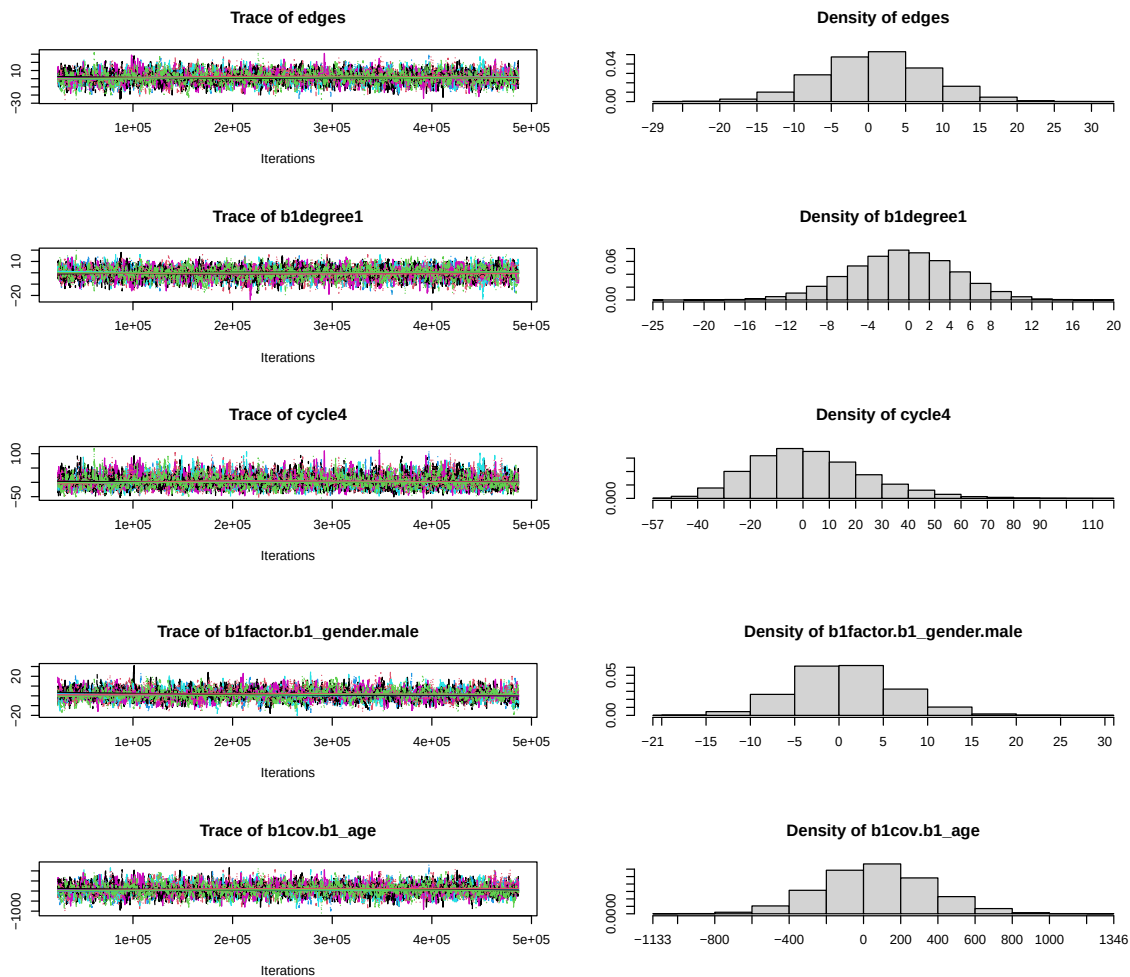
	obs	min	mean	max	MC	p-value
13	1	0	0.000	0		0.00
15	1	0	0.005	1		0.01
18	0	0	0.005	1		1.00
20	0	0	0.005	1		1.00
22	0	0	0.005	1		1.00
23	1	0	0.015	1		0.03
25	0	0	0.030	2		1.00
26	0	0	0.030	1		1.00
27	0	0	0.025	1		1.00
28	1	0	0.040	2		0.07
29	0	0	0.025	1		1.00
30	0	0	0.060	1		1.00
31	0	0	0.050	1		1.00
32	0	0	0.065	2		1.00
33	0	0	0.045	1		1.00
34	0	0	0.055	1		1.00
35	0	0	0.065	1		1.00
36	0	0	0.085	2		1.00
37	0	0	0.120	2		1.00
38	0	0	0.080	2		1.00
39	0	0	0.130	2		1.00
40	0	0	0.120	2		1.00
41	0	0	0.125	2		1.00
42	0	0	0.140	1		1.00
43	1	0	0.175	2		0.33
44	0	0	0.135	1		1.00
45	0	0	0.170	2		1.00
46	0	0	0.215	3		1.00
47	0	0	0.175	2		1.00
48	0	0	0.275	2		1.00
49	0	0	0.185	2		1.00
50	0	0	0.285	2		1.00
51	0	0	0.265	2		1.00
52	0	0	0.220	2		1.00
53	0	0	0.270	2		1.00
54	0	0	0.245	4		1.00
55	1	0	0.245	2		0.46
56	0	0	0.260	2		1.00
57	0	0	0.215	2		1.00
58	0	0	0.245	2		1.00
59	0	0	0.200	2		1.00
60	0	0	0.185	2		1.00
61	0	0	0.150	2		1.00
62	0	0	0.195	2		1.00
63	0	0	0.135	2		1.00
64	0	0	0.170	2		1.00
65	0	0	0.165	2		1.00
66	0	0	0.145	2		1.00
67	0	0	0.115	1		1.00
68	0	0	0.090	2		1.00
69	0	0	0.075	1		1.00
70	0	0	0.115	2		1.00

71	0	0	0.090	2	1.00
72	0	0	0.085	1	1.00
73	0	0	0.125	1	1.00
74	0	0	0.085	1	1.00
75	0	0	0.090	2	1.00
76	0	0	0.070	1	1.00
77	0	0	0.055	1	1.00
78	0	0	0.090	2	1.00
79	0	0	0.085	1	1.00
80	0	0	0.115	2	1.00
81	0	0	0.090	1	1.00
82	0	0	0.075	1	1.00
83	0	0	0.075	1	1.00
84	0	0	0.050	1	1.00
85	0	0	0.070	1	1.00
86	0	0	0.035	1	1.00
87	0	0	0.060	1	1.00
88	0	0	0.050	1	1.00
89	0	0	0.060	1	1.00
90	0	0	0.065	1	1.00
91	0	0	0.070	2	1.00
92	0	0	0.060	1	1.00
93	0	0	0.045	2	1.00
94	1	0	0.055	1	0.11
95	0	0	0.070	1	1.00
96	0	0	0.060	1	1.00
97	0	0	0.050	1	1.00
98	0	0	0.015	1	1.00
99	0	0	0.040	1	1.00
100	0	0	0.025	1	1.00
101	0	0	0.015	1	1.00
102	0	0	0.025	1	1.00
103	0	0	0.030	1	1.00
104	0	0	0.035	1	1.00
105	0	0	0.030	1	1.00
106	0	0	0.010	1	1.00
107	0	0	0.005	1	1.00
108	0	0	0.005	1	1.00
109	0	0	0.025	1	1.00
110	0	0	0.020	1	1.00
111	0	0	0.020	1	1.00
112	0	0	0.010	1	1.00
113	0	0	0.010	1	1.00
114	0	0	0.010	1	1.00
115	0	0	0.020	1	1.00
116	0	0	0.005	1	1.00
118	0	0	0.010	1	1.00
119	0	0	0.010	1	1.00
120	0	0	0.020	1	1.00
121	0	0	0.010	1	1.00
122	0	0	0.005	1	1.00
124	0	0	0.005	1	1.00
127	0	0	0.005	1	1.00
128	0	0	0.005	1	1.00
136	1	0	0.000	0	0.00
139	1	0	0.000	0	0.00

Goodness-of-fit for edgewise shared partner

obs	min	mean	max MC	p-value
546.00	527.00	546.07	570.00	1.00

A.4.5 Model 4



Sample statistics summary:

Iterations = 24397:487041
 Thinning interval = 124
 Number of chains = 9
 Sample size per chain = 3732

1. Empirical mean and standard deviation for each variable,
 plus standard error of the mean:

	Mean	SD	Naive SE	Time-series SE
edges	1.4885	7.342	0.04006	0.13991
b1degree1	-0.2421	5.158	0.02815	0.08707
cycle4	2.9974	21.952	0.11978	0.36758
b1factor.b1_gender.male	1.2371	5.804	0.03167	0.13091
b1cov.b1_age	68.2142	290.100	1.58291	5.53954

2. Quantiles for each variable:

	2.5%	25%	50%	75%	97.5%
edges	-13.0	-3	2	6	16
bidegree1	-11.0	-4	0	3	9
cycle4	-34.0	-13	1	16	52
bifactor.bi_gender.male	-10.0	-3	1	5	13
b1cov.bi_age	-493.3	-128	66	262	642

Are sample statistics significantly different from observed?

	edges	bidegree1	cycle4	bifactor.bi_gender.male
diff.	1.488508e+00	-0.242080505	2.997410e+00	1.237108e+00
test stat.	1.067485e+01	-2.788939985	8.253381e+00	9.459740e+00
P-val.	1.334799e-26	0.005288086	1.539758e-16	3.087086e-21
	b1cov.bi_age	(Omni)		
diff.	6.821421e+01	NA		
test stat.	1.234136e+01	1.836478e+02		
P-val.	5.423515e-35	1.961837e-36		

Sample statistics cross-correlations:

	edges	bidegree1	cycle4
edges	1.0000000	-0.6194906	0.8444657
bidegree1	-0.6194906	1.0000000	-0.5563244
cycle4	0.8444657	-0.5563244	1.0000000
bifactor.bi_gender.male	0.5202118	-0.3085227	0.4001146
b1cov.bi_age	0.9278442	-0.5648680	0.7655343
	bifactor.bi_gender.male	b1cov.bi_age	
edges	0.5202118	0.9278442	
bidegree1	-0.3085227	-0.5648680	
cycle4	0.4001146	0.7655343	
bifactor.bi_gender.male	1.0000000	0.4940175	
b1cov.bi_age	0.4940175	1.0000000	

Sample statistics auto-correlation:

Chain 1					
	edges	bidegree1	cycle4	bifactor.bi_gender.male	b1cov.bi_age
Lag 0	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000
Lag 124	0.8374849	0.7942875	0.7803126	0.8871185	0.8414362
Lag 248	0.6983437	0.6303555	0.6121622	0.7828993	0.7078650
Lag 372	0.5875911	0.5139794	0.4829741	0.6943681	0.5983029
Lag 496	0.4942324	0.4232287	0.3869254	0.6236215	0.5049571
Lag 620	0.4165554	0.3530637	0.3131236	0.5611651	0.4286563
Chain 2					
	edges	bidegree1	cycle4	bifactor.bi_gender.male	b1cov.bi_age
Lag 0	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000
Lag 124	0.8383225	0.7916470	0.7806649	0.8891942	0.8368086
Lag 248	0.7002352	0.6310644	0.6246221	0.7854262	0.6984174
Lag 372	0.5905196	0.5020604	0.5019041	0.6964845	0.5904724
Lag 496	0.5001746	0.4028386	0.4089298	0.6196010	0.5006779
Lag 620	0.4244166	0.3227622	0.3352208	0.5524969	0.4271361
Chain 3					
	edges	bidegree1	cycle4	bifactor.bi_gender.male	b1cov.bi_age
Lag 0	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000
Lag 124	0.8615095	0.8196269	0.7953017	0.8950554	0.8572147
Lag 248	0.7410270	0.6783742	0.6389925	0.8019235	0.7339498
Lag 372	0.6323685	0.5562304	0.5143062	0.7177998	0.6210777
Lag 496	0.5423396	0.4579825	0.4248686	0.6461454	0.5253416
Lag 620	0.4711679	0.3790521	0.3656032	0.5799580	0.4538163
Chain 4					
	edges	bidegree1	cycle4	bifactor.bi_gender.male	b1cov.bi_age
Lag 0	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000
Lag 124	0.8561850	0.8062786	0.7835227	0.8972264	0.8625570
Lag 248	0.7406347	0.6512770	0.6281802	0.8069659	0.7508933
Lag 372	0.6429638	0.5281485	0.5154130	0.7245146	0.6543449
Lag 496	0.5597939	0.4335170	0.4229578	0.6536452	0.5722606
Lag 620	0.4859649	0.3583084	0.3431372	0.5874509	0.4985239
Chain 5					

	edges	bidegree1	cycle4	bifactor.bi_gender.male	bicov.bi_age
Lag 0	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000
Lag 124	0.8483661	0.8106589	0.7941002	0.8895643	0.8523591
Lag 248	0.7166964	0.6615552	0.6418816	0.7910443	0.7234805
Lag 372	0.6070666	0.5521223	0.5189634	0.6987463	0.6181309
Lag 496	0.5112907	0.4615184	0.4173753	0.6120248	0.5242244
Lag 620	0.4308833	0.3952802	0.3465701	0.5350108	0.4454119

Chain 6

	edges	bidegree1	cycle4	bifactor.bi_gender.male	bicov.bi_age
Lag 0	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000
Lag 124	0.8440858	0.8169353	0.7926372	0.8794121	0.8483472
Lag 248	0.7072567	0.6661694	0.6289095	0.7780482	0.7142171
Lag 372	0.5889945	0.5478361	0.5072255	0.6925403	0.5967909
Lag 496	0.4919709	0.4477178	0.4088420	0.6202981	0.4991670
Lag 620	0.4093479	0.3623102	0.3314522	0.5515389	0.4146804

Chain 7

	edges	bidegree1	cycle4	bifactor.bi_gender.male	bicov.bi_age
Lag 0	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000
Lag 124	0.8276469	0.7840563	0.7680361	0.8812961	0.8303218
Lag 248	0.6838055	0.6143846	0.5907489	0.7776281	0.6871530
Lag 372	0.5563137	0.4767232	0.4611630	0.6858163	0.5627388
Lag 496	0.4511159	0.3735897	0.3588736	0.6088053	0.4610083
Lag 620	0.3611492	0.2954504	0.2796229	0.5461082	0.3715808

Chain 8

	edges	bidegree1	cycle4	bifactor.bi_gender.male	bicov.bi_age
Lag 0	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000
Lag 124	0.8502719	0.8158044	0.8033231	0.8961069	0.8551644
Lag 248	0.7212653	0.6720300	0.6556043	0.8067487	0.7284289
Lag 372	0.6089928	0.5499170	0.5400857	0.7258174	0.6171646
Lag 496	0.5151437	0.4553121	0.4498942	0.6572668	0.5236445
Lag 620	0.4360382	0.3809871	0.3782412	0.5912337	0.4455866

Chain 9

	edges	bidegree1	cycle4	bifactor.bi_gender.male	bicov.bi_age
Lag 0	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000
Lag 124	0.8566705	0.8092882	0.7907467	0.8862144	0.8512702
Lag 248	0.7370720	0.6654765	0.6403479	0.7898657	0.7272172
Lag 372	0.6361992	0.5476526	0.5251665	0.7077886	0.6240128
Lag 496	0.5434866	0.4496682	0.4222984	0.6302241	0.5327234
Lag 620	0.4643092	0.3647572	0.3475467	0.5626955	0.4569503

Sample statistics burn-in diagnostic (Geweke):

Chain 1

Fraction in 1st window = 0.1

Fraction in 2nd window = 0.5

	edges	bidegree1	cycle4
	-0.1688130	-0.3143331	0.1829230
bifactor.bi_gender.male		bicov.bi_age	
	0.5842642	0.3253693	

Individual P-values (lower = worse):

	edges	bidegree1	cycle4
	0.8659437	0.7532680	0.8548584
bifactor.bi_gender.male		bicov.bi_age	
	0.5590426	0.7449016	

Joint P-value (lower = worse): 0.8466799

Chain 2

Fraction in 1st window = 0.1

Fraction in 2nd window = 0.5

	edges	bidegree1	cycle4
	0.1634696	-1.7034344	-0.1104156
bifactor.bi_gender.male		bicov.bi_age	
	-1.3277024	-0.3132087	

Individual P-values (lower = worse):

	edges	bidegree1	cycle4
	0.87014867	0.08848681	0.91207977
bifactor.b1_gender.male		b1cov.b1_age	
	0.18427644	0.75412209	

Joint P-value (lower = worse): 0.3843382

Chain 3

Fraction in 1st window = 0.1
 Fraction in 2nd window = 0.5

	edges	bidegree1	cycle4
	-2.7773335	0.7191514	-2.9640630
bifactor.b1_gender.male		b1cov.b1_age	
	-0.2985897	-1.4002474	

Individual P-values (lower = worse):

	edges	bidegree1	cycle4
	0.005480690	0.472047664	0.003036062
bifactor.b1_gender.male		b1cov.b1_age	
	0.765253095	0.161439250	

Joint P-value (lower = worse): 0.02025937

Chain 4

Fraction in 1st window = 0.1
 Fraction in 2nd window = 0.5

	edges	bidegree1	cycle4
	-0.9135621	0.9209371	-1.9148485
bifactor.b1_gender.male		b1cov.b1_age	
	2.0238862	-0.7310445	

Individual P-values (lower = worse):

	edges	bidegree1	cycle4
	0.36094699	0.35708327	0.05551182
bifactor.b1_gender.male		b1cov.b1_age	
	0.04298186	0.46475196	

Joint P-value (lower = worse): 0.1181432

Chain 5

Fraction in 1st window = 0.1
 Fraction in 2nd window = 0.5

	edges	bidegree1	cycle4
	-0.83100175	2.14295138	-1.38371681
bifactor.b1_gender.male		b1cov.b1_age	
	0.91877545	-0.06611033	

Individual P-values (lower = worse):

	edges	bidegree1	cycle4
	0.4059726	0.0321170	0.1664452
bifactor.b1_gender.male		b1cov.b1_age	
	0.3582130	0.9472900	

Joint P-value (lower = worse): 0.4623169

Chain 6

Fraction in 1st window = 0.1
 Fraction in 2nd window = 0.5

	edges	bidegree1	cycle4
	-0.81548604	0.71614808	-1.40917569
bifactor.b1_gender.male		b1cov.b1_age	
	0.71236910	0.07078785	

Individual P-values (lower = worse):

	edges	bidegree1	cycle4
	0.4147942	0.4738999	0.1587832
bifactor.b1_gender.male		b1cov.b1_age	
	0.4762362	0.9435666	

Joint P-value (lower = worse): 0.4325175
Chain 7

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

	edges	b1degree1	cycle4
	0.5264058	-0.9639856	0.4135393
bifactor.b1_gender.male	2.4152942	b1cov.b1_age	
		0.4394015	

Individual P-values (lower = worse):

	edges	b1degree1	cycle4
	0.59860632	0.33505316	0.67921157
bifactor.b1_gender.male		b1cov.b1_age	
	0.01572251	0.66037066	

Joint P-value (lower = worse): 0.379943
Chain 8

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

	edges	b1degree1	cycle4
	-2.865211	1.354826	-3.222967
bifactor.b1_gender.male		b1cov.b1_age	
	-1.976794	-3.141841	

Individual P-values (lower = worse):

	edges	b1degree1	cycle4
	0.004167309	0.175472883	0.001268703
bifactor.b1_gender.male		b1cov.b1_age	
	0.048064957	0.001678894	

Joint P-value (lower = worse): 0.04766405
Chain 9

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

	edges	b1degree1	cycle4
	-0.9140193	0.1936344	-0.2219224
bifactor.b1_gender.male		b1cov.b1_age	
	-1.5686701	-0.7949470	

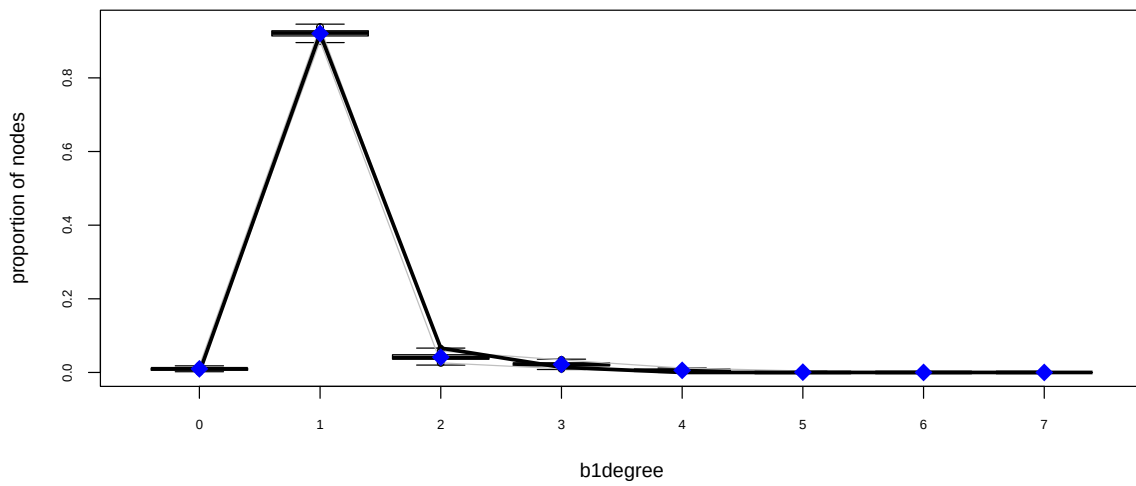
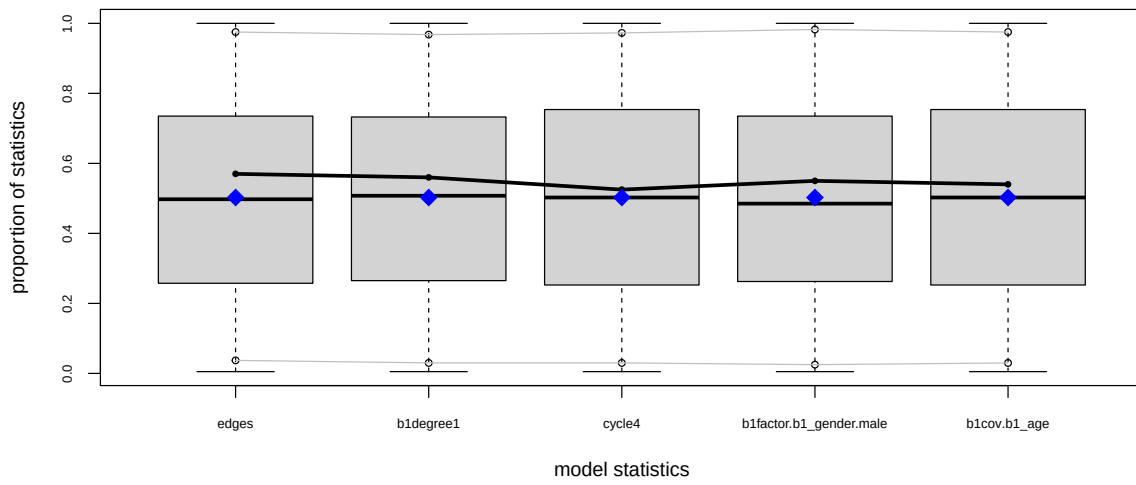
Individual P-values (lower = worse):

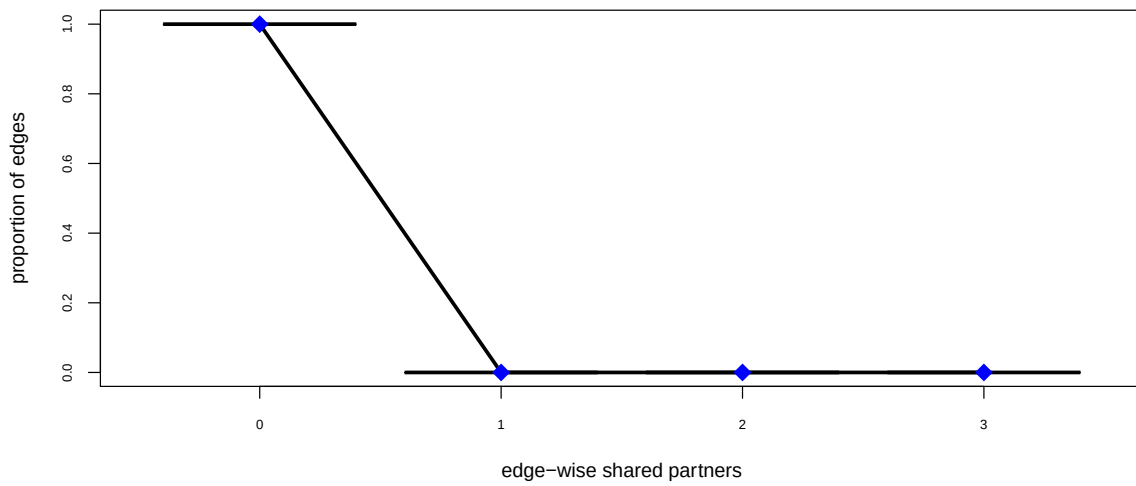
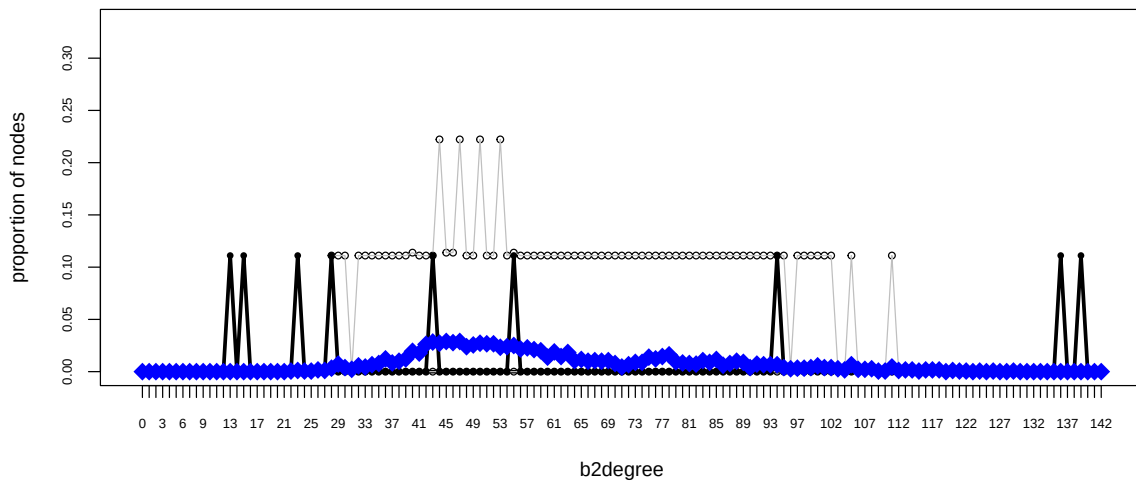
	edges	b1degree1	cycle4
	0.3607067	0.8464621	0.8243743
bifactor.b1_gender.male		b1cov.b1_age	
	0.1167248	0.4266443	

Joint P-value (lower = worse): 0.2069806

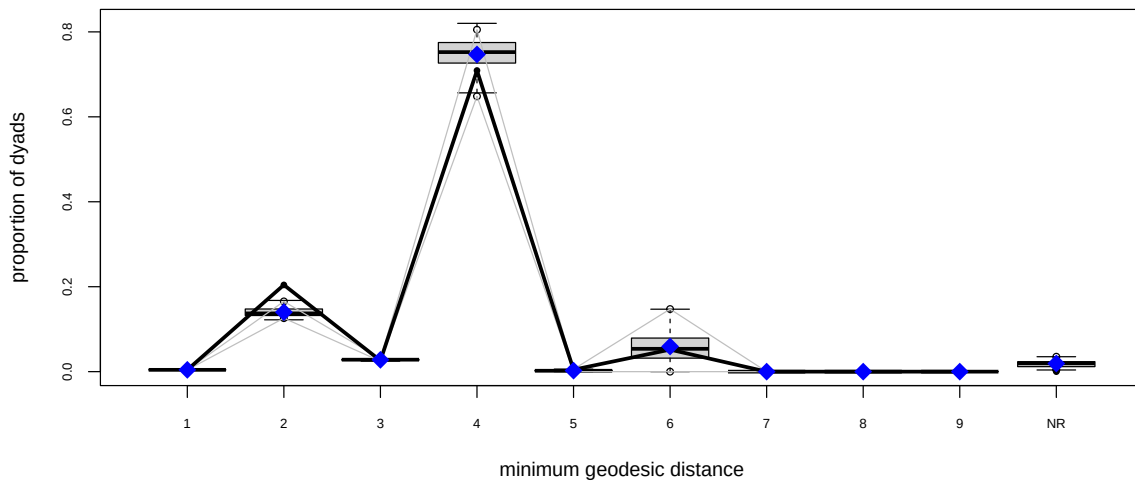
Note: To save space, only one in every 4 iterations of the MCMC sample used for estimation was stored for diagnostics. Sample size per chain was originally around 14928 with thinning interval 31.

Note: MCMC diagnostics shown here are from the last round of simulation, prior to computation of final parameter estimates. Because the final estimates are refinements of those used for this simulation run, these diagnostics may understate model performance. To directly assess the performance of the final model on in-model statistics, please use the GOF command: `gof(ergmFitObject, GOF=-model)`.





Goodness-of-fit diagnostics



Goodness-of-fit for model statistics

	obs	min	mean	max	MC p-value
edges	546	528	546.755	567	0.93
b1degree1	459	445	459.310	472	0.95
cycle4	67	29	68.370	148	0.98
b1factor.b1_gender.male	257	244	257.430	280	1.00
b1cov.b1_age	20959	20177	20992.900	21751	0.92

Goodness-of-fit for minimum geodesic distance

	obs	min	mean	max	MC p-value
1	546	528	546.755	567	0.93
2	26301	15719	18052.580	22225	0.00
3	3351	2939	3586.890	3925	0.27
4	91309	78517	96179.115	105613	0.32
5	594	0	313.345	948	0.16
6	6677	0	7631.630	24797	0.96
Inf	0	507	2467.685	6513	0.00

Goodness-of-fit for bipartition 1 degree

	obs	min	mean	max	MC p-value
0	0	1	4.890	13	0.00
1	459	445	459.310	472	0.95
2	33	10	20.650	33	0.01
3	7	3	10.900	18	0.22
4	0	0	2.840	9	0.11
5	0	0	0.380	3	1.00
6	0	0	0.025	1	1.00
7	0	0	0.005	1	1.00

Goodness-of-fit for bipartition 2 degree

	obs	min	mean	max	MC p-value
13	1	0	0.000	0	0.00
15	1	0	0.000	0	0.00
22	0	0	0.005	1	1.00
23	1	0	0.010	1	0.02
24	0	0	0.005	1	1.00
25	0	0	0.005	1	1.00
26	0	0	0.015	1	1.00

27	0	0	0.010	1	1.00
28	1	0	0.030	1	0.06
29	0	0	0.065	2	1.00
30	0	0	0.035	2	1.00
31	0	0	0.020	1	1.00
32	0	0	0.050	1	1.00
33	0	0	0.040	1	1.00
34	0	0	0.060	2	1.00
35	0	0	0.070	2	1.00
36	0	0	0.110	2	1.00
37	0	0	0.075	2	1.00
38	0	0	0.090	1	1.00
39	0	0	0.110	2	1.00
40	0	0	0.175	2	1.00
41	0	0	0.160	2	1.00
42	0	0	0.235	2	1.00
43	1	0	0.255	2	0.48
44	0	0	0.245	2	1.00
45	0	0	0.260	2	1.00
46	0	0	0.250	2	1.00
47	0	0	0.260	2	1.00
48	0	0	0.215	2	1.00
49	0	0	0.230	2	1.00
50	0	0	0.245	2	1.00
51	0	0	0.240	2	1.00
52	0	0	0.240	2	1.00
53	0	0	0.210	3	1.00
54	0	0	0.215	2	1.00
55	1	0	0.225	2	0.40
56	0	0	0.195	2	1.00
57	0	0	0.205	2	1.00
58	0	0	0.190	2	1.00
59	0	0	0.180	2	1.00
60	0	0	0.125	2	1.00
61	0	0	0.170	2	1.00
62	0	0	0.130	2	1.00
63	0	0	0.165	2	1.00
64	0	0	0.095	1	1.00
65	0	0	0.105	3	1.00
66	0	0	0.090	1	1.00
67	0	0	0.095	2	1.00
68	0	0	0.090	2	1.00
69	0	0	0.095	2	1.00
70	0	0	0.070	1	1.00
71	0	0	0.040	1	1.00
72	0	0	0.060	1	1.00
73	0	0	0.080	2	1.00
74	0	0	0.080	1	1.00
75	0	0	0.125	2	1.00
76	0	0	0.110	2	1.00
77	0	0	0.130	2	1.00
78	0	0	0.145	1	1.00
79	0	0	0.085	1	1.00
80	0	0	0.075	1	1.00
81	0	0	0.070	1	1.00
82	0	0	0.065	1	1.00
83	0	0	0.095	1	1.00
84	0	0	0.080	2	1.00
85	0	0	0.105	2	1.00
86	0	0	0.055	1	1.00
87	0	0	0.070	1	1.00
88	0	0	0.090	1	1.00
89	0	0	0.080	1	1.00
90	0	0	0.035	1	1.00
91	0	0	0.065	1	1.00
92	0	0	0.060	1	1.00
93	0	0	0.060	1	1.00
94	1	0	0.060	1	0.12

95	0	0	0.035	2	1.00
96	0	0	0.025	1	1.00
97	0	0	0.030	1	1.00
98	0	0	0.030	1	1.00
99	0	0	0.035	1	1.00
100	0	0	0.050	2	1.00
101	0	0	0.035	1	1.00
102	0	0	0.035	1	1.00
103	0	0	0.025	1	1.00
104	0	0	0.015	1	1.00
105	0	0	0.060	2	1.00
106	0	0	0.020	1	1.00
107	0	0	0.020	1	1.00
108	0	0	0.025	1	1.00
109	0	0	0.005	1	1.00
110	0	0	0.005	1	1.00
111	0	0	0.040	2	1.00
112	0	0	0.010	1	1.00
113	0	0	0.015	1	1.00
114	0	0	0.015	1	1.00
115	0	0	0.005	1	1.00
116	0	0	0.015	1	1.00
117	0	0	0.015	1	1.00
118	0	0	0.015	1	1.00
120	0	0	0.010	1	1.00
121	0	0	0.005	1	1.00
122	0	0	0.005	1	1.00
124	0	0	0.005	1	1.00
126	0	0	0.005	1	1.00
129	0	0	0.005	1	1.00
136	1	0	0.000	0	0.00
139	1	0	0.000	0	0.00

Goodness-of-fit for edgewise shared partner

obs	min	mean	max MC	p-value
546.000	528.000	546.755	567.000	0.930

A.5 Source Code - Data Preprocessing

```

1 # ----- #
2 # If not already installed
3 install.packages("viridis") # For Colours
4 install.packages("here")    # To locate files from RProj
5
6 # Ensure repeatability
7 set.seed(42)
8 # ----- #
9 # Import the Bondora P2P Dataset
10 obj_paths = "resources/objects/"
11 bondora_raw <- read.csv("dataset/LoanData_Bondora.csv")
12 raw_cols <- colnames(bondora_raw)
13 # ----- #
14 # Select Columns to Keep
15 keep_cols <- c("LoanId", "UserName", "Age", "Gender",
16               "Country", "Amount", "Interest", "LoanDuration",
17               "UseOfLoan", "Rating", "Restructured", "MonthlyPayment",
18               "OccupationArea", "BiddingStartedOn", "Country")
19
20 bondora <- bondora_raw[keep_cols]
21
22 # Change date format to the correct one
23 bondora$BiddingStartedOn <- as.POSIXct(bondora$BiddingStartedOn,
24                                       format = "%Y-%m-%d %H:%M:%S")
25 bondora$year <- as.numeric(format(bondora$BiddingStartedOn, "%Y"))
26
27 # Remove Rows with any NAs -> Complete Dataset Preferred
28 print(paste("NA Count |", sum(is.na(bondora)), "rows"))
29 bondora <- na.omit(bondora)
30 print(paste("NA Count |", sum(is.na(bondora)), "rows"))
31
32 # Observe the distribution of Loan Use over Time
33 unique_counts <- tapply(bondora$UserName,
34                         list(bondora$year, bondora$UseOfLoan),
35                         function(x) length(unique(x)))
36 unique_counts[is.na(unique_counts)] <- 0
37
38 barplot(t(unique_counts),
39         beside = TRUE,
40         col = viridis::viridis(ncol(unique_counts)),
41         legend.text = colnames(unique_counts),
42         args.legend = list(x = "topright", cex = 0.8),
43         xlab = "Year",
44         ylab = "Number of Unique Users")
45 user_counts_plt <- recordPlot()
46 saveRDS(user_counts_plt,
47         here::here("resources", "objects", "preprocessing", "user_cnt_plt.Rds"))
48
49 # Restrict to Recent Time Period
50 bondora_test <- bondora[bondora$year > 2013 & bondora$year < 2017, ]
51
52 # See the distribution of LoanUses
53 see_counts <- function(var1, var2) {
54   counts <- table(var1, var2)
55   total_counts <- colSums(counts)
56   prop <- total_counts / sum(total_counts)
57
58   return(list(total_counts, round(prop, 2)))
59 }
60 before_counts <- see_counts(bondora_test$UserName, bondora_test$UseOfLoan)
61 barplot(before_counts[[1]])
62
63 # Number of unique users in the subsample
64 unique_users <- unique(bondora_test$UserName)
65 sample_users <- sample(unique_users, 500)

```

```

66 bondora_clean <- bondora_test[bondora_test$UserName %in% sample_users, ]
67
68 after_counts <- see_counts(bondora_clean$UserName, bondora_clean$UseOfLoan)
69 barplot(after_counts[[1]])
70
71 # DEPRECATED METHOD
72 # Filter People with 2+ Loans
73 #user_counts <- table(bondora_test$UserName)
74 #multi_users <- names(user_counts[user_counts > 2])
75 #bondora_clean <- bondora_test[bondora_test$UserName %in% multi_users, ]
76
77 # DEPRECATED METHOD
78 # Remove Users with only One Loan
79 #user_counts <- table(bondora_clean$UserName)
80 #multi_users <- names(user_counts[user_counts > 5])
81 #bondora_clean <- bondora_clean[bondora_clean$UserName %in% multi_users, ]
82 #bondora_test <- bondora_raw[bondora_raw$UserName %in% multi_users, ]
83
84 # See if Ratings are Properly Encoded
85 unique(bondora_clean$Rating)
86 bondora_clean <- bondora_clean[bondora_clean$Rating != "", ]
87
88 # Extract UseofLoan Types and Turn into Factor
89 bondora_clean$UseOfLoan_factor <- as.factor(bondora_clean$UseOfLoan)
90 unique(bondora_clean$UseOfLoan_factor)
91
92 loan_use_labels <- c(
93   "-1" = "NA",
94   "0" = "Loan Consolidation",
95   "1" = "Real Estate",
96   "2" = "Home Improvement",
97   "3" = "Business",
98   "4" = "Education",
99   "5" = "Travel",
100  "6" = "Vehicle",
101  "7" = "Other",
102  "8" = "Health",
103  "110" = "Other Business",
104  "102" = "Undefined Business",
105  "108" = "Undefined Business"
106 )
107 # Change Labels for Cleaned Dataset
108 bondora_clean$UseOfLoan_factor <- loan_use_labels[
109   as.character(bondora_clean$UseOfLoan)]
110
111 # Change Labels for Uncleaned Dataset
112 bondora_test$UseOfLoan_factor <- loan_use_labels[
113   as.character(bondora_test$UseOfLoan)]
114
115 # Add labels to the OccupationArea Variable
116 levels(as.factor(bondora_clean$OccupationArea)) # view codes
117
118 occupation_labels <- c(
119   "-1" = "NA",
120   "1" = "Other",
121   "2" = "Mining",
122   "3" = "Processing",
123   "4" = "Energy",
124   "5" = "Utilities",
125   "6" = "Construction",
126   "7" = "Retail",
127   "8" = "Transport",
128   "9" = "Hospitality",
129   "10" = "Info/Telcom",
130   "11" = "Finance",
131   "12" = "Realestate",

```



```

132 "13" = "Research",
133 "14" = "Admin",
134 "15" = "Civil/Mil",
135 "16" = "Education",
136 "17" = "Healthcare",
137 "18" = "Social",
138 "19" = "Art/Ent",
139 "20" = "Agriculture",
140 "21" = "Forestry/Fish"
141 )
142 # store original
143 bondora_clean$occupation_code <- bondora_clean$OccupationArea
144 bondora_clean$occupation_label <- occupation_labels[
145   as.character(bondora_clean$OccupationArea)]
146
147 bondora_test$occupation_code <- bondora_test$OccupationArea
148 bondora_test$occupation_label <- occupation_labels[
149   as.character(bondora_test$OccupationArea)]
150 # ----- #
151 # Observe Descriptive Statistics
152
153 cols <- viridis::viridis(30)
154
155 # Make function to save plots
156 save_plot <- function(plt_nam) {
157   plt <- recordPlot()
158   saveRDS(plt, here::here("resources", "objects", "preprocessing",
159     paste0(plt_nam, ".Rds")))
160 }
161
162 # Make function to consistently plot comparisons
163 plot_desc_hists <- function(df1, df2, col_name, type) {
164
165   par(mfrow=c(1,3))
166
167   hist(df1[[col_name]], xlab=type, col=cols[15], main="", breaks=10)
168   mtext("Initial Sample", side=3, adj=0, line=0.25, cex=1, font=2)
169
170   hist(df2[[col_name]], xlab=type, col=cols[15], main="", breaks=10)
171   mtext("Reduced Sample", side=3, adj=0, line=0.25, cex=1, font=2)
172
173   qqplot(df1[[col_name]], df2[[col_name]], main="", cex=1,
174     xlab="Full Sample", ylab="Subsample", line=0.25)
175   abline(0, 1, lty=2)
176
177   mtext("QQ Plot", side=3, adj=0, line=0.25, cex=1, font=2)
178
179   mtext(type, outer = TRUE, line = -2, side=3, cex = 1.3, font = 2)
180
181   # Reset plot window
182   par(mfrow=c(1,1), mar=c(5,4,4,2)+0.1)
183 }
184
185 plot_desc_bar <- function(df1, df2, col_name, type) {
186
187   par(mfrow=c(1,2), mar=c(5,10,4,2))
188
189   barplot(sort(table(df1[[col_name]]), decreasing = F),
190     xlab=type, col=cols[15], horiz=TRUE, las=1)
191   mtext("Initial Sample", side=3, adj=0, line=0.25, cex=1, font=2)
192
193   barplot(sort(table(df2[[col_name]]), decreasing = F),
194     xlab=type, col=cols[15], horiz=TRUE, las=1)
195   mtext("Reduced Sample", side=3, adj=0, line=0.25, cex=1, font=2)
196
197   mtext(type, outer = TRUE, line = -2, side=3, cex = 1.3, font = 2)

```

```

198
199 # Reset plot window
200 par(mfrow=c(1,1), mar=c(5,4,4,2)+0.1)
201 }
202
203 plot_desc_hists(bondora_test, bondora_clean, "Amount", "Amount")
204 save_plot("hist_amt")
205
206 plot_desc_hists(bondora_test, bondora_clean, "Interest", "Interest")
207 save_plot("hist_int")
208
209 plot_desc_hists(bondora_test, bondora_clean, "LoanDuration", "Loan Duration")
210 save_plot("hist_loan_dur")
211
212 plot_desc_hists(bondora_test, bondora_clean, "MonthlyPayment", "Monthly Payment")
213 save_plot("hist_monpmt")
214
215 plot_desc_hists(bondora_test, bondora_clean, "Age", "Age")
216 save_plot("hist_age")
217
218 plot_desc_bar(bondora_test, bondora_clean, "UseOfLoan_factor", "Loan Purpose")
219 save_plot("bar_loanuse")
220
221 plot_desc_bar(bondora_test, bondora_clean, "Rating", "Credit Rating")
222 save_plot("bar_rating")
223
224 plot_desc_bar(bondora_test, bondora_clean, "occupation_label", "Occupation")
225 save_plot("bar_occupation")
226
227 plot_desc_bar(bondora_test, bondora_clean, "Country", "Country")
228 save_plot("bar_country")
229
230 # Get Tabular Summary Statistics
231 tab_comps <- function(df1, df2, cols) {
232   stats <- c("Mean"=mean, "Median"=median, "Std. Dev."=sd, "Min"=min, "Max"=max)
233
234   get_stats <- function(d) {
235     t(sapply(d[cols], function(x)
236       sapply(stats, function(f) round(f(x, na.rm=TRUE), 2))
237     ))
238   }
239
240   df1_stats <- get_stats(df1)
241   df2_stats <- get_stats(df2)
242
243   out <- rbind(
244     cbind(DataFrame = "Initial Sample",
245           Variable = rownames(df1_stats), df1_stats),
246     cbind(DataFrame = "Reduced Sample",
247           Variable = rownames(df2_stats), df2_stats)
248   )
249   rownames(out) <- NULL
250   as.data.frame(out)
251 }
252
253 tab_results <- tab_comps(bondora_test, bondora_clean,
254   c("Amount", "Interest", "Age", "LoanDuration"))
255
256 knitr::kable(tab_results)
257
258 # Save table for use in the Report
259 saveRDS(tab_results,
260   file=paste0(obj_paths, "preprocessing/", "summary_table.Rds"))
261 # ----- #
262 # Convert Dataset into Incidence Matrix to form Network Object (for ERGM)
263 bondora_slim <- bondora_clean

```

```

264
265 # Create the Incidence Matrix for Use of Loan
266 bondora_matrix <- table(
267   bondora_slim$UserName, bondora_slim$UseOfLoan)
268 bondora_matrix[bondora_matrix > 0] <- 1 # Given that ergm.counts fails with GOF
269
270 # Create network object with counts as Edge attribute
271 bondora_net <- network::network(
272   bondora_matrix, directed=FALSE, bipartite=nrow(bondora_matrix),
273   ignore.eval = FALSE, names.eval="frequency", loops=FALSE)
274
275 # Set the bipartite Attribute - UNNECESSARY GIVEN bipartite=length(n)
276 len <- dim(bondora_matrix)[1]
277 len_b2 <- dim(bondora_matrix)[2]
278 b_indicator <- c(rep(1,len),rep(2,len_b2))
279 network::set.vertex.attribute(bondora_net,
280                               "bipartite",value=b_indicator)
281
282 # Extract Partition 2 Labels
283 loan_use <- levels(bondora_clean$UseOfLoan_factor)
284
285 # Create Loan Type Attribute for Partition 2
286 b2_loantype <- rep(NA, len)
287 b2_loantype <- c(b2_loantype, loan_use)
288
289 if (length(b2_loantype) == network::network.size(bondora_net)) {
290   network::set.vertex.attribute(
291     bondora_net, "b2_loantype", value = b2_loantype)
292 }
293
294 # Add Age Vertex Attribute to B1
295 age <- bondora_clean$Age[match(
296   rownames(bondora_matrix), bondora_clean$UserName)]
297 b1_age <- c(age, rep(NA, len_b2))
298 network::set.vertex.attribute(
299   bondora_net, "b1_age", value=b1_age
300 )
301
302 # Add Gender Vertex Attribute to B1
303 gender <- bondora_clean$Gender[match(
304   rownames(bondora_matrix), bondora_clean$UserName)]
305 unique(gender) # Check to see if encoded properly
306 gender <- ifelse(gender == 0, "male", "female")
307 b1_gender <- c(gender, rep(NA, len_b2))
308 network::set.vertex.attribute(
309   bondora_net, "b1_gender", value = b1_gender
310 )
311
312 # Save the network and data frame object
313 saveRDS(bondora_slim, file=paste0(obj_paths, "preprocessing/", "bondora_df.Rds"))
314 saveRDS(bondora_net, file=paste0(obj_paths, "preprocessing/", "bondora_net.Rds"))
315 # ----- #
316 # First, create Incidence Matrix again but with Frequency Counts
317 loan_use_matrix <- table(
318   bondora_slim$UserName, bondora_slim$UseOfLoan)
319
320 # Get the Adjacency Matrix for Loan Use Similarity (Dependent QAP Variable)
321 adj_mat_loan_use <- loan_use_matrix %>% t(loan_use_matrix)
322 # Remove self weights to remove any loops
323 diag(adj_mat_loan_use) <- 0
324
325 # Get the Adjacency Matrix for Credit Rating Similarity (Predictor in QAP)
326 incidence_rating <- table(bondora_slim$UserName, bondora_slim$Rating)
327 adj_mat_rating <- incidence_rating %>% t(incidence_rating)
328 diag(adj_mat_rating) <- 0
329

```

```

330 # Get Adjacency Matrix for Occupation Similarity (Predictor in QAP, Binary)
331 incidence_occupation <- table(bondora_slim$UserName,
332                               bondora_slim$occupation_label)
333 adj_mat_occupation <- incidence_occupation %*% t(incidence_occupation)
334 adj_mat_occupation[adj_mat_occupation > 0] <- 1
335 diag(adj_mat_occupation) <- 0
336
337 # Get the Adjacency Matrix for Loan Amount (Control in QAP) - BINARY
338 # First, bin the Loan Amounts
339 bondora_slim$Amount_bins <- cut(
340   bondora_slim$Amount, breaks=c(0,2000,4000,6000,8000,10000),
341   labels = c(1:5)
342 )
343 incidence_amount_bins <- table(bondora_slim$UserName, bondora_slim$Amount_bins)
344 adj_mat_amount_bins <- incidence_amount_bins %*% t(incidence_amount_bins)
345 diag(adj_mat_amount_bins) <- 0
346
347 # Continous Absolute Difference Approach
348 avg_amount <- tapply(bondora_slim$Amount, bondora_slim$UserName, mean)
349 adj_mat_amount_diff <- outer(avg_amount, avg_amount,
350                               FUN = function(x,y) abs(x - y))
351 diag(adj_mat_amount_diff) <- 0
352
353 # Get Matrix for Differences in Age
354 incidence_age <- table(bondora_slim$UserName, bondora_slim$Age)
355 borrower_ages <- as.numeric(colnames(incidence_age)[max.col(incidence_age)])
356 names(borrower_ages) <- rownames(incidence_age)
357 adj_mat_age <- outer(borrower_ages, borrower_ages,
358                     FUN = function(x, y) abs(x - y))
359 rownames(adj_mat_age) <- colnames(adj_mat_age) <- names(borrower_ages)
360
361 # Get Adjacency Matrix for (same) Gender
362 incidence_gender <- table(bondora_slim$UserName, bondora_slim$Gender)
363 adj_mat_gender <- incidence_gender %*% t(incidence_gender)
364 # Make the matrix binary for homophily
365 adj_mat_gender <- ifelse(adj_mat_gender > 0, 1, 0)
366 diag(adj_mat_gender) <- 0
367
368 # Get Adjacency Matrix for Differences in Average Loan Duration
369 borrower_loandur <- sapply(tapply(bondora_slim$LoanDuration,
370                                   bondora_slim$UserName, unique),
371                             mean)
372 adj_mat_loandur_diff <- outer(borrower_loandur, borrower_loandur,
373                               FUN=function(x,y) abs(x-y))
374 diag(adj_mat_loandur_diff) <- 0
375
376 # Get Adjacency Matrix for Homophily in Restructure of Loans
377 incidence_restructure <- table(bondora_slim$UserName, bondora_slim$Restructured)
378 adj_mat_rest <- incidence_restructure %*% t(incidence_restructure)
379 diag(adj_mat_rest) <- 0
380
381 # Save the objects for the QAP Regression in different Script
382 qap_paths = paste0(obj_paths, "/qap/")
383 saveRDS(b_indicator, file=paste0(
384   "resources/objects/preprocessing/indicator.Rds"))
385
386 saveRDS(adj_mat_loan_use, file=paste0(qap_paths, "adj_mat_loanuse.Rds"))
387 saveRDS(adj_mat_occupation, file=paste0(qap_paths, "adj_mat_occup.Rds"))
388 saveRDS(adj_mat_rating, file=paste0(qap_paths, "adj_mat_rating.Rds"))
389 saveRDS(adj_mat_amount_diff, file=paste0(qap_paths, "adj_mat_amtdiffs.Rds"))
390 saveRDS(adj_mat_age, file=paste0(qap_paths, "adj_mat_agediffs.Rds"))
391 saveRDS(adj_mat_gender, file=paste0(qap_paths, "adj_mat_gender.Rds"))
392 saveRDS(adj_mat_loandur_diff, file=paste0(qap_paths, "adj_mat_loandurdiffs.Rds"))
393 saveRDS(adj_mat_rest, file=paste0(qap_paths, "adj_mat_rest.Rds"))
394 # ----- #

```

scripts/bondora_preprocessing.R

A.6 Source Code - QAP Linear Regression

```
1 # ----- #
2 # If not already Installed
3 install.packages("viridis") # For Colours
4 install.packages("here") # To locate files from RProj
5 install.packages("gridExtra") # For multiple ggplot2 plots
6
7 # Set colour palette
8 cols <- viridis::viridis(30)
9
10 # Ensure repeatability
11 set.seed(42)
12 # ----- #
13 # Load Relevant Files
14 qap_path="resources/objects/qap/"
15
16 loan_use_mat <- readRDS(paste0(qap_path, "adj_mat_loanuse.RDS"))
17 rating_mat <- readRDS(paste0(qap_path, "adj_mat_rating.RDS"))
18 amt_diffs_mat <- readRDS(paste0(qap_path, "adj_mat_amtdiffs.RDS"))
19 age_diffs_mat <- readRDS(paste0(qap_path, "adj_mat_agediffs.RDS"))
20 gender_mat <- readRDS(paste0(qap_path, "adj_mat_gender.RDS"))
21 loandur_diffs_mat <- readRDS(paste0(qap_path, "adj_mat_loandurdiffs.RDS"))
22 rest_mat <- readRDS(paste0(qap_path, "adj_mat_rest.RDS"))
23 occup_mat <- readRDS(paste0(qap_path, "adj_mat_occup.Rds"))
24 # ----- #
25 # Create function to determine significance from t-value statistic
26 t_to_stars <- function(t) {
27   stars <- rep("", length(t))
28   stars[abs(t) >= 1.96] <- "*"
29   stars[abs(t) >= 2.576] <- "**"
30   stars[abs(t) >= 3.291] <- "***"
31
32   return(stars)
33 }
34
35 # Make function to save plots
36 save_qap_plot <- function(plt_nam) {
37   plt <- recordPlot()
38   saveRDS(plt, here::here("resources", "objects", "qap",
39     paste0(plt_nam, ".Rds")))
40 }
41
42 # Make function to run QAPs automatically
43 run_QAP <- function(y_mat, xlist, varnames, model_name) {
44
45   # Run QAP Linear Regression
46   model <- sna::netlm(y = y_mat, x = xlist, nullhyp = "qapspp", reps = 3000)
47   model$names <- varnames
48   summary(model)
49
50   # Obtain results from Model and Name them
51   results <- model$coefficients
52   names(results) <- varnames
53
54   # Add significance stars to model results
55   results_sig <- paste(round(results,3), t_to_stars(model$tstat))
56
57   # Save Model
58   saveRDS(model, here::here("resources", "objects", "qap",
59     paste0(model_name, ".Rds")))
60
61   # Prepare function output
62   list_names <- c("model", "results", "results_sig")
63   items <- list(model, results, results_sig)
64   names(items) <- list_names
65 }
```

```

66 |   return(items)
67 | }
68 |
69 | # Make function to Plot QAP Coefficients
70 | plot_coef_qap <- function(model,
71 |                           model_name = "Model Estimates",
72 |                           sub = "Subtitle",
73 |                           x_axis_label = "Coefficient Estimate") {
74 |
75 |   # Extract Coefficients and SEs from ERM
76 |   coefs <- model$coefficients
77 |   ses <- model$coefficients / model$tstat
78 |
79 |   # Make DF
80 |   df <- data.frame(
81 |     term = model$names,
82 |     estimate = coefs,
83 |     se = ses
84 |   )
85 |
86 |   df$lower <- df$estimate - 1.96 * df$se
87 |   df$upper <- df$estimate + 1.96 * df$se
88 |
89 |   df$sig <- df$lower * df$upper > 0
90 |   df$sig <- as.character(df$sig) # Convert to character to make it work
91 |
92 |   order_index <- order(df$estimate)
93 |   df$term <- factor(df$term, levels = df$term[order_index])
94 |
95 |   ggplot2::ggplot(df, ggplot2::aes(x = estimate, y = term)) +
96 |     ggplot2::geom_vline(xintercept = 0, linetype = "dashed", color = "grey40") +
97 |     ggplot2::geom_errorbarh(ggplot2::aes(xmin = lower, xmax = upper),
98 |                           height = 0.2) +
99 |     ggplot2::geom_point(size = 3) +
100 |     ggplot2::theme_bw() +
101 |     ggplot2::theme(
102 |       panel.grid.major.x = ggplot2::element_blank(),
103 |       panel.grid.minor.x = ggplot2::element_blank()) +
104 |     ggplot2::labs(
105 |       x = x_axis_label,
106 |       y = "Variable",
107 |       title = model_name,
108 |       subtitle = sub
109 |       #color = "Significance"
110 |     )
111 | }
112 | # Make Function to Plot Diagnostics
113 | # DEPRECATED SINCE PROGRESS MEETING 3
114 | qap_diags <- function(model, title, filename) {
115 |   resid <- model$model$residuals
116 |   fitted <- model$model$fitted.values
117 |
118 |   # Set viewing window to two plots
119 |   par(mfrow=c(1,2))
120 |
121 |   # Residuals Plot
122 |   hist(resid, xlab="Residuals",
123 |        main=title)
124 |
125 |   # Fitted vs Residuals
126 |   plot(fitted, resid, xlab="Fitted Values", ylab="Residuals",
127 |        main=title)
128 |
129 |   # Capture and Save the Plot
130 |   save_qap_plot(filename)
131 |

```

```

132 # Reset plot view
133 par(mfrow=c(1,1))
134 }
135
136 var_names <- c("Intercept", "Rating", "Occupation", "Loan Amount", "Age",
137               "Gender", "Loan Duration", "Restructured")
138 main_pred_names <- c("Intercept", "Rating", "Occupation")
139 main_pred_vars <- list(rating_mat, occup_mat)
140 all_pred_vars <- list(rating_mat, occup_mat, amt_diffs_mat, age_diffs_mat,
141                      gender_mat, loandur_diffs_mat, rest_mat)
142 # ----- #
143 # Basic QAP Linear Regression 1 - Unstandardised + No Controls
144 qap_m1 <- run_QAP(loan_use_mat, main_pred_vars, main_pred_names, "qap_m1")
145 summary(qap_m1$model)
146
147 # Plot the QAP Model Coefficients and Save it
148 qap_m1_coef_plot <- plot_coef_qap(qap_m1$model,
149                                  model_name = "MRQAP Model 1",
150                                  sub = "No Controls")
151 saveRDS(qap_m1_coef_plot,
152         here::here("resources", "objects", "qap", "qap_m1_coef_plot.Rds"))
153
154 # Run Diagnostics Plots - DEPRECATED SINCE PROGRESS MEETING 3 SUGGESTIONS
155 #qap_diags(qap_m1, "QAP LR - Unstandardised + No Controls", "qap_m1_diags")
156
157 # ----- #
158 # Basic QAP Linear Regression 2 - Unstandardised + Controls
159 qap_m3 <- run_QAP(loan_use_mat, all_pred_vars, var_names, "qap_m3")
160
161 summary(qap_m3$model)
162
163 # Plot the QAP Model Coefficients and Save it
164 qap_m3_coef_plot <- plot_coef_qap(qap_m3$model,
165                                  model_name = "MRQAP Model 2",
166                                  sub = "With Controls")
167 saveRDS(qap_m3_coef_plot,
168         here::here("resources", "objects", "qap", "qap_m3_coef_plot.Rds"))
169
170 # Arrange the Plots together
171 qap_coef_plots <- gridExtra::grid.arrange(qap_m1_coef_plot, qap_m3_coef_plot,
172                                           ncol = 2)
173
174 # Save the Plots
175 saveRDS(qap_coef_plots,
176         here::here("resources", "objects", "qap", "qap_coef_plots.Rds"))
177
178 # Run Diagnostics Plots - DEPRECATED SINCE PROGRESS MEETING 3 SUGGESTIONS
179 #qap_diags(qap_m3, "QAP LR - Unstandardised + Controls", "qap_m3_diags")
180
181 # ----- #
182 # Collect Results in Table
183
184 # Extract R2 from the netlm object from the Console output
185 # It was too difficult to calculate R2 from scratch and netlm does not
186 # provide this as an output directly from the netlm object
187 calc_r2 <- function(model) {
188   out <- capture.output(model)
189   r2_line <- grep("Multiple R-squared", out, value = TRUE)
190   r2 <- as.numeric(
191     sub(".*Multiple R-squared:\\\\s*([0-9\\.]+).*", "\\1", r2_line))
192   return(r2)
193 }
194
195
196 # Make function to get details of netlm manually
197 extract_netlm <- function(model) {

```

```

198   tr <- texreg::createTexreg(
199     coef.names = model$names,
200     coef = model$coefficients,
201     se = model$coefficients / model$tstat,
202     pvalues = model$pgreqabs,
203     gof.names = c("R-squared"),
204     gof = (calc_r2(model)),
205     gof.decimal = c(TRUE)
206   )
207   return(tr)
208 }
209
210 # View the Table of Results
211 titles <- c("M1 | No Controls", "M2 | Controls")
212 extracted_models <- lapply(list(qap_m1$model, qap_m3$model), extract.netlm)
213 texreg::screenreg(extracted_models, custom.model.names = titles, digits = 3)
214
215 # Save the table for use in the Report
216 saveRDS(list(extracted_models, titles),
217         here::here("resources", "objects", "qap", "qap_tables.Rds"))
218 # ----- #

```

scripts/qap_network_analysis.R

A.7 Source Code - ERGM Network Analysis

```
1 # ----- #
2 # If not already Installed
3 install.packages("viridis")      # For Colours
4 install.packages("Rglpk")       # Additional solver for ERGMs
5 install.packages("here")       # To locate files from RProj
6 install.packages("gridExtra")  # To display multiple ggplots side by side
7
8 # Import the Network and Other Object
9 ergm_path <- "resources/objects/ergm/"
10 bondora_net <- readRDS(here::here(
11   "resources", "objects", "preprocessing", "bondora_net.Rds"))
12 b_indicator <- readRDS(here::here(
13   "resources", "objects", "preprocessing", "indicator.Rds"))
14
15 # Set colour palette
16 cols <- viridis::viridis(30)
17
18 # Determine acceptable core count
19 n_cores <- parallel::detectCores() - 3 # Leave some out for other processes
20 print(paste("You have", n_cores, "usable cores"))
21
22 # Repeatability
23 set.seed(42)
24
25 # Save plots
26 save_ergm_plot <- function(plt_nam) {
27   plt <- recordPlot()
28   saveRDS(plt, here::here("resources", "objects", "ergm",
29     paste0(plt_nam, ".Rds")))
30 }
31 # ----- #
32 # Copy network for plotting
33 bondora_plot <- bondora_net
34
35 # Get node type for plotting
36 type_indicator <- ifelse(b_indicator == 2, TRUE, FALSE)
37 shape <- ifelse(type_indicator, "square", "circle")
38 network::set.vertex.attribute(bondora_plot, "shape", shape)
39
40 # Get Category Count for Vertex Size
41 counts <- sna::degree(bondora_plot)
42 counts_att <- ifelse(type_indicator, log(counts)*4, counts*2.5)
43 network::set.vertex.attribute(bondora_plot, "size", counts_att)
44
45 # Colours for the Node Types
46 plot_cols <- ifelse(type_indicator, cols[5], cols[30])
47 network::set.vertex.attribute(bondora_plot, "color", plot_cols)
48
49 # Legend Plotting
50 type_legend <- ifelse(type_indicator, "Borrowers", "Loan Type")
51 type_legend <- as.factor(type_legend)
52
53 # Plot the Network
54 plot(snafun::to_igraph(bondora_plot),
55   #main = "Bipartite User-LoanUse",
56   edge.arrow.size = 0.3,
57   edge.color = rgb(0,0,0, alpha = 0.35),
58   vertex.frame.color = "black",
59   vertex.label = NA,
60   vertex.frame.size = 3,
61   edge.curved = FALSE,
62   layout=igraph::layout_fruchterman_reingold)
63 legend("bottomleft",
64   legend = levels(type_legend),
65   inset = c(0.15, 0.01),
```

```

66     col = c(cols[30], cols[5]),
67     pch = c(16, 15),
68     title = "Node Partitions",
69     title.font = 2,
70     cex = 1,           # Increase the text size
71     pt.cex = 2,        # Increase the point symbol size
72     box.lwd = 1,        # Thin box border
73     box.col = "black",  # Box color
74     bty = "o"          # Use a box around legend
75 )
76 save_ergm_plot("network_plot")
77
78 # Summary Statistics and Save Them
79 density <- snafun::g_density(bondora_net)[1]
80 centralization <- snafun::g_centralize(bondora_net)[1]
81 vertices <- snafun::count_vertices(bondora_net)[1]
82 edges <- snafun::count_edges(bondora_net)[1]
83 dist <- snafun::g_mean_distance(bondora_net)[1]
84
85 net_names <- c("Vertex Count", "Edge Count", "Density", "Centralization",
86               "Mean Distance")
87 net_stats <- c(vertices, edges, density, centralization, dist)
88 net_stats <- sapply(net_stats, function(x) round(as.numeric(x), 2))
89
90 net_summary <- data.frame(Statistic = net_names,
91                           "Measure" = net_stats)
92 knitr::kable(net_summary)
93 saveRDS(net_summary, here::here("resources", "objects", "ergm", "net_summary.Rds"))
94
95 # Plot Network Summary Statistics
96 snafun::plot_centralities(bondora_net)
97 save_ergm_plot("network_plots")
98 # ----- #
99 # Make function to calculate probabilities from log odds
100 lodds_to_prob <- function(l_odd) {
101   return(exp(l_odd) / (1 + exp(l_odd)))
102 }
103 # Make function to save ERGM object
104 save_ergm <- function(object, id) {
105   saveRDS(object, file=here::here(
106     "resources", "objects", "ergm", paste0(id, ".Rds")))
107 }
108 # Make function to conduct ERGMs automatically
109 auto_ergm <- function(model, mcmc, name) {
110
111   # Conducts the GOF Diagnostics and then saves the model,
112   # mcmc diagnostics and gof object in a list.
113   # This list can be imported as an .RDS object into the R environment
114
115   # Diagnostics
116   if (mcmc) {
117     ergm::mcmc.diagnostics(model)
118   }
119
120   # The GOF must be adjusted otherwise it takes too long
121   # We do not limit the GOF by changing its range of parameters
122   gof <- ergm::gof(model,
123                     control = ergm::control.gof.ergm(
124                       nsim = 200,
125                       MCMC.burnin = 5000,
126                       MCMC.interval = 1000,
127                       parallel = n_cores,
128                       parallel.type = "PSOCK"
129                     ))
130
131   # Return List to view each item separately

```

```

132 result <- list(model, gof)
133 names(result) <- c("model", "gof")
134 save_ergm(result, paste0(name, "_panel"))
135
136 return(result)
137 }
138 # Make Function to do Coefficient Plot
139 plot_coef <- function(model,
140                       model_name = "Model Estimates",
141                       sub = "Subtitle",
142                       x_axis_label = "Coefficient Estimate") {
143
144   # Extract Coefficients and SEs from ERGM
145   coefs <- coef(model)
146   ses <- sqrt(diag(vcov(model)))
147
148   # Make DF
149   df <- data.frame(
150     term = names(coefs),
151     estimate = coefs,
152     se = ses
153   )
154
155   df$lower <- df$estimate - 1.96 * df$se
156   df$upper <- df$estimate + 1.96 * df$se
157
158   df$sig <- df$lower * df$upper > 0
159   df$sig <- as.character(df$sig) # Convert to character to make it work
160
161   order_index <- order(df$estimate)
162   df$term <- factor(df$term, levels = df$term[order_index])
163
164   ggplot2::ggplot(df, ggplot2::aes(x = estimate, y = term)) +
165     ggplot2::geom_vline(xintercept = 0, linetype = "dashed", color = "grey40") +
166     ggplot2::geom_errorbarh(ggplot2::aes(xmin = lower, xmax = upper),
167                           height = 0.2) +
168     ggplot2::geom_point(size = 3) +
169     ggplot2::theme_bw() +
170     ggplot2::theme(
171       panel.grid.major.x = ggplot2::element_blank(),
172       panel.grid.minor.x = ggplot2::element_blank()) +
173     ggplot2::labs(
174       x = x_axis_label,
175       y = "Variable",
176       title = model_name,
177       subtitle = sub
178       #color = "Significance"
179   )
180 }
181 # ----- #
182 # Find max degree
183 (max_deg <- max(summary(bondora_net ~ b2factor("b2_loantype"))))
184 # ----- #
185 # Base Model + GOF
186 formula_base_model <- bondora_net ~ edges
187 base_ergm <- ergm::ergm(formula_base_model)
188
189 base_ergm_panel <- auto_ergm(base_ergm, mcmc = FALSE, name = "ergm_base")
190
191 snafun::stat_plot_gof(base_ergm_panel$gof)
192 models = list(base_ergm)
193 texreg::screenreg(models)
194 # ----- #
195 # Iteration 1 + MCMC Diagnostics + GOF
196 model_1_params <- bondora_net ~ edges +
197   # See if individuals tend to pick one loan type

```

```

198   b1degree(1)
199
200 model_1 <- ergm::ergm(
201   model_1_params,
202   #constraints = ~ bd(minout = 0, maxout = 80),
203
204   control = ergm::control.ergm(
205     # Greater burn-in for cleaner result
206     MCMC.burnin = 20000,
207     # Greater sample size for greater stability
208     MCMC.samplesize = 100000,
209     seed = 42,
210     MCMC.interval = 1000,
211     # Only needed for convergence pvals to improve
212     MCMLE.maxit = 45,
213     # Smaller steps for stability
214     MCMLE.steplength = 0.25,
215     parallel = n_cores,
216     parallel.type = "PSOCK"
217   )
218 )
219
220 model_1_panel <- auto_ergm(model=model_1, mcmc=TRUE, name="ergm_m1")
221 model_1_panel$gof
222
223 snafun::stat_plot_gof(model_1_panel$gof)
224 save_ergm_plot("m1_gof")
225
226 texreg::screenreg(list(base_ergm, model_1))
227
228 # Plot Coefficients
229 m1_coef_plot <- plot_coef(model_1, model_name = "Model 1",
230                           sub = "b1degree")
231 saveRDS(m1_coef_plot,
232         here::here("resources", "objects", "ergm", "m1_coef_plot.Rds"))
233 # ----- #
234 # Iteration 2 + MCMC Diagnostics + GOF
235 model_2_params <- bondora_net ~ edges + b1degree(1) +
236   # See if there is clustering around pairs of loan types
237   cycle(4)
238
239 model_2 <- ergm::ergm(
240   model_2_params,
241   #constraints = ~ bd(minout = 0, maxout = 80),
242
243   control = ergm::control.ergm(
244     # Greater burn-in for cleaner result
245     MCMC.burnin = 20000,
246     # Greater sample size for greater stability
247     MCMC.samplesize = 100000,
248     seed = 42,
249     MCMC.interval = 1000,
250     # Only needed for convergence pvals to improve
251     MCMLE.maxit = 45,
252     # Smaller steps for stability
253     MCMLE.steplength = 0.25,
254     parallel = n_cores,
255     parallel.type = "PSOCK"
256   )
257 )
258
259 model_2_panel <- auto_ergm(model=model_2, mcmc=TRUE, name="ergm_m2")
260 snafun::stat_plot_gof(model_2_panel$gof)
261 model_2_panel$gof
262 models <- list(base_ergm, model_1, model_2)
263 texreg::screenreg(models)

```

```

264
265 # Plot Coefficients
266 m2_coef_plot <- plot_coef(model_2_panel$model,
267                           model_name = "Model 2",
268                           sub = "b1degree + cycle(4)")
269 saveRDS(m2_coef_plot,
270         here::here("resources", "objects", "ergm", "m2_coef_plot.Rds"))
271 # ----- #
272 # Iteration 3 + MCMC Diagnostics + GOF
273 model_3_params <- bondora_net ~ edges + b1degree(1) + cycle(4) +
274
275 # See if there is gender effect
276 b1factor("b1_gender")
277
278 model_3 <- ergm::ergm(
279   model_3_params,
280   #constraints = ~ bd(minout = 0, maxout = 80),
281
282   control = ergm::control.ergm(
283     # Greater burn-in for cleaner result
284     # Cycles mixed with the rest of the terms requires more burnin samples
285     MCMC.burnin = 20000,
286     # Greater sample size for greater stability
287     MCMC.samplesize = 100000,
288     seed = 42,
289     # Reducing interval to reduce complexity between intervals
290     MCMC.interval = 1000,
291     # Only needed for convergence pvals to improve
292     MCMLE.maxit = 25,
293     # Smaller steps for stability
294     MCMLE.steplength = 0.25,
295     parallel = n_cores,
296     parallel.type = "PSOCK"
297   )
298 )
299
300 model_3_panel <- auto_ergm(model=model_3, mcmc=TRUE, name="ergm_m3")
301 snafun::stat_plot_gof(model_3_panel$gof)
302 model_3_panel$gof
303 models <- list(base_ergm, model_1, model_2, model_3)
304 texreg::screenreg(models)
305
306 # Plot Coefficients
307 m3_coef_plot <- plot_coef(model_3_panel$model,
308                           model_name = "Model 3",
309                           sub = "b1degree + cycle(4) + b1factor(gender)")
310 # Save Plot
311 saveRDS(m3_coef_plot,
312         here::here("resources", "objects", "ergm", "m3_coef_plot.Rds"))
313 # ----- #
314 # Iteration 4 + MCMC Diagnostics + GOF
315 model_4_params <- bondora_net ~ edges + b1degree(1) + cycle(4) +
316   b1factor("b1_gender") +
317
318 # See if higher ages make a difference
319 b1cov("b1_age")
320
321 model_4 <- ergm::ergm(
322   model_4_params,
323   #constraints = ~ bd(minout = 0, maxout = 80),
324
325   control = ergm::control.ergm(
326     # Greater burn-in for cleaner result
327     MCMC.burnin = 20000,
328     # Greater sample size for greater stability
329     MCMC.samplesize = 100000,

```

```

330     seed = 42,
331     MCMC.interval = 1000,
332     # Only needed for convergence pvals to improve
333     MCMC.maxit = 45,
334     # Smaller steps for stability
335     MCMC.steplength = 0.25,
336     parallel = n_cores,
337     parallel.type = "PSOCK"
338 )
339 )
340
341 model_4_panel <- auto_ergm(model=model_4, mcmc=TRUE, name="ergm_m4")
342 model_4_panel$gof
343 snafun::stat_plot_gof(model_4_panel$gof)
344 models <- list(base_ergm, model_1, model_2, model_3, model_4)
345 texreg::screenreg(models)
346
347 # Plot Coefficients
348 m4_ergm_coef_plot <- plot_coef(model_4_panel$model,
349                                model_name = "Model 4",
350                                sub = "Unrestricted Model")
351
352 # Save Plot
353 saveRDS(m4_ergm_coef_plot,
354         here::here("resources", "objects", "ergm", "m4_coef_plot.Rds"))
355 # ----- #
356 # Collect Plots Together
357 gridExtra::grid.arrange(m1_coef_plot, m2_coef_plot,
358                          m3_coef_plot, m4_ergm_coef_plot,
359                          nrow = 2, ncol = 2)
360
361 # Collect Models in Table 1: Raw results
362 headers <- c("Base ERGM", "b1degree(1)", "cycle(4)", "Age", "Gender")
363 texreg::knitreg(models, custom.model.names = headers,
364                 digits = 3)
365
366 # Save the Table to use in Report
367 saveRDS(list(models, headers),
368         here::here("resources", "objects", "ergm", "ergm_table.Rds"))
369
370 # Padding
371 pad <- function(x, max) {
372   return(c(unname(x), rep("-", max-length(x))))
373 }
374
375 # Collect Models in Table 2: Log Odds Results
376 max_ergms <- length(model_4$coefficients)
377 df_prob <- data.frame(
378   Terms = names(model_4$coefficients),
379   "Base" = pad(round(lodds_to_prob(base_ergm$coefficients), 3), max_ergms),
380   "b1degree(1)" = pad(round(lodds_to_prob(model_1$coefficients), 3), max_ergms),
381   "cycle(4)" = pad(round(lodds_to_prob(model_2$coefficients), 3), max_ergms),
382   "Age" = pad(round(lodds_to_prob(model_3$coefficients), 3), max_ergms),
383   "Gender" = pad(round(lodds_to_prob(model_4$coefficients), 3), max_ergms)
384 )
385 knitr::kable(df_prob)
386
387 # Save Table to use in Report
388 saveRDS(df_prob,
389         here::here("resources", "objects", "ergm", "ergm_prob_tables.Rds"))
390 # ----- #

```

scripts/ergm_network_analysis.R

B Technology Statement

During the preparation of this work, we used ChatGPT in order to generate select parts of the R script utilised to process the dataset. Specifically, the tool was used to transform the processed dataset into a format that igraph would accept as a network object. Additionally, some R functions written by the group were improved with ChatGPT's suggestions. No AI tool was utilised to independently write parts of the report. The following parts of the assignment were affected/generated by AI tool usage: **INTRODUCTION**; The tool was utilised to evaluate the validity of certain theoretical concepts and refine them. **DATASET**; the data described within this section was processed partly by some code drafted by ChatGPT and edited by the group. After using this tool/service, **Group 7** evaluated the validity of the tool's outputs, including the sources that generative AI tools have used, and edited the content as needed. As a consequence, **Group 7** takes full responsibility for the content of their work.

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